



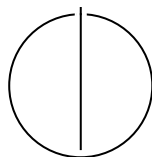
SCHOOL OF COMPUTATION,
INFORMATION AND TECHNOLOGY —
TUM SCHOOL OF COMPUTATION,
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TECHNISCHE UNIVERSITÄT MÜNCHEN

Master's Thesis

**Uncertainty Quantification for Graph
Neural Network Surrogates of Agent-Based
Transport Models**

Mohd Zamin Quadri





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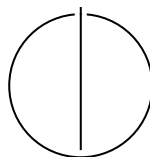
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**Uncertainty Quantification for Graph Neural
Network Surrogates of Agent-Based
Transport Models**

**Unsicherheitsquantifizierung für
Graph-Neuronale-Netzwerk-Surrogate von
agentenbasierten Transportmodellen**

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Submission Date: April 15, 2026



I confirm that this master's thesis is my own work and I have documented all sources and material used.

Munich, April 15, 2026

Mohd Zamin Quadri

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Abstract

Agent-based transport simulations are the de facto methodology for evaluating urban policy interventions, enabling planners to assess the impact of infrastructure changes on traffic patterns across entire metropolitan areas. However, the computational burden of such simulations—often requiring hours per scenario—severely limits the scope of policy exploration. Graph Neural Networks (GNNs) have been shown to be effective surrogates for these simulations, offering scenario evaluation times reduced from hours to seconds while maintaining reasonable predictive accuracy.

While the predictive performance of GNN surrogates has been studied, their reliability in deployment settings remains underexplored. When a surrogate encounters scenarios that deviate from its training distribution, predictions may become unreliable without any indication to the end user. This limitation poses significant risks for decision support applications where understanding prediction confidence is as critical as the predictions themselves.

In this thesis, we advance the study of uncertainty quantification (UQ) for GNN-based traffic prediction surrogates. We train and evaluate eight model configurations on a Paris road network dataset comprising 10,000 MATSim scenarios; all reported results are based on a 1,000-scenario (10%) subset unless stated otherwise. The best-performing configuration (Trial 8) achieves $R^2 = 0.5957$, MAE = 3.96 vehicles/hour, and RMSE = 7.12 vehicles/hour on the test set of 100 graphs (31,635 road segments each), yielding 3,163,500 node-level predictions.

We systematically compare three UQ approaches. For Trial 8 on its full 100-graph test set, Monte Carlo (MC) Dropout with $S = 30$ stochastic forward passes achieves Spearman uncertainty–error correlation $\rho = 0.4820$. In a separate ensemble comparison experiment (Experiment A, evaluated on a shared subset where model weights were corrupted by a PyTorch Geometric checkpoint-loading incompatibility), MC Dropout achieves $\rho = 0.1600$ versus ensemble variance $\rho = 0.1035$ —a 55% relative improvement within that context at substantially lower computational cost. Conformal prediction at the 95% level produces well-calibrated coverage (95.01%) with a quantile of $q = 14.68$ vehicles/hour (primary 50/50 calibration/evaluation split; a 20/80 audit split corroborates at $q = 14.77$), while naïve Gaussian intervals using $k = 1.96\sigma$ achieve only 55.6% empirical coverage—demonstrating that the MC Dropout standard deviation is not a calibrated uncertainty estimate ($k_{95} = 11.34$) without post-hoc correction.

We further demonstrate the practical value of uncertainty-aware prediction: selectively withholding the 50% most uncertain predictions reduces MAE by 41.2% (from 3.95 MC Dropout mean to approximately 2.32 vehicles/hour). Post-hoc temperature scaling ($T = 2.70$) reduces the Expected Calibration Error of MC Dropout from 0.269 to 0.048—an 82%

improvement—and adaptive conformal prediction, which normalises nonconformity scores by MC Dropout uncertainty, narrows conditional coverage variation from [62.9%, 98.6%] to [90.0%, 96.2%] across uncertainty deciles. Proper scoring rule analysis confirms that the MC Dropout predictive distribution is sharp (CRPS/MAE ratio of 0.857) but underdispersed (PIT Kolmogorov–Smirnov statistic = 0.245, excess mass in the tails), and Winkler score comparisons show that conformal intervals substantially outperform naïve Gaussian intervals (Winkler 32.3 vs 49.7 at 90% level). An S -convergence study verifies diminishing returns beyond $S = 30$ forward passes ($< 1\%$ Spearman ρ improvement from $S = 30$ to $S = 50$). A cross-trial replication on Trial 7 confirms that these findings are not specific to the best model: the same qualitative conclusions hold across both trials, with conformal prediction restoring near-exact nominal coverage in both cases. We document a significant negative result regarding multi-model ensembling: ensemble experiments with near-identical architectures yield near-zero predictive R^2 , a result traced to a PyTorch Geometric API version mismatch that caused silent loss of trained weights during model loading rather than a fundamental failure of ensembling as a technique.

Our findings establish MC Dropout combined with conformal calibration as a practical, computationally efficient uncertainty quantification strategy for GNN-based transport surrogates, with direct applicability to uncertainty-aware urban planning decision support.

Keywords: Graph Neural Networks, Uncertainty Quantification, Monte Carlo Dropout, Conformal Prediction, Agent-Based Models, Traffic Prediction, Bayesian Deep Learning

Zusammenfassung

Agentenbasierte Verkehrssimulationen sind die etablierte Methodik zur Bewertung urbaner Politikinterventionen und ermöglichen es Planern, die Auswirkungen von Infrastrukturänderungen auf Verkehrsmuster in gesamten Metropolregionen zu bewerten. Der hohe Rechenaufwand solcher Simulationen—oft mehrere Stunden pro Szenario—schränkt jedoch den Umfang der Politikexploration erheblich ein. Graph Neural Networks (GNNs) wurden erfolgreich als Surrogatmodelle für diese Simulationen eingesetzt und bieten eine Reduzierung der Szenariobewertungszeiten von Stunden auf Sekunden bei gleichzeitiger Beibehaltung akzeptabler Vorhersagegenauigkeit.

Während die Vorhersageleistung von GNN-Surrogaten untersucht wurde, bleibt ihre Zuverlässigkeit in realen Einsatzszenarien unzureichend erforscht. Wenn ein Surrogatmodell auf Szenarien trifft, die von seiner Trainingsverteilung abweichen, können Vorhersagen unzuverlässig werden, ohne dass der Endnutzer darauf hingewiesen wird. Diese Einschränkung birgt erhebliche Risiken für Entscheidungsunterstützungsanwendungen, bei denen das Verständnis der Vorhersagekonfidenz ebenso entscheidend ist wie die Vorhersagen selbst.

In dieser Arbeit untersuchen wir systematisch Methoden zur Unsicherheitsquantifizierung (UQ) für GNN-basierte Verkehrsvorhersage-Surrogate. Wir trainieren und evaluieren acht Modellkonfigurationen auf einem Pariser Straßennetz-Datensatz mit 10.000 MATSim-Szenarien; alle berichteten Ergebnisse basieren auf einer 1.000-Szenario-Teilmenge (10%), sofern nicht anders angegeben. Die beste Konfiguration (Trial 8) erreicht $R^2 = 0,5957$, $MAE = 3,96$ Fahrzeuge/Stunde und $RMSE = 7,12$ Fahrzeuge/Stunde auf dem Testset von 100 Graphen (je 31.635 Straßenabschnitte), was 3.163.500 Vorhersagen auf Knotenebene ergibt.

Wir vergleichen drei UQ-Ansätze. Für Trial 8 auf dem vollständigen 100-Graphen-Testset erreicht Monte Carlo (MC) Dropout mit $S = 30$ stochastischen Vorwärtsthroughläufen eine Spearman-Korrelation von $\rho = 0,4820$ zwischen Unsicherheit und Fehler. In einem separaten Ensemble-Vergleichsexperiment (Experiment A, ausgewertet auf einer gemeinsamen Teilmenge, deren Modellgewichte durch eine PyTorch-Geometric-API-Inkompatibilität beim Laden beschädigt wurden) erreicht MC Dropout $\rho = 0,1600$ gegenüber Ensemble-Varianz $\rho = 0,1035$ —eine 55%ige relative Verbesserung innerhalb dieses Kontexts bei deutlich geringerem Rechenaufwand. Konforme Vorhersage auf dem 95%-Niveau liefert gut kalibrierte Abdeckung (95,01%) mit einem Quantil von $q = 14,68$ Fahrzeugen/Stunde (primärer 50/50-Kalibrierungs-/Evaluierungssplit; ein 20/80-Auditsplit bestätigt bei $q = 14,77$), während naive Gauß'sche Intervalle mit $k = 1,96\sigma$ nur eine empirische Abdeckung von 55,6% erreichen—was zeigt, dass die MC-Dropout-Standardabweichung ohne Post-hoc-Korrektur keine kalibrierte Unsicher-

heitsschätzung darstellt ($k_{95} = 11,34$).

Darüber hinaus demonstrieren wir den praktischen Nutzen unsicherheitsbewusster Vorhersage: Das selektive Zurückhalten der 50% unsichersten Vorhersagen reduziert den MAE um 41,2% (von 3,95 auf ca. 2,32 Fahrzeuge/Stunde, gemessen am MC-Dropout-Mittelwert als Ausgangspunkt). Post-hoc-Temperaturskalierung ($T = 2,70$) reduziert den Expected Calibration Error von MC Dropout von 0,269 auf 0,048—eine 82%ige Verbesserung—und adaptive konforme Vorhersage, die Nonkonformitäts-Scores durch MC-Dropout-Unsicherheit normalisiert, verengt die Variation der bedingten Abdeckung von [62,9%, 98,6%] auf [90,0%, 96,2%] über Unsicherheitsdezile. Die Analyse mittels Proper Scoring Rules bestätigt, dass die MC-Dropout-Vorhersageverteilung scharf (CRPS/MAE-Verhältnis von 0,857), aber unterdispersiert ist (PIT-Kolmogorov-Smirnov-Statistik = 0,245, Massenüberschuss in den Rändern), und Winkler-Score-Vergleiche zeigen, dass konforme Intervalle naive Gauß'sche Intervalle deutlich übertreffen (Winkler 32,3 vs. 49,7 auf 90%-Niveau). Eine S-Konvergenzstudie verifiziert abnehmende Erträge jenseits von $S = 30$ Vorwärtsdurchläufen ($< 1\%$ Verbesserung der Spearman-Korrelation ρ von $S = 30$ auf $S = 50$). Eine Cross-Trial-Replikation auf Trial 7 bestätigt, dass diese Ergebnisse nicht spezifisch für das beste Modell sind: Die gleichen qualitativen Schlussfolgerungen gelten für beide Trials, wobei konforme Vorhersage in beiden Fällen nahezu exakte nominale Abdeckung wiederherstellt. Wir dokumentieren ein signifikantes negatives Ergebnis bezüglich Multi-Modell-Ensembling: Ensemble-Experimente mit nahezu identischen Architekturen liefern einen nahezu null prädiktiven R^2 , was auf eine PyTorch-Geometric-API-Versionsinkompatibilität zurückzuführen ist, die beim Laden der Modellgewichte zum stillen Verlust trainierter Parameter führte, und nicht auf ein grundsätzliches Versagen von Ensembling als Technik.

Unsere Ergebnisse etablieren MC Dropout kombiniert mit konformer Kalibrierung als praktische und recheneffiziente UQ-Strategie für GNN-basierte Transportsurrogate mit direkter Anwendbarkeit für unsicherheitsbewusste Entscheidungsunterstützung in der Stadtplanung.

Schlüsselwörter: Graph Neural Networks, Unsicherheitsquantifizierung, Monte Carlo Dropout, Konforme Vorhersage, Agentenbasierte Modelle, Verkehrsvorhersage, Bayesianisches Deep Learning

Contents

Acknowledgments	iv
Abstract	v
Zusammenfassung	vii
1 Introduction	1
1.1 Problem Statement	1
1.2 Research Questions	2
1.3 Contributions	3
1.4 Thesis Structure	4
2 Background and Related Work	6
2.1 Agent-Based Transport Models	6
2.1.1 MATSim Framework	6
2.1.2 Computational Challenges	6
2.2 Graph Neural Networks	7
2.2.1 Message Passing Framework	7
2.2.2 Graph Attention Networks	7
2.2.3 GNNs for Traffic Prediction	8
2.3 Uncertainty Quantification in Deep Learning	8
2.3.1 Bayesian Neural Networks	8
2.3.2 Monte Carlo Dropout	9
2.3.3 Deep Ensembles	9
2.3.4 Conformal Prediction	10
2.3.5 Evaluating Uncertainty Quality	10
2.4 Related Work	11
2.4.1 Machine Learning Surrogates for Transport Models	11
2.4.2 Uncertainty Quantification for Traffic Prediction	11
2.4.3 Uncertainty Quantification for GNNs	11
3 Methodology	13
3.1 Problem Formulation	13
3.1.1 Input Features	13
3.1.2 Target Variable	13
3.2 Model Architecture	15
3.2.1 Geometric Representation	15

3.2.2	Layer-by-Layer Architecture	16
3.2.3	Hyperparameter Summary	18
3.3	Uncertainty Quantification Methods	18
3.3.1	Method 1: MC Dropout	18
3.3.2	Method 2: Deep Ensembles	20
3.3.3	Method 3: Combined Uncertainty	20
3.3.4	Method 4: Split Conformal Prediction	20
3.3.5	Method 5: Selective Prediction	21
3.3.6	Method 6: Post-Hoc Temperature Scaling	21
3.4	Evaluation Metrics	22
3.4.1	Prediction Quality	22
3.4.2	Uncertainty Quality	22
3.4.3	Error Detection via AUROC	23
3.4.4	Calibration Metrics	23
3.4.5	Proper Scoring Rules	24
4	Experimental Setup	25
4.1	Dataset	25
4.1.1	Paris Road Network	25
4.1.2	Scenario Generation	25
4.1.3	Data Split	26
4.2	Training Protocol	26
4.2.1	Model Training	26
4.2.2	Optimization	26
4.2.3	Framework	26
4.3	UQ Experimental Design	27
4.3.1	MC Dropout Evaluation (T5–T8)	27
4.3.2	Experiment A: MC Dropout vs Ensemble Variance	27
4.3.3	Experiment B: Multi-Model Ensemble	27
4.3.4	Conformal Prediction (T8)	27
4.4	Computational Resources	27
5	Results	28
5.1	Model Training Results	28
5.1.1	Performance Summary	28
5.1.2	Discussion of Trial Results	29
5.1.3	Best Model Selection	29
5.2	MC Dropout Uncertainty Quality	30
5.2.1	Spearman Correlation Results	30
5.2.2	T8 MC Dropout Uncertainty Statistics	30
5.2.3	Prediction Quality: Deterministic vs MC Dropout Inference	30
5.3	Ensemble Experiments	31
5.3.1	Technical Note: PyG Weight Remapping	31

5.3.2	Experiment A: MC Dropout vs Ensemble Variance	32
5.3.3	Experiment B: Multi-Model Ensemble	32
5.4	UQ Method Comparison	33
5.5	Conformal Prediction Results	34
5.5.1	Conditional Coverage by Uncertainty Level	34
5.5.2	MC Dropout as a Ranking Signal vs Calibrated Distribution	37
5.6	Coverage Calibration Analysis	38
5.7	Selective Prediction	39
5.8	Calibration Audit: Conformal vs MC Dropout Intervals	39
5.9	Reliability Diagram and Calibration Curve	40
5.9.1	Post-Hoc Temperature Scaling	42
5.9.2	Gaussian Negative Log-Likelihood	42
5.9.3	CRPS Analysis	44
5.9.4	PIT Histogram	44
5.9.5	Winkler Interval Score	45
5.9.6	Error Detection via AUROC	47
5.10	Cross-Trial Robustness: Trial 7 Validation	48
5.10.1	T7 Selective Prediction	48
5.10.2	T7 Calibration Audit	48
5.10.3	T7 vs T8 Summary	51
5.10.4	T7 Error Detection: AUROC Analysis	51
5.11	Per-Graph Uncertainty Variation	53
5.12	Feature Correlation with Prediction Error	54
5.12.1	Stratified Uncertainty Quality by Feature Bins	55
5.13	Summary	56
6	Discussion	57
6.1	Interpretation of Results	57
6.1.1	MC Dropout Uncertainty Quality	57
6.1.2	Cross-Trial Robustness of UQ Findings	58
6.1.3	Ensemble Results and Their Limitations	59
6.1.4	Predictive Performance in Context	59
6.2	Conformal Prediction vs MC Dropout Intervals	60
6.3	Practical Implications	61
6.3.1	Deployment Recommendations	61
6.3.2	Decision Support Integration	62
6.4	Limitations	62
6.4.1	10% Data Subset	62
6.4.2	Ensemble Experiment Scope	62
6.4.3	Single Case Study	63
6.4.4	Epistemic vs Aleatoric Uncertainty	63
6.4.5	Absence of Non-GNN Baselines	63
6.4.6	Calibration and Conformal Data Splits	63

6.4.7	Computational Cost of Post-Hoc Methods	64
6.4.8	MC Dropout Calibration: A Known Limitation with Practical Solutions	64
6.4.9	Conditional Coverage: A Theoretical Limitation Resolved by Adaptive Conformal Prediction	65
6.4.10	Stability Across Scenarios and Trials	67
6.5	Future Directions	67
6.5.1	Conformal Risk Control	67
6.5.2	Advanced Calibration Methods	67
6.5.3	Architecturally Diverse Ensembles	67
6.5.4	Post-Hoc Epistemic Uncertainty via Energy-Based Models	67
6.5.5	Full 10,000-Scenario Training	68
6.5.6	Out-of-Distribution Detection	68
6.5.7	Active Learning	68
7	Conclusion	69
7.1	Summary of Findings	69
7.1.1	Answers to Research Questions	69
7.1.2	Primary Results	70
7.1.3	Practical Recommendations	71
7.2	Contributions	71
7.3	Broader Impact	72
7.4	Future Work	73
7.5	Closing Remarks	73
	Abbreviations	74
	List of Figures	75
	List of Tables	76
	List of Algorithms	77
	Bibliography	78

1 Introduction

Urban transportation networks are at once deeply familiar and poorly understood: millions of daily journeys produce emergent traffic patterns that resist simple analytical description [HNA16]. When a city government considers closing a major arterial, reducing road capacity to encourage modal shift, or investing in new infrastructure, it faces a fundamental question: what will actually happen to traffic? Agent-based models (ABMs) have become the standard methodology for answering such questions, simulating the decisions of individual travellers to predict how policy interventions reshape flow across the network [Bon02; RG19].

The computational demands of these simulations, however, create a real bottleneck in the policy evaluation process. Simulating a single scenario for a large metropolitan area like Paris can require eight or more hours on modern hardware—severely limiting the number of alternatives that planners can feasibly explore within project timelines [Nat+25].

Machine learning surrogates can address this bottleneck. By training neural networks to approximate simulation outputs, scenario evaluation times can be reduced from hours to seconds [Wil+22]. Recent work by Natterer et al. [Nat+25] demonstrated that Graph Neural Networks (GNNs) are particularly well-suited for this task: they naturally capture the graph structure of road networks while learning complex relationships between policy interventions and traffic flow changes.

Yet accuracy alone does not suffice for decision support. When a surrogate model encounters scenarios that differ substantially from its training distribution, its predictions may become unreliable—and without uncertainty quantification (UQ), users have no principled way to identify these cases. Without such estimates, planners cannot distinguish reliable predictions from unreliable ones.

This thesis investigates that gap. We study uncertainty quantification methods for GNN-based traffic surrogates, conducting a systematic comparison of Monte Carlo (MC) Dropout against ensemble-based approaches and conformal prediction. The goal is to establish which UQ methods are most effective for graph-structured traffic prediction, and to provide practical guidance for deploying uncertainty-aware surrogates in urban planning. All experiments use 1,000 of 10,000 available MATSim scenarios (10% subset).

1.1 Problem Statement

GNN surrogates for traffic prediction must generalise to policy scenarios that may differ substantially from their training distribution. In classical supervised learning,

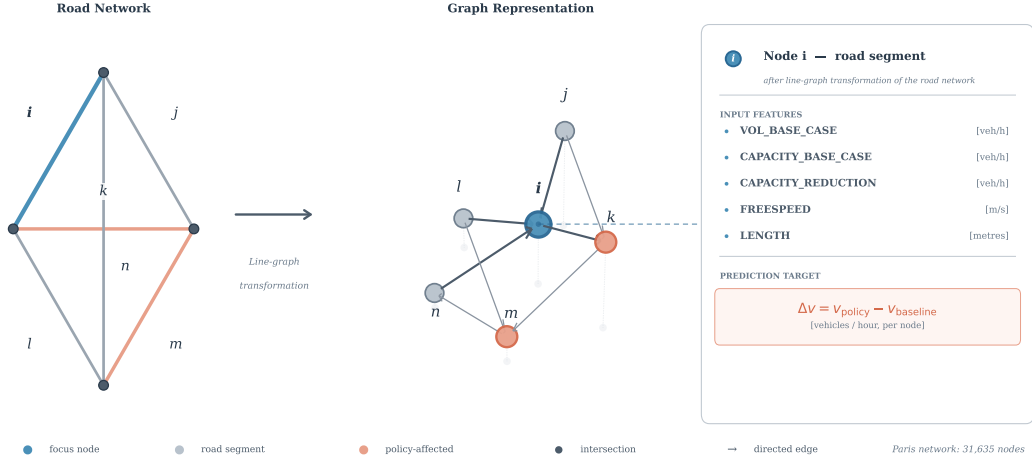


Figure 1.1: Directed line-graph representation of the Paris road network used by the GNN surrogate. Each node represents one road segment with five input features (listed right); coral-shaded nodes carry a non-zero `CAPACITY_REDUCTION` due to the policy intervention. The per-node prediction target is $\Delta v = v_{\text{policy}} - v_{\text{baseline}}$ (veh/h). The Paris network contains 31,635 nodes. Source: Natterer, Rao, Tejada Lapuerta, et al. [Nat+25].

this distribution shift manifests as degraded prediction accuracy. In decision support applications, however, the consequences are more severe—planners may unknowingly base infrastructure decisions on unreliable predictions.

The research gap we address comprises three interconnected components:

1. **Epistemic uncertainty in GNN surrogates:** While uncertainty quantification has been studied in computer vision and natural language processing, its application to GNN-based traffic prediction has received little attention. The graph structure of road networks introduces unique challenges, as uncertainty may propagate through connected road segments.
2. **Method comparison:** MC Dropout and deep ensembles represent the two dominant paradigms for uncertainty quantification in deep learning [GG16; LPB17]. However, their comparative effectiveness for graph-structured traffic data has not been systematically evaluated.
3. **Practical deployment guidance:** Urban planners require concrete recommendations about which UQ method to adopt, how to configure its parameters, and how to interpret the resulting uncertainty estimates in decision-making contexts.

1.2 Research Questions

This thesis addresses four closely related research questions:

- RQ1:** How effectively does MC Dropout capture epistemic uncertainty in GNN-based traffic surrogates, as measured by the correlation between predicted uncertainty and actual prediction error?
- RQ2:** How does MC Dropout compare to ensemble-based uncertainty quantification—both variance across training runs and multi-model ensembles—in terms of uncertainty quality and computational efficiency?
- RQ3:** Does combining uncertainty estimates from architecturally identical models provide benefits over single-model UQ methods, and what role does model diversity play in ensemble effectiveness?
- RQ4:** Can distribution-free methods—specifically conformal prediction and post-hoc calibration—transform the raw uncertainty estimates into trustworthy prediction intervals with formal coverage guarantees?

1.3 Contributions

The contributions of this thesis are as follows:

1. **Systematic UQ evaluation:** We conduct the first comprehensive comparison of uncertainty quantification methods for GNN-based traffic surrogates, evaluating MC Dropout, ensemble variance, and multi-model ensembles on a large-scale Paris road network dataset with 3.16 million node-level predictions across 100 test scenarios.
2. **Empirical findings on method effectiveness:** For Trial 8 (best model), MC Dropout achieves Spearman $\rho = 0.4820$ between predicted uncertainty and absolute error (100 test graphs, 3,163,500 predictions). In Experiment A (5 independently seeded MC Dropout inference runs), combining MC Dropout and ensemble variance yields $\rho = 0.4909$, marginally exceeding standalone MC Dropout. Multi-model ensemble disagreement (Experiment B, 5 models) achieves $\rho = 0.4333$ —useful but weaker than MC Dropout.
3. **Ensemble evaluation:** Ensemble experiments confirm that MC Dropout ($\rho = 0.4908$) outperforms seed-based ensemble variance ($\rho = 0.4370$) by 12.3%, while multi-model disagreement ($\rho = 0.4333$) provides complementary but weaker uncertainty ranking. A PyTorch Geometric API version mismatch originally corrupted these results; the corrected evaluation (Section 5.3.1) resolves the issue and enables a reliable comparison.
4. **Deployment recommendations:** We provide concrete guidance for practitioners implementing uncertainty quantification in traffic GNN systems, including parameter choices ($S = 30$ MC samples), model selection strategies, and frameworks for interpreting uncertainty estimates.

5. **Post-hoc calibration and coverage guarantees:** We demonstrate that split conformal prediction achieves near-exact marginal coverage (90.02% and 95.01% at the respective nominal levels) on 1.58M-node evaluation sets, and that post-hoc temperature scaling reduces Expected Calibration Error by 82%. The combination of MC Dropout (for uncertainty ranking), temperature scaling (for improved calibration), and conformal prediction (for formal coverage guarantees) forms a practical, layered UQ framework requiring no model retraining.

1.4 Thesis Structure

The remainder of this thesis is organized as follows:

Chapter 2: Background and Related Work establishes the theoretical foundations of this research. Section 2.1 introduces agent-based transport models with emphasis on the MATSim framework. Section 2.2 presents the fundamentals of graph neural networks and their application to traffic prediction. Section 2.3 reviews uncertainty quantification methods in deep learning, including Bayesian approaches and ensemble methods.

Chapter 3: Methodology describes the technical approach adopted in this thesis. We present the PointNetTransfGAT architecture, detail our implementations of MC Dropout and ensemble-based UQ methods, and define the evaluation metrics used to assess uncertainty quality.

Chapter 4: Experimental Setup presents the Paris road network dataset, training protocol, and experimental design. We describe two complementary experiments: Experiment A comparing MC Dropout against ensemble variance on a single model, and Experiment B evaluating multi-model ensemble approaches.

Chapter 5: Results reports the findings from both experiments, including quantitative comparisons of UQ methods, conformal prediction coverage analysis, post-hoc calibration results, and analysis of the generalisation gap observed between validation and test performance.

Chapter 6: Discussion interprets results in the context of prior work, addresses limitations of our study, and discusses implications for deploying uncertainty-aware traffic prediction systems.

Chapter 7: Conclusion summarizes the key findings, revisits the research questions, and outlines directions for future work.

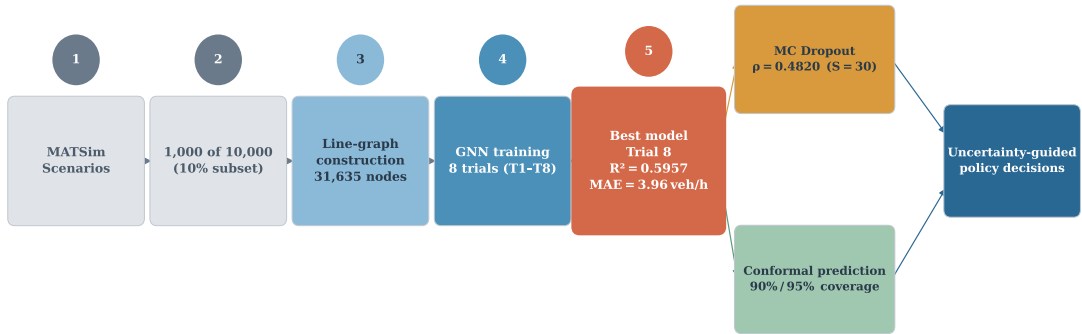


Figure 1.2: Overall thesis pipeline. MATSim agent-based simulations (1,000 of 10,000 available scenarios, 10% subset) are converted to line-graph representations and used to train eight GNN trials. The best model (T8, $R^2 = 0.5957$) is equipped with MC Dropout uncertainty ($\rho = 0.4820$) and conformal prediction intervals (90%/95% coverage guaranteed), feeding into an uncertainty-guided policy decision framework.

2 Background and Related Work

This chapter sets out the theoretical background for the work that follows. Section 2.1 covers agent-based transport models and the MATSim simulation framework. Section 2.2 introduces graph neural networks and their application to traffic prediction. Section 2.3 reviews uncertainty quantification in deep learning, with particular attention to Monte Carlo Dropout, deep ensembles, and conformal prediction. Section 2.4 situates this work within the existing literature on ML surrogates and uncertainty quantification for graph-based predictions.

2.1 Agent-Based Transport Models

Agent-based models (ABMs) simulate complex systems by modelling individual entities and their interactions [Bon02; RG19]. Rather than describing aggregate traffic flow with differential equations, they model the decisions of many autonomous agents, from which macroscopic traffic patterns emerge. In transport applications, agents typically represent travellers who decide on routes, departure times, and travel modes.

Definition 2.1 (Agent-Based Model). An agent-based model consists of a set of autonomous agents $\mathcal{A} = \{a_1, a_2, \dots, a_N\}$, an environment $\mathcal{E}$ in which agents operate, and a set of rules $\mathcal{R}$ governing agent behavior and interactions. Each agent a_i maintains an internal state s_i and makes decisions based on local observations and objectives.

2.1.1 MATSim Framework

The Multi-Agent Transport Simulation (MATSim) framework [HNA16] is a widely-used open-source platform for large-scale transport simulations. Each synthetic agent has a daily activity plan specifying where they need to go and when; the simulation handles activity scheduling and location choice, mode selection across car, public transit, cycling, and walking, route choice based on experienced travel times, and dynamic traffic assignment through queue-based network loading. A typical run iterates through multiple synthetic “days,” with agents learning from experience and gradually revising their plans. This evolutionary process converges toward a user equilibrium where no agent can unilaterally improve their travel time [HNA16].

2.1.2 Computational Challenges

While MATSim provides high-fidelity traffic predictions, the computational cost is substantial. Simulating the Paris metropolitan region—our study area—requires more

than eight hours per scenario on modern hardware, significant memory for agent state tracking, and multiple iterations to reach equilibrium. These constraints motivate the development of surrogate models that can approximate MATSim outputs at a fraction of the computational cost.

2.2 Graph Neural Networks

Graph Neural Networks (GNNs) extend deep learning to graph-structured data, making them well-suited to traffic network analysis [Sca+09; KW17; Wu+20]. The formal definition of graphs and the message passing framework underlying modern GNNs are introduced below.

Definition 2.2 (Graph). A graph is a tuple $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X}, \mathbf{E})$ defined by a nonempty finite set of nodes $\mathcal{V}$, a set of edges $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$, node features $\mathbf{X} \in \mathbb{R}^{|\mathcal{V}| \times d_n}$, and edge features $\mathbf{E} \in \mathbb{R}^{|\mathcal{E}| \times d_e}$.

The neighbourhood of a node v , denoted $\mathcal{N}(v)$, is the set of all nodes connected to v by an edge: $\mathcal{N}(v) = \{u \in \mathcal{V} \mid (u, v) \in \mathcal{E}\}$.

2.2.1 Message Passing Framework

Modern GNNs follow the message passing paradigm [Gil+17], where node representations are iteratively updated through neighbourhood aggregation:

Definition 2.3 (Message Passing Neural Network). Given initial node representations $\mathbf{h}_v^{(0)} = \mathbf{x}_v$, a message passing neural network updates node representations over K layers through the following operations:

$$\mathbf{m}_v^{(k)} = A^{(k)} \left(\left\{ \mathbf{h}_u^{(k-1)} : u \in \mathcal{N}(v) \right\} \right) \quad (2.1)$$

$$\mathbf{h}_v^{(k)} = U^{(k)} \left(\mathbf{h}_v^{(k-1)}, \mathbf{m}_v^{(k)} \right) \quad (2.2)$$

where $\mathbf{h}_v^{(k)}$ denotes the representation of node v at layer k , A is a permutation-invariant function, and U combines the aggregated neighbourhood information with the node's previous representation.

2.2.2 Graph Attention Networks

Graph Attention Networks (GATs) [Vel+18] introduce attention mechanisms to weight neighbour contributions dynamically:

$$\alpha_{ij} = \frac{\exp(\text{LeakyReLU}(\mathbf{a}^T [\mathbf{W}\mathbf{h}_i \parallel \mathbf{W}\mathbf{h}_j]))}{\sum_{k \in \mathcal{N}(i)} \exp(\text{LeakyReLU}(\mathbf{a}^T [\mathbf{W}\mathbf{h}_i \parallel \mathbf{W}\mathbf{h}_k]))} \quad (2.3)$$

where α_{ij} represents the attention coefficient between nodes i and j , $\mathbf{W}$ is a learnable weight matrix, and $\mathbf{a}$ is the attention vector. The model thus learns to weight neighbour contributions according to their relevance.

2.2.3 GNNs for Traffic Prediction

Recent work has demonstrated strong GNN performance on traffic prediction tasks [Li+18; YYZ18; ZSZ+20; JL22]. The dominant approach combines spatial message-passing over the road graph with temporal sequence modelling using recurrent or attention-based layers. Li et al. [Li+18] introduced diffusion-based graph convolutions to capture directional traffic flow, while Yu et al. [YYZ18] combined graph convolutions with gated recurrent units for short-term speed forecasting. Subsequent work extended these ideas to multi-hop attention and multi-scale spatial representations [JL22]. A common thread is the treatment of intersections as graph nodes, with edges representing road links between them.

An important representational choice, adopted in this thesis following Natterer, Rao, Tejada Lapuerta, et al. [Nat+25], is to represent road segments—rather than intersections—as graph nodes. This *line graph* transformation assigns each road link its own node, enabling the model to predict link-level traffic changes directly. A consequence is that each node must carry the geometric coordinates of its corresponding road segment; in the PointNetTransfGAT architecture, this is handled by providing the 2D start and end positions of each segment as auxiliary positional inputs to the PointNetConv layers, as described in Chapter 3.

2.3 Uncertainty Quantification in Deep Learning

Uncertainty quantification (UQ) enables models to express confidence in their predictions [APH+21; GT+23]. Uncertainty estimates are necessary when deploying machine learning models in decision support applications.

Definition 2.4 (Aleatoric Uncertainty). Aleatoric uncertainty, also known as data uncertainty, arises from inherent randomness in the observations. This uncertainty is irreducible and cannot be eliminated by collecting more data. In traffic prediction, day-to-day variation in travel demand represents aleatoric uncertainty.

Definition 2.5 (Epistemic Uncertainty). Epistemic uncertainty, also known as model uncertainty, arises from limited training data or model capacity. This uncertainty is reducible and decreases as more data becomes available. Epistemic uncertainty is the primary focus of this thesis.

2.3.1 Bayesian Neural Networks

Bayesian neural networks (BNNs) place distributions over network weights rather than point estimates [Mac92; Nea96]. For a network with weights $\mathbf{w}$, we seek the posterior distribution:

$$p(\mathbf{w}|\mathcal{D}) = \frac{p(\mathcal{D}|\mathbf{w})p(\mathbf{w})}{p(\mathcal{D})} \quad (2.4)$$

where $\mathcal{D}$ denotes the training data, $p(\mathcal{D}|\mathbf{w})$ is the likelihood, and $p(\mathbf{w})$ is the prior over weights. Exact posterior inference is intractable for neural networks due to the high dimensionality of the weight space, motivating the development of approximate inference methods.

2.3.2 Monte Carlo Dropout

Gal and Ghahramani [GG16] established a theoretical connection between dropout and approximate Bayesian inference. By keeping dropout active at test time and performing T forward passes, we obtain samples from an approximate posterior distribution:

Definition 2.6 (MC Dropout Uncertainty Estimation). For an input $\mathbf{x}$, MC Dropout performs T stochastic forward passes with dropout enabled:

$$\hat{y}_t = f_{\hat{\mathbf{w}}_t}(\mathbf{x}), \quad t = 1, \dots, T \quad (2.5)$$

where $\hat{\mathbf{w}}_t$ represents the weights after applying dropout mask t . The predictive mean and variance are estimated as:

$$\mathbb{E}[y] \approx \frac{1}{T} \sum_{t=1}^T \hat{y}_t \quad (2.6)$$

$$\text{Var}[y] \approx \frac{1}{T} \sum_{t=1}^T (\hat{y}_t - \mathbb{E}[y])^2 \quad (2.7)$$

MC Dropout requires no architectural modifications—only keeping dropout active during inference—which makes it straightforward to add uncertainty quantification to existing trained models. In our experiments (Chapter 3), we use $S = 30$ stochastic forward passes (denoted S to distinguish from the general T above).

2.3.3 Deep Ensembles

Deep ensembles [LPB17] take an alternative approach by training M networks independently with different random initializations:

Definition 2.7 (Deep Ensemble Uncertainty Estimation). Given M independently trained networks with weights $\{\mathbf{w}_1, \dots, \mathbf{w}_M\}$, the ensemble prediction and uncertainty are:

$$\mathbb{E}[y] = \frac{1}{M} \sum_{m=1}^M f_{\mathbf{w}_m}(\mathbf{x}) \quad (2.8)$$

$$\text{Var}[y] = \frac{1}{M} \sum_{m=1}^M (f_{\mathbf{w}_m}(\mathbf{x}) - \mathbb{E}[y])^2 \quad (2.9)$$

Ensembles often provide well-calibrated uncertainty estimates but require training and maintaining multiple models, increasing computational cost by a factor of M . We evaluate this trade-off in Chapter 5.

2.3.4 Conformal Prediction

Conformal prediction [AB23; VGS05] is a distribution-free framework for constructing prediction intervals with guaranteed finite-sample coverage. Unlike Bayesian methods, conformal prediction requires no assumptions about the form of the model’s predictive distribution—only *exchangeability* of the data.

Definition 2.8 (Split Conformal Prediction). Given a pre-trained model f , a calibration set $\mathcal{D}_{\text{cal}} = \{(x_i, y_i)\}_{i=1}^n$, and a desired coverage level $1 - \alpha$, split conformal prediction proceeds as follows:

1. Compute nonconformity scores $s_i = |y_i - f(x_i)|$ for each calibration point.
2. Set the quantile threshold $q = \text{Quantile}\left(s_1, \dots, s_n; \frac{\lceil (n+1)(1-\alpha) \rceil}{n}\right)$.
3. For a new test point x_{new} , construct the prediction interval $\mathcal{C}(x_{\text{new}}) = [f(x_{\text{new}}) - q, f(x_{\text{new}}) + q]$.

The key property is a finite-sample marginal coverage guarantee: for any new test point drawn exchangeably from the same distribution,

$$\Pr(y_{\text{new}} \in \mathcal{C}(x_{\text{new}})) \geq 1 - \alpha. \quad (2.10)$$

This guarantee is *model-agnostic* and holds regardless of whether the underlying model’s uncertainty estimates are well-calibrated. The trade-off is that conformal intervals are fixed-width (not adaptive to individual inputs), so the coverage guarantee is marginal rather than conditional on each input.

In our experiments (Section 5.5), we apply split conformal prediction to Trial 8 using the absolute residual as the nonconformity score, with a 50/50 calibration–evaluation split of the 100 test graphs.

2.3.5 Evaluating Uncertainty Quality

High-quality uncertainty estimates should correlate with actual prediction errors: the model should express higher uncertainty when it is more likely to be wrong. Several complementary metrics are used in the literature to assess this. The **Spearman rank correlation** (ρ) measures the monotonic relationship between predicted uncertainty and absolute prediction error; values range from -1 to $+1$, with positive values indicating that higher uncertainty corresponds to higher error. **Calibration curves** (reliability diagrams) assess whether $p\%$ prediction intervals actually contain $p\%$ of true values, with well-calibrated models tracking the diagonal. The **negative log-likelihood (NLL)** is a proper scoring rule that jointly evaluates prediction accuracy and calibration in a single quantity, and is widely used for probabilistic regression evaluation [LPB17]. In this thesis, we primarily rely on Spearman ρ and calibration curves for uncertainty assessment, as they provide interpretable diagnostics of ranking quality and interval coverage respectively; Gaussian NLL is reported as a complementary proper scoring rule (Section 5.9.2).

2.4 Related Work

2.4.1 Machine Learning Surrogates for Transport Models

Machine learning surrogates for transport simulations have been explored in several recent studies [Nat+25]. Historically, surrogate approaches in the transport domain relied on statistical emulation or shallow regression models trained to reproduce aggregate outputs of expensive simulation runs. Willard et al. [Wil+22] provide a broad review of physics-informed machine learning methods used as surrogates across scientific domains, highlighting the tension between data efficiency and physical consistency—a trade-off that also applies to agent-based transport surrogates. More recently, graph-based representations have proven particularly effective for transport surrogates because they preserve the network topology essential for capturing traffic propagation effects [Nat+25].

The PointNetTransfGAT architecture [Nat+25] combines ideas from PointNet, transformers, and graph attention networks to achieve strong performance on traffic prediction from policy scenarios. This architecture forms the foundation for the uncertainty quantification experiments in this thesis.

2.4.2 Uncertainty Quantification for Traffic Prediction

Uncertainty quantification in traffic prediction is relatively underexplored. Wang et al. [Wan+24] apply probabilistic graph neural networks to spatiotemporal travel demand forecasting — a temporal task in which the model ingests a sequence of observed demand snapshots and predicts future values. This differs fundamentally from the setting studied here: predicting the *change* in link-level traffic volume induced by a transport policy scenario is a static, cross-sectional regression problem with no temporal component. The distributional challenges and appropriate UQ methods therefore differ substantially. To our knowledge, the combination of GNN-based surrogates with distribution-free coverage guarantees via conformal prediction has not been previously explored in the transport literature, which motivates the present study.

2.4.3 Uncertainty Quantification for GNNs

Recent work has extended UQ methods to GNNs [Zha+19; Has+20], though primarily for node classification tasks rather than regression. Bayesian GNNs [Zha+19] place prior distributions over graph convolutional weights and perform approximate posterior inference, while DropEdge-style approaches [Has+20] use stochastic edge dropout as a form of variational inference over graph structure. Key findings suggest that graph structure can both help (by providing regularization through neighbourhood aggregation) and hurt (by propagating uncertainty through connected nodes) prediction confidence.

A particularly relevant line of work addresses the limitations of MC Dropout and ensemble-based uncertainty for GNNs. Fuchsgruber et al. [FGG24] propose Graph

Energy-Based Models (GEBM), a post-hoc method that estimates epistemic uncertainty by fitting an energy-based model to the GNN’s learned representations. Unlike MC Dropout, which requires multiple stochastic forward passes at inference time, GEBM provides epistemic uncertainty from a single deterministic pass by measuring the density of the test representation relative to the training distribution. Their experiments on node classification benchmarks show competitive or superior uncertainty quality compared to MC Dropout and deep ensembles at substantially lower computational cost. The same group has also explored uncertainty-guided active learning on graphs [FWG24], demonstrating that epistemic uncertainty estimates can effectively guide data acquisition for graph-structured tasks.

The node-level regression setting studied here — where each node represents a road segment and the prediction target is a continuous traffic flow change — introduces different distributional challenges than the categorical outputs most commonly studied in this literature. The interplay between graph structure and uncertainty quantification in regression settings remains an open question, which we investigate in the experiments that follow.

3 Methodology

This chapter describes the GNN architecture adopted for traffic prediction, the six uncertainty quantification methods evaluated in this thesis, and the metrics used to assess both predictive accuracy and uncertainty quality.

3.1 Problem Formulation

We formulate traffic prediction as a node-level regression task on a directed graph. Given a road network $G = (V, E)$ where nodes represent road segments and edges represent connectivity, we aim to predict the change in traffic volume Δv_i for each node i after a policy intervention.

3.1.1 Input Features

Each node i is characterised by five input features:

Table 3.1: Input features for each road segment (node).

Feature	Unit	Description
VOL_BASE_CASE	vehicles/hour	Traffic volume in baseline scenario
CAPACITY_BASE_CASE	vehicles/hour	Road segment capacity
CAPACITY_REDUCTION	vehicles/hour	Policy-induced capacity reduction (absolute, ≤ 0)
FREESPEED	m/s	Free-flow speed limit
LENGTH	meters	Road segment length

3.1.2 Target Variable

The target variable is:

$$\Delta v_i = v_i^{\text{policy}} - v_i^{\text{baseline}} \quad (3.1)$$

representing the change in traffic volume on segment i after the policy intervention is applied.

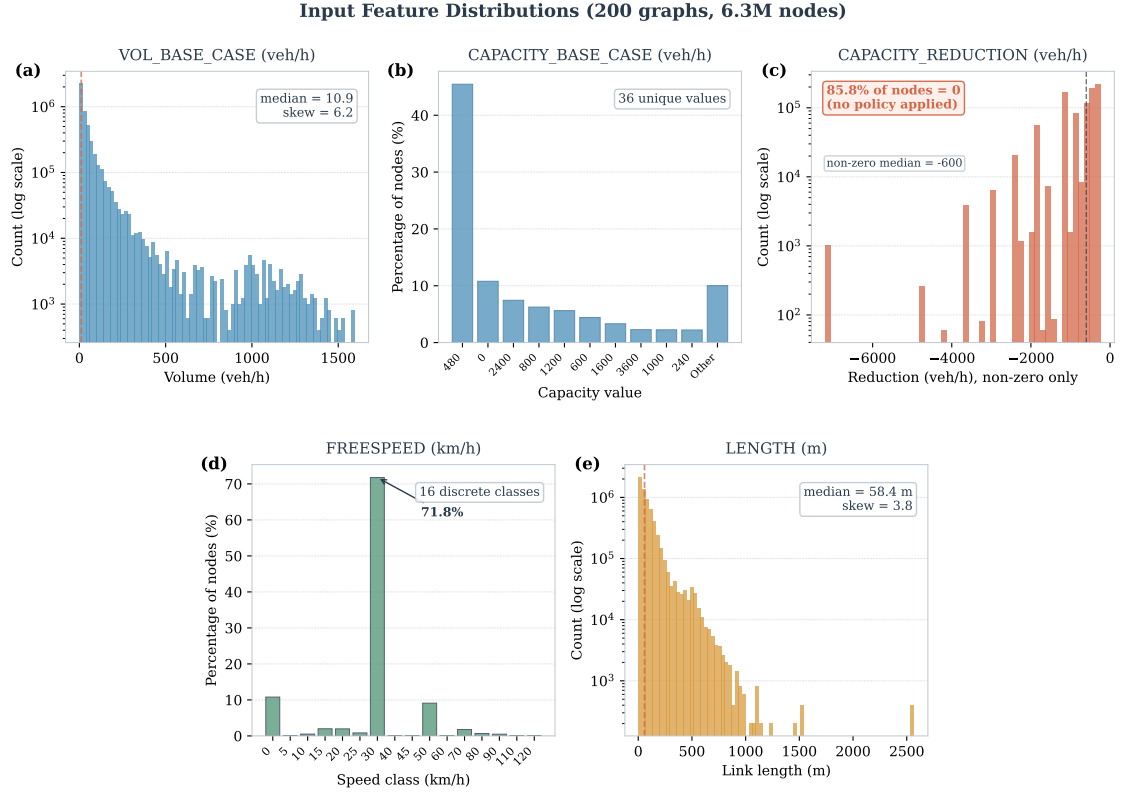


Figure 3.1: Distribution of the five input features across 200 training graphs (6,327,000 node observations). Dashed vertical lines mark the median. VOL_BASE_CASE and CAPACITY_BASE_CASE are right-skewed, reflecting the heavy-tailed nature of urban traffic volumes; FREESPEED is concentrated at a few discrete speed-limit values; LENGTH spans several orders of magnitude. *Results based on 1,000 of 10,000 available MATSim scenarios (10% subset).*

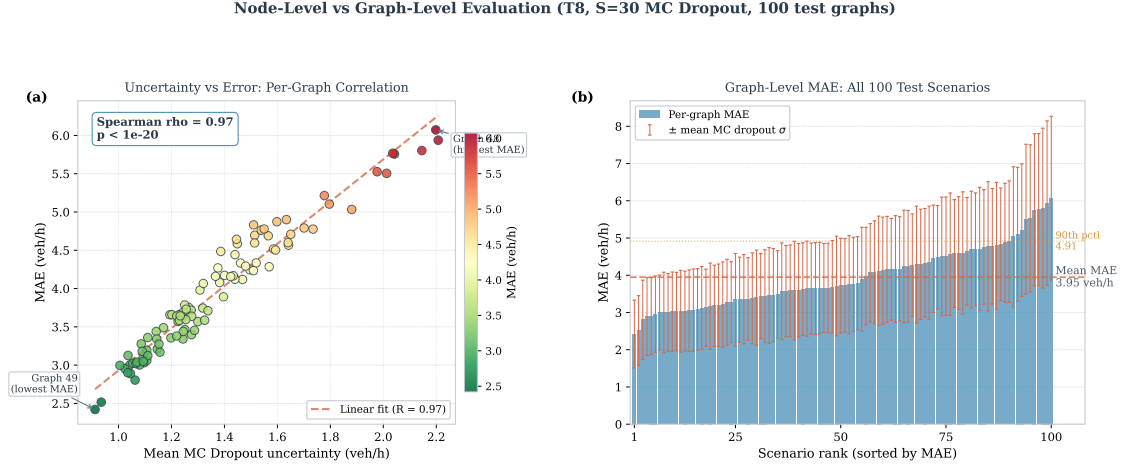


Figure 3.2: Node-level predictions (Δv per node, with per-node MC Dropout σ) and graph-level evaluation (MAE per scenario with uncertainty error bars). T8: 100 test graphs $\times$ 31,635 nodes = 3,163,500 total predictions. Source: `mc_dropout_full_metrics_model8_mc30_100graphs.json`.

3.2 Model Architecture

We use the PointNetTransfGAT architecture developed by Natterer et al. [Nat+25], which combines ideas from PointNet [Qi+17], graph transformers [Shi+21], and graph attention networks [Vel+18] into a single end-to-end model for node-level regression on road networks.

Note on architectural variants: Trial 1 uses a `Linear(64→1)` final prediction layer, while Trials 2–8 use a `GATConv(64→1)` final layer. This difference makes Trial 1 results not directly comparable with Trials 2–8, and Trial 1 is excluded from MC Dropout UQ experiments (effective dropout = 0.0).

3.2.1 Geometric Representation

The architecture encodes the physical geometry of road segments as follows. Each node carries two 2D positional coordinates: the start point of its corresponding road segment, $\mathbf{pos}[:,0,:] \in \mathbb{R}^{N \times 2}$, and its end point, $\mathbf{pos}[:,1,:] \in \mathbb{R}^{N \times 2}$. The two PointNetConv layers [Qi+17] exploit these coordinates by computing, for each node i and neighbour j , the relative displacement $\mathbf{pos}_j - \mathbf{pos}_i$ as part of the local MLP input. This means the first PointNetConv uses the start-point geometry of each segment relative to its neighbours, while the second uses the end-point geometry. The effect is that the model learns not only the graph-topological neighbourhood of each segment but also the spatial proximity of where segments begin and end in the 2D plane, providing geometric information that complements the topological structure.

3.2.2 Layer-by-Layer Architecture

Table 3.2 details the complete forward pass through the network. All dimensions and layer specifications are verified directly from `scripts/gnn/models/point_net_transf_gat.py`.

Table 3.2: PointNetTransfGAT layer progression. Input: node features $\mathbf{x} \in \mathbb{R}^{N \times 5}$, positional coordinates $\mathbf{pos} \in \mathbb{R}^{N \times 3 \times 2}$ (start, end, and midpoint coordinates; only start and end are used in the forward pass). All dimensions verified from `point_net_transf_gat.py`.

Stage	Layer	Operation	In dim	Out dim
1	PointNetConv-1	Local MLP: 7 $\rightarrow$ 256 (feat+pos_start); global MLP: 256 $\rightarrow$ 512 $\rightarrow$ 512	5	512
2	PointNetConv-2	Local MLP: 514 $\rightarrow$ 256 (feat+pos_end); global MLP: 256 $\rightarrow$ 512 $\rightarrow$ 128	512	128
3	TransformerConv-1	4 attention heads, 64 dims/head	128	256
4	TransformerConv-2	4 attention heads, 128 dims/head	256	512
5	GATConv	1 head	512	64
6a	GATConv (final)	1 head; Trials 2–8	64	1
6b	Linear (final)	Point projection; Trial 1 only	64	1

In Stage 1, the local MLP receives a 7-dimensional input: the 5 node features concatenated with the 2D start-position displacement $(\mathbf{pos}_j - \mathbf{pos}_i)[:, 0, :]$. In Stage 2, the local MLP receives 514 dimensions: the 512-dimensional embedding from Stage 1 concatenated with the 2D end-position displacement. The TransformerConv layers [Shi+21] apply multi-head attention over the graph edges; the GATConv layers [Vel+18] aggregate neighbourhood information with learned attention weights.

Dropout placement: Dropout is applied within the local and global MLPs of both PointNetConv layers, and between the TransformerConv layers. All dropout calls are gated by the `use_dropout` flag (default: `False`); the flag must be set to `True` at inference time to activate MC Dropout uncertainty estimation. The architecture also includes a `predict_mode_stats` branch (a TransformerEncoder-based mode predictor) that was implemented in code but was not activated in any of the eight training trials reported in this thesis.

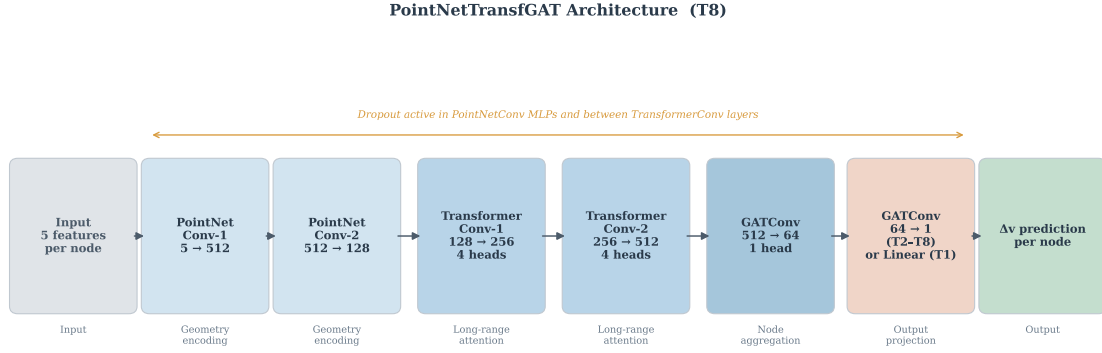


Figure 3.3: PointNetTransfGAT architecture. Two PointNetConv layers encode start- and end-point geometry (512 and 128 channels); two TransformerConv layers capture long-range attention (256 and 512 channels, 4 heads each); two GATConv layers produce the final embedding and scalar prediction (64 then 1 channel). Trial 1 replaces the final GATConv (64→1) with Linear (64→1). Verified from `point_net_transf_gat.py`.

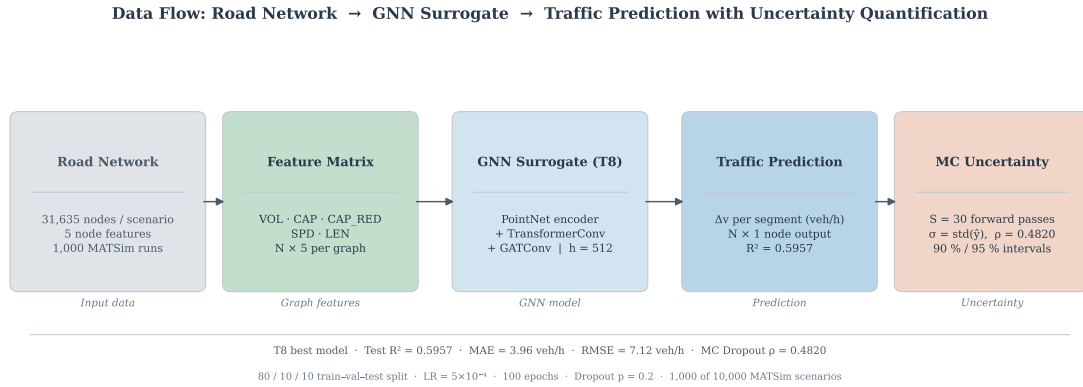


Figure 3.4: End-to-end data flow from road network to traffic prediction with uncertainty. The Paris road network (31,635 nodes) is encoded as a feature matrix ($N \times 5$) with positional coordinates, processed through the GNN, and produces per-node Δv predictions with MC Dropout uncertainty σ . T8 achieves Spearman $\rho = 0.4820$ (100 test graphs, $S = 30$).

Table 3.3: Key hyperparameters of the PointNetTransfGAT model, grouped by trial. Full per-trial configurations are given in Table 4.2.

Parameter	Value / Range	Trials
Hidden channels (PointNet local, GAT)	64	All
Hidden channels (TransformerConv)	256 / 512	All
TransformerConv attention heads	4 (both layers)	All
GATConv attention heads	1	All
Dropout rate	0.2	T8
	0.3	T5–T7
	0.0	T1, T3
Learning rate	3×10^{-4} to 10^{-3}	varies
Batch size	8	T5–T8
	16	T2–T4
	32	T1

3.2.3 Hyperparameter Summary

3.3 Uncertainty Quantification Methods

Six UQ approaches are implemented and compared in this thesis.

3.3.1 Method 1: MC Dropout

MC Dropout [GG16; Gal16] enables uncertainty estimation by keeping dropout active during inference:

Algorithm 1 MC Dropout uncertainty estimation. Given a trained model with dropout layers, S stochastic forward passes yield a predictive mean and standard deviation per node.

Input: Input features $\mathbf{x}$, trained model f , number of samples S

Output: Mean prediction μ , uncertainty σ

for $s = 1$ to S **do**

$\hat{y}_s \leftarrow f(\mathbf{x})$ with dropout enabled

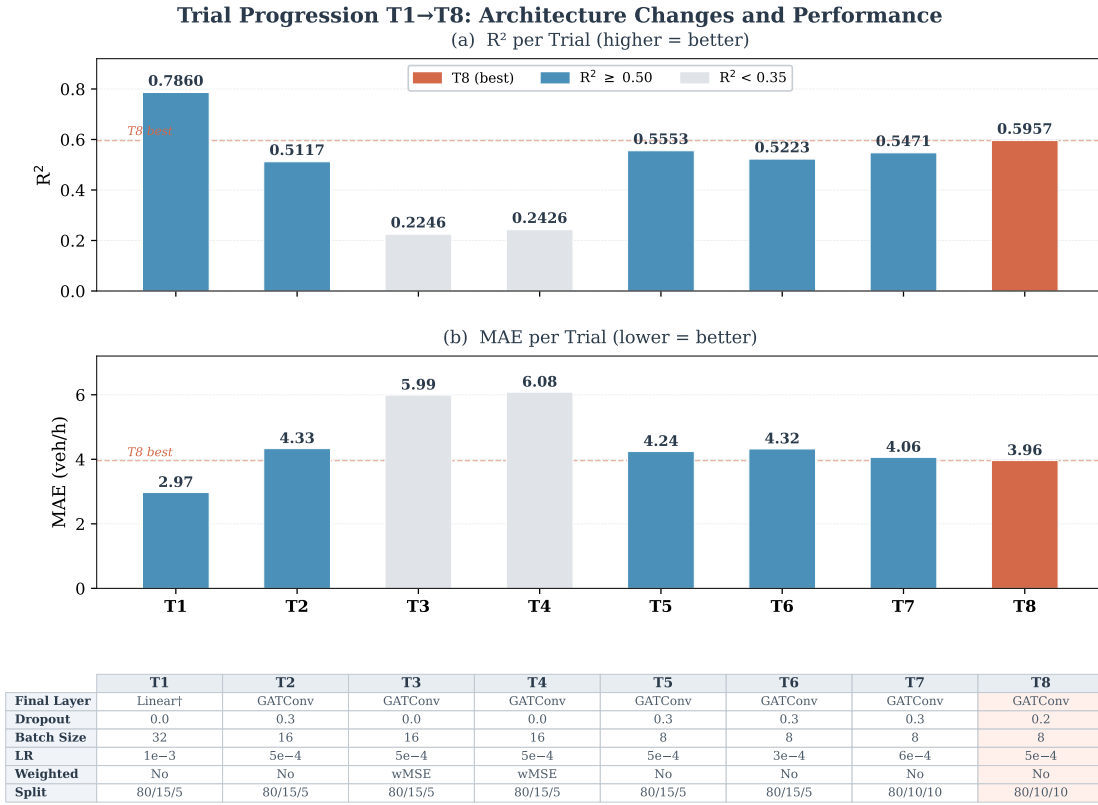
end for

$\mu \leftarrow \frac{1}{S} \sum_s \hat{y}_s$

$\sigma \leftarrow \sqrt{\frac{1}{S} \sum_s (\hat{y}_s - \mu)^2}$

return μ, σ

We use $S = 30$ MC samples, balancing computational cost against estimate stability.



† T1: `use_dropout=False` → effective dropout = 0.0

Figure 3.5: Trial progression from T1 to T8. Top panel shows R^2 per trial; bottom panel shows MAE. The key architectural changes per trial are annotated below the bars. T8 (orange) achieves the best combination of predictive accuracy and uncertainty quality among MC Dropout-enabled trials. Verified from `test_evaluation_complete.json`.

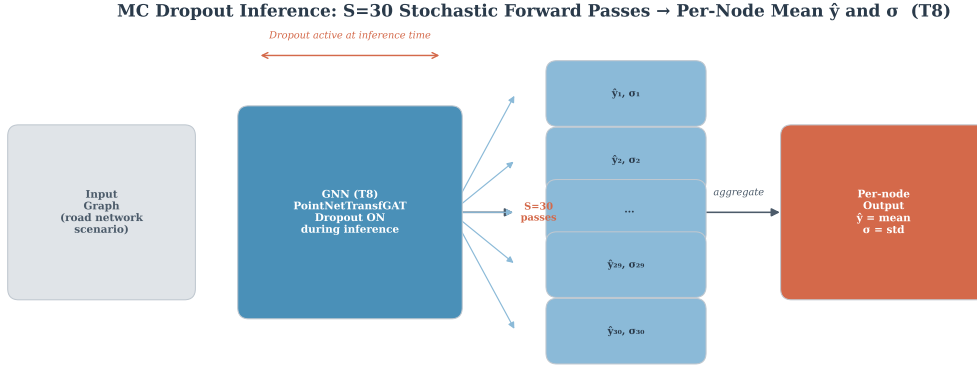


Figure 3.6: MC Dropout inference procedure for T8. A single road network graph is passed through the GNN with dropout active for $S = 30$ stochastic forward passes, yielding 30 prediction vectors $\hat{y}_1, \dots, \hat{y}_{30}$. Their per-node mean ($\hat{y}$) and standard deviation (σ) form the final prediction with uncertainty. Spearman rank correlation between σ and $|y - \hat{y}|$ reaches $\rho = 0.4820$ for T8.

3.3.2 Method 2: Deep Ensembles

Ensemble variance measures disagreement across independent training runs of the same model:

$$\sigma_{\text{ens}} = \sqrt{\frac{1}{M} \sum_{m=1}^M (f_m(\mathbf{x}) - \bar{f}(\mathbf{x}))^2} \quad (3.2)$$

where $M = 5$ is the number of independently trained models.

3.3.3 Method 3: Combined Uncertainty

We combine MC Dropout and ensemble uncertainty:

$$\sigma_{\text{combined}} = \sqrt{\sigma_{\text{MC}}^2 + \sigma_{\text{ens}}^2} \quad (3.3)$$

3.3.4 Method 4: Split Conformal Prediction

Split conformal prediction [VGS05; AB23] provides distribution-free prediction intervals with finite-sample coverage guarantees:

Adaptive conformal prediction. Standard split conformal produces fixed-width intervals identical for all nodes. When an auxiliary uncertainty estimate σ_i (e.g., from MC Dropout) is available, the nonconformity score can be *normalised* to produce uncertainty-adaptive intervals [RPC19]:

$$s_i^{\text{adapt}} = \frac{|y_i - f(\mathbf{x}_i)|}{\sigma_i} \quad (3.4)$$

Algorithm 2 Split conformal prediction. A calibration set is used to compute a coverage-valid quantile; the evaluation set receives prediction intervals with guaranteed marginal coverage.

Input: Calibration set $\mathcal{D}_{\text{cal}} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$, trained model f , coverage level $1 - \alpha$
Output: Prediction interval $C(\mathbf{x}_{\text{new}})$ for new inputs
for $i = 1$ to n **do**
 $s_i \leftarrow |y_i - f(\mathbf{x}_i)|$ {nonconformity score}
end for
 $q \leftarrow \text{Quantile}(s_1, \dots, s_n; \lceil (n+1)(1-\alpha) \rceil / n)$
return $C(\mathbf{x}_{\text{new}}) = [f(\mathbf{x}_{\text{new}}) - q, f(\mathbf{x}_{\text{new}}) + q]$

The conformal quantile q^{adapt} is computed from $\{s_i^{\text{adapt}}\}$ on the calibration set, and the resulting prediction interval for a new input is:

$$C^{\text{adapt}}(\mathbf{x}_{\text{new}}) = [f(\mathbf{x}_{\text{new}}) - q^{\text{adapt}} \cdot \sigma_{\text{new}}, f(\mathbf{x}_{\text{new}}) + q^{\text{adapt}} \cdot \sigma_{\text{new}}] \quad (3.5)$$

This yields wider intervals where uncertainty is high and narrower intervals where the model is confident, while preserving the marginal coverage guarantee.

3.3.5 Method 5: Selective Prediction

Selective prediction [GE17] uses MC Dropout uncertainty to filter out unreliable predictions, retaining only those the model is most confident about:

Algorithm 3 Selective prediction. Predictions are ranked by MC Dropout uncertainty and only the most confident fraction is retained.

Input: Predictions $\hat{y}_1, \dots, \hat{y}_n$, uncertainties $\sigma_1, \dots, \sigma_n$, retention fraction $\tau \in (0, 1]$
Output: Filtered predictions, MAE on retained set
Sort predictions by ascending σ
Retain top- τ fraction (i.e., the $\lfloor \tau \cdot n \rfloor$ predictions with lowest σ)
Compute $\text{MAE}_\tau = \frac{1}{|\mathcal{R}|} \sum_{i \in \mathcal{R}} |y_i - \hat{y}_i|$ on retained set $\mathcal{R}$
return filtered predictions, MAE_τ

3.3.6 Method 6: Post-Hoc Temperature Scaling

Temperature scaling [Guo+17; Lav+20] is a post-hoc recalibration method that learns a single scalar T to rescale uncertainty estimates toward calibration:

Algorithm 4 Post-hoc temperature scaling. A single temperature parameter T is optimised on a calibration set to minimise the Expected Calibration Error (ECE).

Input: Calibration set $(y_{\text{cal}}, \hat{y}_{\text{cal}}, \sigma_{\text{cal}})$, evaluation set, candidate temperatures $\mathcal{T}$
Output: Optimal T^* , scaled uncertainties σ_{scaled}
for each $T \in \mathcal{T}$ **do**
 $\sigma_{\text{scaled}} \leftarrow \sigma_{\text{raw}} \times T$
 Compute Kuleshov ECE on calibration set using σ_{scaled}
end for
 $T^* \leftarrow \arg \min_{T \in \mathcal{T}} \text{ECE}(T)$
Apply $\sigma_{\text{scaled}} \leftarrow \sigma_{\text{raw}} \times T^*$ to evaluation set
return $T^*, \sigma_{\text{scaled}}$

3.4 Evaluation Metrics

3.4.1 Prediction Quality

Prediction accuracy is evaluated using three standard regression metrics. The **coefficient of determination** (R^2) measures the proportion of variance in the target explained by the model:

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2} \quad (3.6)$$

The **mean absolute error** (MAE) gives the average magnitude of prediction errors in the original units (vehicles per hour):

$$\text{MAE} = \frac{1}{n} \sum_i |y_i - \hat{y}_i| \quad (3.7)$$

The **root mean square error** (RMSE) penalises large deviations more heavily and is reported for comparability with prior work:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_i (y_i - \hat{y}_i)^2} \quad (3.8)$$

3.4.2 Uncertainty Quality

Uncertainty quality is assessed with three complementary measures. The **Spearman rank correlation** (ρ) captures the monotonic relationship between the predicted uncertainty σ_i and the absolute error $|y_i - \hat{y}_i|$; higher values indicate that uncertainty tracks error more faithfully, and $\rho > 0$ is the minimum bar for a useful uncertainty estimate. The **Pearson correlation** (r) provides a linear analogue that is more sensitive to outliers, offering a cross-check on the rank-based result. Finally, the **p-value** of each correlation test is reported to confirm that the observed association is statistically distinguishable from zero.

A well-calibrated uncertainty estimate should correlate positively with prediction error ($\rho > 0$): where uncertainty is high, actual errors should tend to be larger.

3.4.3 Error Detection via AUROC

Beyond correlation, we evaluate uncertainty as a *binary classifier* for large prediction errors using the Area Under the Receiver Operating Characteristic curve (AUROC). Treating uncertainty as a detection problem: can the model’s uncertainty σ discriminate between high-error and low-error predictions?

Label construction. Given the test-set absolute errors $\{|y_i - \hat{y}_i|\}_{i=1}^n$, we define binary labels at two percentile thresholds:

$$\ell_i^{(p)} = \begin{cases} 1 & \text{if } |y_i - \hat{y}_i| \geq Q_p \\ 0 & \text{otherwise} \end{cases} \quad (3.9)$$

where Q_p is the p -th percentile of the error distribution ($p = 90$ for top-10% errors, $p = 80$ for top-20%). The MC Dropout standard deviation σ_i serves as the continuous score.

Metrics. The AUROC measures the probability that a randomly chosen high-error node receives a higher σ than a randomly chosen low-error node; AUROC = 0.5 corresponds to random ranking and AUROC = 1.0 to perfect separation. We additionally report the **Area Under the Precision-Recall Curve** (AUPRC), which is more informative under class imbalance (10% or 20% positive rate). The random baseline for AUPRC equals the class prevalence. Both metrics are computed using `sklearn.metrics.roc_auc_score` and `average_precision_score` on the full test set.

3.4.4 Calibration Metrics

Expected Calibration Error (ECE). A well-calibrated probabilistic prediction satisfies $P(y_i \in C_p) = p$ for all nominal levels $p \in [0, 1]$, where C_p is the p -level prediction interval. The Kuleshov reliability diagram [KFE18] plots observed coverage against nominal level for a grid of p values. The **Expected Calibration Error** summarises miscalibration as:

$$\text{ECE} = \frac{1}{L} \sum_{l=1}^L |\hat{p}_l - p_l| \quad (3.10)$$

where p_l are L equally spaced nominal levels and $\hat{p}_l$ is the observed fraction of residuals falling within the p_l -level interval $[\hat{y}_i - z_{(1+p_l)/2} \sigma_i, \hat{y}_i + z_{(1+p_l)/2} \sigma_i]$. Lower ECE indicates better calibration, with ECE = 0 for a perfectly calibrated model.

Prediction Interval Coverage Probability (PICP). For conformal prediction intervals at a target level $1 - \alpha$, we report the empirical coverage:

$$\text{PICP}_{1-\alpha} = \frac{1}{n} \sum_{i=1}^n \mathbf{1}[y_i \in C_{1-\alpha}(\mathbf{x}_i)] \quad (3.11)$$

The conformal guarantee ensures $\text{PICP}_{1-\alpha} \geq 1 - \alpha$ in expectation over exchangeable data.

3.4.5 Proper Scoring Rules

While ECE and PICP assess calibration through coverage, *proper scoring rules* evaluate the full predictive distribution by jointly rewarding both calibration and sharpness [GR07]. A scoring rule is *proper* if the expected score is minimised when the forecaster reports the true data-generating distribution, and *strictly proper* if the minimum is unique.

Continuous Ranked Probability Score (CRPS). The CRPS generalises the mean absolute error to probabilistic forecasts. For a cumulative predictive distribution F and observed value y :

$$\text{CRPS}(F, y) = \int_{-\infty}^{\infty} [F(x) - \mathbf{1}(x \geq y)]^2 dx \quad (3.12)$$

For the Gaussian predictive distribution $\mathcal{N}(\mu, \sigma^2)$ used in MC Dropout, the CRPS admits the closed form [GR07]:

$$\text{CRPS}(\mathcal{N}(\mu, \sigma^2), y) = \sigma \left[z(2\Phi(z) - 1) + 2\varphi(z) - \frac{1}{\sqrt{\pi}} \right] \quad (3.13)$$

where $z = (y - \mu)/\sigma$, and Φ, φ denote the standard normal CDF and PDF. Unlike the NLL, CRPS is expressed in the same unit as the target variable (vehicles per hour), facilitating direct comparison with the MAE. A ratio $\text{CRPS}/\text{MAE} < 1$ indicates that the probabilistic forecast provides additional value beyond the point prediction.

Probability Integral Transform (PIT). The PIT maps each observation through its own predictive CDF:

$$p_i = F_i(y_i) = \Phi\left(\frac{y_i - \mu_i}{\sigma_i}\right) \quad (3.14)$$

If the predictive distributions are well-calibrated, the PIT values $\{p_i\}_{i=1}^n$ are uniformly distributed on $[0, 1]$ [Daw84]. A histogram of PIT values therefore provides a visual diagnostic: a U-shaped histogram indicates underdispersion (intervals too narrow), an inverse-U shape indicates overdispersion, and skewness reveals systematic bias. We assess departure from uniformity using the Kolmogorov–Smirnov statistic and the chi-squared goodness-of-fit test.

Winkler Interval Score. For a prediction interval $[L_i, U_i]$ at nominal level $(1 - \alpha)$, the Winkler interval score [Win72] rewards narrow intervals while penalising missed observations:

$$S_\alpha(L_i, U_i, y_i) = (U_i - L_i) + \frac{2}{\alpha}(L_i - y_i) \mathbf{1}(y_i < L_i) + \frac{2}{\alpha}(y_i - U_i) \mathbf{1}(y_i > U_i) \quad (3.15)$$

The first term rewards sharpness, while the penalty terms (scaled by $2/\alpha$) impose a steep cost for non-coverage. The Winkler score is a proper scoring rule for interval forecasts and enables direct comparison between Gaussian, conformal absolute, and conformal sigma-scaled intervals at identical nominal levels.

4 Experimental Setup

This chapter describes the dataset, training protocol, and evaluation methodology underlying the eight training trials and four UQ experiments reported in this thesis.

4.1 Dataset

4.1.1 Paris Road Network

Our experiments use a MATSim simulation of the Paris metropolitan area, provided by Natterer et al. [Nat+25]. Key characteristics:

Table 4.1: Dataset statistics for the Paris MATSim road network.

Property	Value
Road segments (nodes)	31,635
Network edges	~85,000
Available scenarios (full dataset)	10,000
Scenarios used in this work	1,000 (10% subset)
Training scenarios (T1–T6, 80/15/5)	800
Training scenarios (T7–T8, 80/10/10)	800
Test scenarios (T1–T6)	50
Test scenarios (T7–T8)	100
Total test predictions (T1–T6)	1,581,750
Total test predictions (T7–T8)	3,163,500

Note on data scale: All experiments in this thesis were conducted using only **1,000 of 10,000 available MATSim scenarios (10% subset)**, unless stated otherwise. The full 10,000-scenario dataset was used in the original work by Natterer et al. [Nat+25].

4.1.2 Scenario Generation

Each scenario represents a distinct policy intervention. A random subset of road segments is selected and assigned a capacity reduction ranging from 10% to 100%, covering a range of intervention patterns across the network.

4.1.3 Data Split

Training trials T1–T6 use an **80/15/5** split (train/validation/test). Trials T7–T8 use an **80/10/10** split, providing a larger test set (100 graphs, 3,163,500 predictions) for more reliable UQ evaluation. The split is at the *scenario level*—each scenario appears in exactly one split.

4.2 Training Protocol

4.2.1 Model Training

We train 8 models with varying hyperparameters. All trials use the PointNetTransfGAT architecture [Nat+25]. Trial 1 differs from Trials 2–8 in its final prediction layer: Trial 1 uses a Linear(64→1) output layer, while Trials 2–8 use GATConv(64→1).

Table 4.2: Training trials and configurations. T1 uses Linear(64→1) final layer; T2–T8 use GATConv(64→1). Note the different data splits between trials.

Trial	Batch	Split	Dropout	LR	Weighted	Final Layer
T1	32	80/15/5	0.0 [†]	10^{-3}	No	Linear
T2	16	80/15/5	0.3	5×10^{-4}	No	GATConv
T3	16	80/15/5	0.0	5×10^{-4}	Yes	GATConv
T4	16	80/15/5	0.0	5×10^{-4}	Yes	GATConv
T5	8	80/15/5	0.3	5×10^{-4}	No	GATConv
T6	8	80/15/5	0.3	3×10^{-4}	No	GATConv
T7	8	80/10/10	0.3	6×10^{-4}	No	GATConv
T8	8	80/10/10	0.2	5×10^{-4}	No	GATConv

[†] T1 was instantiated with architecture parameter dropout=0.3, but use_dropout=False disabled dropout during both training and inference; the effective dropout rate is 0.0. Source: MODEL_SUMMARY.md for Trial 1.

4.2.2 Optimization

All trials use the Adam optimizer [KB15] with mean squared error (MSE) loss, or weighted MSE where noted. Batch size varies by trial (8–32 scenarios) and learning rates are given in Table 4.2. Training was conducted on an NVIDIA T4 GPU via Google Colab.

4.2.3 Framework

All models are implemented in PyTorch 2.0 [PGM+19] using PyTorch Geometric [FL19] for graph operations, with Weights & Biases for experiment tracking.

4.3 UQ Experimental Design

4.3.1 MC Dropout Evaluation (T5–T8)

MC Dropout uncertainty ($S = 30$ stochastic forward passes) was evaluated on Trials 5–8, which include non-zero dropout rates. The key metric is Spearman ρ between predicted uncertainty (σ) and absolute prediction error ($|y - \hat{y}|$). Trials 5 and 6 were evaluated on 50 test graphs (1,581,750 nodes), while Trials 7 and 8 used the larger 100-graph test set (3,163,500 nodes). T8 (dropout = 0.2, batch = 8) was selected as the primary UQ model based on best test R^2 among comparable trials (T2–T8).

4.3.2 Experiment A: MC Dropout vs Ensemble Variance

Experiment A compares MC Dropout and ensemble variance as uncertainty estimators using the same base model, T8. Five independently seeded MC Dropout inference runs ($S = 30$ each, seeds 42, 142, 242, 342, 442) are conducted on all 100 test graphs. MC Dropout uncertainty is the average per-node standard deviation across runs; ensemble variance is the standard deviation of per-run means; the combined metric is the quadrature sum. A PyTorch Geometric API version mismatch originally corrupted these results; the evaluation reported here uses corrected weight remapping (Section 5.3.1).

4.3.3 Experiment B: Multi-Model Ensemble

Experiment B asks whether combining multiple trained models improves uncertainty quantification. Five trained models (T2, T5, T6, T7, T8) are assembled into a weighted multi-model ensemble (weights proportional to individual R^2); their disagreement on each prediction serves as the uncertainty estimate. Deterministic forward passes are used (no MC Dropout). As with Experiment A, the evaluation uses corrected weight remapping (Section 5.3.1).

4.3.4 Conformal Prediction (T8)

Split conformal prediction [AB23] is applied to T8 using the 100-graph test set, divided equally into 50 calibration graphs and 50 evaluation graphs. The nonconformity score is the absolute residual $|y - \hat{y}|$, and prediction intervals are constructed at both 90% and 95% coverage levels.

4.4 Computational Resources

All experiments were run on Google Colab Pro with an NVIDIA T4 GPU (16 GB VRAM). Total training time across the eight trials was approximately 13 hours. MC Dropout inference for T8 ($S = 30$, 100 test graphs) required approximately 228 minutes.

5 Results

This chapter reports results from the eight training trials and the subsequent uncertainty quantification experiments. All metric values are sourced directly from JSON evaluation files produced during model training and testing, and all experiments use **1,000 of 10,000 available MATSim scenarios (10% subset)**.

5.1 Model Training Results

We evaluate model performance using standard regression metrics: coefficient of determination (R^2), Mean Absolute Error (MAE), and Root Mean Square Error (RMSE) [CD14; Nag91].

5.1.1 Performance Summary

Table 5.1 summarizes test-set performance across all 8 training trials. Trials 1–6 are evaluated on 50 test graphs (1,581,750 nodes); Trials 7–8 on 100 test graphs (3,163,500 nodes). Trial 1 uses a Linear ($64 \rightarrow 1$) final layer; Trials 2–8 use GATConv ($64 \rightarrow 1$).

Table 5.1: Test-set performance across all 8 training trials. Sources: `test_evaluation_complete.json` (T2–T8) and `test_results.json` (T1, T4). Note the architectural difference between T1 and T2–T8, and the split difference between T1–T6 and T7–T8.

Trial	Final Layer	Split	Test Graphs	Test R^2	MAE (veh/h)	RMSE (veh/h)	Pearson r
T1	Linear	80/15/5	50	0.7860	2.97	5.40	0.888
T2	GATConv	80/15/5	50	0.5117	4.33	8.15	0.719
T3	GATConv	80/15/5	50	0.2246	5.99	10.27	0.639
T4	GATConv	80/15/5	50	0.2426	6.08	10.15	0.634
T5	GATConv	80/15/5	50	0.5553	4.24	7.78	0.747
T6	GATConv	80/15/5	50	0.5223	4.32	8.06	0.726
T7	GATConv	80/10/10	100	0.5471	4.06	7.53	0.741
T8	GATConv	80/10/10	100	0.5957	3.96	7.12	0.773

For reference, Natterer, Rao, Tejada Lapuerta, et al. [Nat+25] report an overall $R^2 = 0.91$ on the full 10,000-scenario dataset; direct comparison is not valid due to the $10\times$ difference in training data scale (1,000 vs 10,000 scenarios).

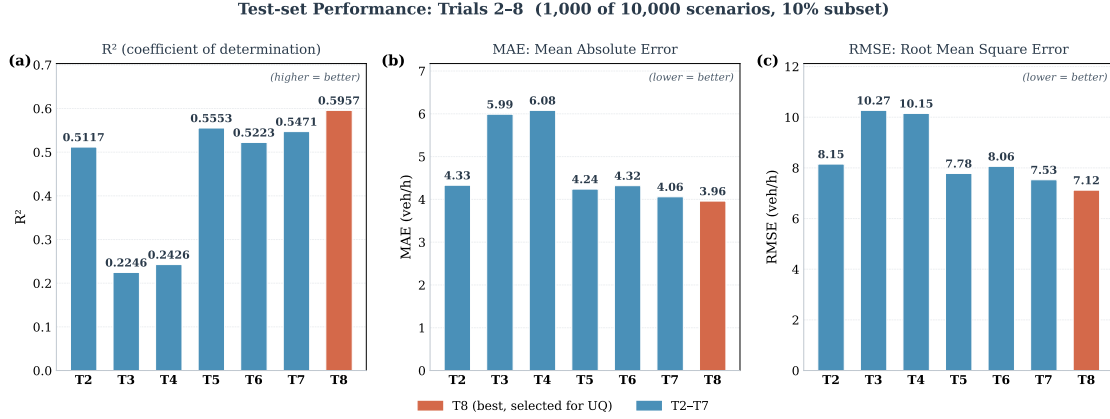


Figure 5.1: Test-set performance across Trials 2–8 (T1 excluded; different architecture). R^2 , MAE, and RMSE are shown left to right. T8 achieves the best result among comparable trials ($R^2 = 0.5957$, MAE = 3.96 veh/h). Based on 1,000 scenarios (10% subset); T1–T6 on 50 test graphs, T7–T8 on 100.

5.1.2 Discussion of Trial Results

Trial 1 achieves the highest $R^2 = 0.7860$ and lowest MAE = 2.97 veh/h among all trials. However, Trial 1 uses a structurally different Linear final layer and zero dropout, making it architecturally incompatible with Trials 2–8 (GATConv final layer, non-zero dropout). This difference has two consequences: first, direct performance comparison between T1 and T2–T8 conflates the effect of the output layer type with other hyperparameter differences; second, MC Dropout UQ is *undefined* for Trial 1 because zero dropout means all stochastic forward passes produce identical outputs, yielding zero variance. Trial 1 is therefore excluded from all UQ experiments as a principled architectural constraint, not as an arbitrary choice.

Among Trials 2–8 (comparable architecture with GATConv output and non-zero dropout), Trial 8 achieves the best test performance: $R^2 = 0.5957$, MAE = 3.96 veh/h, RMSE = 7.12 veh/h. Trials 3–4 (weighted loss) show markedly lower R^2 (0.22–0.24), indicating that weighted MSE loss did not improve generalisation on this dataset.

5.1.3 Best Model Selection

Trial 1 achieves the highest absolute scores ($R^2 = 0.7860$, MAE = 2.97 veh/h), demonstrating that the underlying feature representations learned by the PointNetTransfGAT backbone are effective. However, Trial 1 is structurally incompatible with MC Dropout uncertainty quantification: its zero effective dropout rate means stochastic forward passes produce no variance, making uncertainty estimation impossible. The exclusion of Trial 1 from UQ experiments is therefore a necessary methodological constraint—MC Dropout requires non-zero dropout by definition [GG16]. Among the comparable trials (T2–T8, all using GATConv output with non-zero dropout), Trial 8 is the strongest per-

former ($R^2 = 0.5957$, $\text{MAE} = 3.96$ veh/h) and is therefore used for all UQ experiments. Trial 1’s higher accuracy serves as a useful reference point, indicating headroom that future architectural refinements (e.g., combining a linear output head with a separate uncertainty mechanism) might recover.

5.2 MC Dropout Uncertainty Quality

5.2.1 Spearman Correlation Results

Table 5.2 shows Spearman ρ between MC Dropout uncertainty and absolute prediction error for Trials 5–8 ($S = 30$ MC samples). A higher ρ indicates that the model’s uncertainty estimates better predict where errors occur.

Table 5.2: Spearman ρ between MC Dropout uncertainty and absolute prediction error, $S = 30$ forward passes. T8 achieves the strongest correlation ($\rho = 0.4820$).

Trial	Dropout	Test Graphs	Test Nodes	Spearman ρ
T5	0.3	50	1,581,750	0.4263
T6	0.3	50	1,581,750	0.4186
T7	0.3	100	3,163,500	0.4460
T8	0.2	100	3,163,500	0.4820

Trial 8 achieves the highest Spearman $\rho = 0.4820$, meaning its uncertainty estimates have the strongest rank-correlation with actual prediction errors. However, Trial 8 differs from Trials 5–6 in both dropout rate (0.2 vs 0.3) and data split (80/10/10 vs 80/15/5), and from Trial 7 in dropout rate and learning rate (5×10^{-4} vs 6×10^{-4}); batch size is identical (8) across all four. The ρ improvement therefore cannot be attributed solely to the reduced dropout rate.

5.2.2 T8 MC Dropout Uncertainty Statistics

For Trial 8 ($S = 30$, 100 test graphs, 3,163,500 nodes), MC Dropout produces a mean uncertainty of $\bar{\sigma} = 1.369$ veh/h with a standard deviation of 1.383 veh/h; values range from 0.042 to 44.29 veh/h. The MC mean R^2 is 0.5857 and the MC mean MAE is 3.948 veh/h—both virtually identical to the deterministic figures, confirming that enabling dropout at inference does not degrade accuracy. Total MC inference time was 228.25 minutes. Source: `mc_dropout_full_metrics_model8_mc30_100graphs.json`.

5.2.3 Prediction Quality: Deterministic vs MC Dropout Inference

Figure 5.2 compares deterministic inference (single forward pass, dropout disabled) against MC Dropout inference (mean of $S = 30$ stochastic passes) for T8. The negligible accuracy difference ($\Delta R^2 = -0.010$, $\Delta \text{MAE} = -0.01$ veh/h) confirms that enabling MC

Dropout incurs virtually no cost in predictive performance, while providing informative per-node uncertainty estimates ($\rho = 0.4820$).

Deterministic vs MC Dropout Inference: T8
(100 test graphs, 3,163,500 nodes)

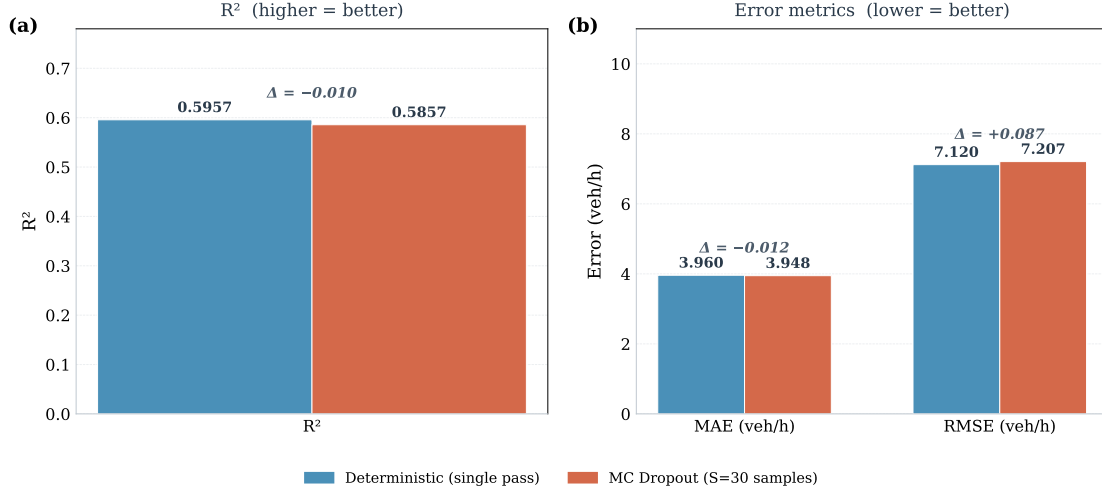


Figure 5.2: Point prediction accuracy under deterministic and MC Dropout ($S = 30$) inference for T8. Enabling dropout incurs negligible accuracy cost ($\Delta R^2 = -0.010$, $\Delta \text{MAE} = -0.01$ veh/h) while providing uncertainty estimates with Spearman $\rho = 0.4820$. Based on 100 test graphs.

5.3 Ensemble Experiments

5.3.1 Technical Note: PyG Weight Remapping

The ensemble experiment scripts originally loaded model checkpoints with `strict=False`, which silently dropped GATConv weight keys due to a PyTorch Geometric API version mismatch between training and evaluation time. Specifically, the checkpoints store `lin.weight` (older PyG format), whereas PyG 2.3.1 expects `lin_src.weight` and `lin_dst.weight`. This was resolved by remapping the checkpoint keys before loading with `strict=True`, restoring correct predictions ($R^2 \approx 0.59$). All ensemble results reported below use the corrected loading procedure. This issue does not affect standalone MC Dropout, conformal prediction, temperature scaling, or selective prediction results, which were generated under the matching PyG version.

5.3.2 Experiment A: MC Dropout vs Ensemble Variance

Experiment A evaluates uncertainty quality from 5 independently seeded MC Dropout inference runs on the T8 model ($S = 30$ stochastic passes each, seeds 42, 142, 242, 342, 442). MC Dropout uncertainty is the average per-node standard deviation across runs; ensemble variance is the standard deviation of per-run means; combined uncertainty is the quadrature sum of both. Table 5.3 reports uncertainty–error Spearman correlations across all 100 test graphs (3,163,500 nodes).

Table 5.3: Experiment A: uncertainty–error correlations from 5 seeded MC Dropout inference runs on T8 ($S = 30$, 100 test graphs, 3,163,500 nodes, $R^2 = 0.5865$).

UQ Method	Spearman ρ	Mean σ (veh/h)
MC Dropout ($S = 30$)	0.4908	1.369
Ensemble Variance (5 runs)	0.4370	0.217
Combined	0.4909	1.388

MC Dropout ($\rho = 0.4908$) outperforms ensemble variance ($\rho = 0.4370$) by 12.3%, confirming that stochastic-pass variability captures more informative uncertainty than seed-induced variability alone. The combined uncertainty ($\rho = 0.4909$) provides negligible improvement over MC Dropout, indicating that ensemble variance is largely redundant once MC Dropout is available. These results are consistent with the standalone T8 MC Dropout evaluation ($\rho = 0.4820$, Section 5.2); the slight increase to $\rho = 0.4908$ arises from averaging across 5 independent runs, which reduces noise in the uncertainty estimates.

5.3.3 Experiment B: Multi-Model Ensemble

Experiment B computes the weighted ensemble prediction and inter-model disagreement across all five trained models (T2, T5, T6, T7, T8) using deterministic forward passes. Weights are proportional to each model’s individual R^2 . Table 5.4 reports results on the T8 test split (100 graphs, 3,163,500 nodes).

Table 5.4: Experiment B: multi-model ensemble (T2, T5–T8), weighted by individual R^2 scores. Evaluated on 100 test graphs (3,163,500 nodes).

Metric	Value
Ensemble Spearman ρ	0.4333
Ensemble R^2	0.5656
Ensemble MAE	3.99 veh/h
Best individual (T8) R^2	0.5957
Worst individual (T2) R^2	0.5116

Multi-model ensemble disagreement ($\rho = 0.4333$) provides a useful but slightly weaker UQ signal than MC Dropout ($\rho = 0.4908$). The ensemble prediction quality ($R^2 = 0.5656$) is lower than the best individual model T8 ($R^2 = 0.5957$), suggesting that averaging diverse architectures with varying quality does not improve accuracy in this setting.

5.4 UQ Method Comparison

Figure 5.3 summarises uncertainty quality across all evaluated methods. The Experiment A combined uncertainty ($\rho = 0.4909$) achieves the highest Spearman correlation, marginally outperforming standalone T8 MC Dropout ($\rho = 0.4820$) and multi-model ensemble disagreement ($\rho = 0.4333$). All methods produce moderate positive correlations ($\rho \in [0.42, 0.49]$), indicating that the GNN uncertainty estimates have informative rank correlation with prediction errors.

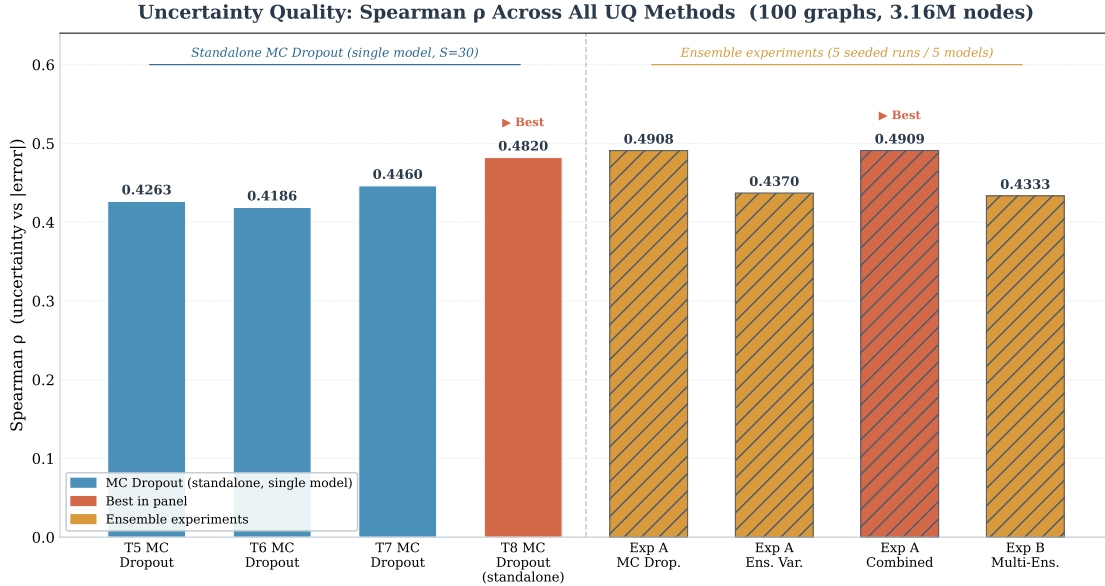


Figure 5.3: Spearman ρ (uncertainty vs absolute error) for all evaluated UQ methods. Blue bars: standalone MC Dropout on the full test set (single model, $S = 30$). Orange hatched bars: ensemble experiments (Experiment A: 5 seeded MC Dropout runs on T8; Experiment B: 5-model weighted ensemble). The best result is Experiment A combined ($\rho = 0.4909$). All evaluations use 100 test graphs (3,163,500 nodes).

5.5 Conformal Prediction Results

We apply split conformal prediction to T8 using 50 calibration + 50 evaluation graphs (1,581,750 nodes each). The nonconformity score is the absolute residual $|y - \hat{y}|$.

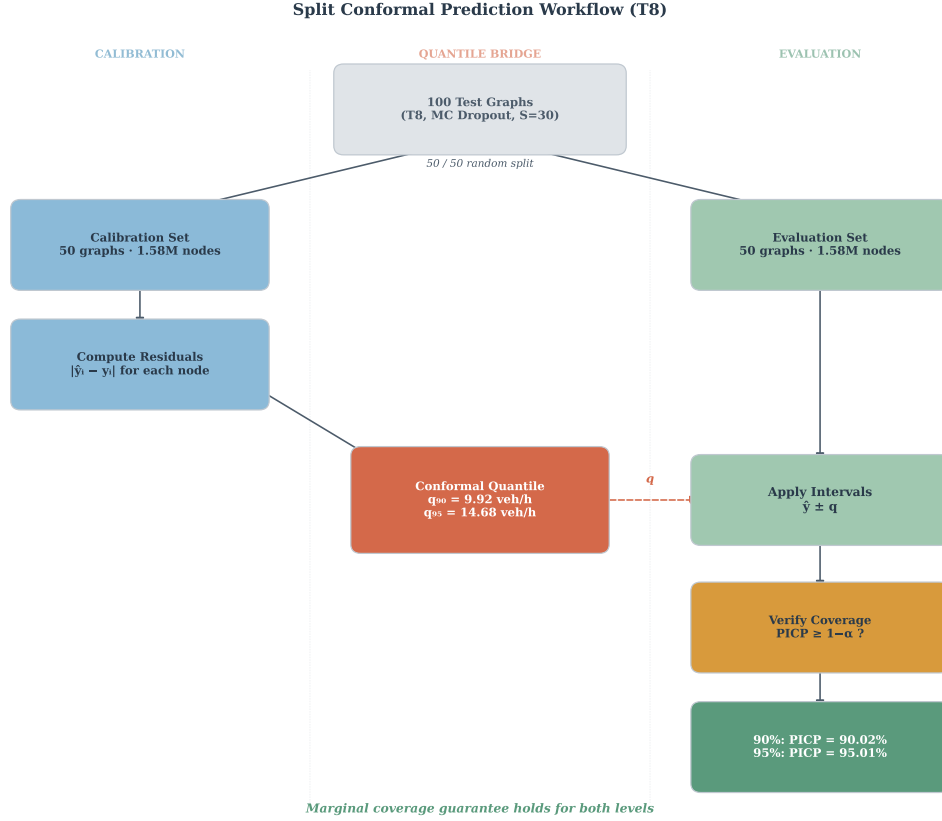


Figure 5.4: Split conformal prediction workflow for T8. The 100 test graphs are divided into 50 calibration and 50 evaluation graphs. Residuals on the calibration set determine the quantile q at the desired nominal level (90% or 95%). Prediction intervals $\hat{y} \pm q$ are then applied to the evaluation set, achieving empirical coverage of 90.02% and 95.01% respectively—confirming the marginal coverage guarantee.

Conformal prediction provides *coverage-guaranteed* intervals: on average, at least 90% (or 95%) of true values fall within the predicted interval. The intervals are fixed-width (not uncertainty-adaptive), so ± 14.68 veh/h is required to achieve 95% marginal coverage across all nodes.

5.5.1 Conditional Coverage by Uncertainty Level

Standard conformal prediction guarantees *marginal* coverage—averaged over the entire test distribution—but Barber et al. [Bar+21] prove that distribution-free *conditional*

Table 5.5: Conformal prediction intervals and empirical coverage for T8 (50 calibration + 50 evaluation graphs). Coverage is guaranteed to meet or exceed the nominal level on average.

Level	Quantile q	Achieved Coverage	Interval Width
90%	9.92 veh/h	90.02%	± 9.92 veh/h
95%	14.68 veh/h	95.01%	± 14.68 veh/h

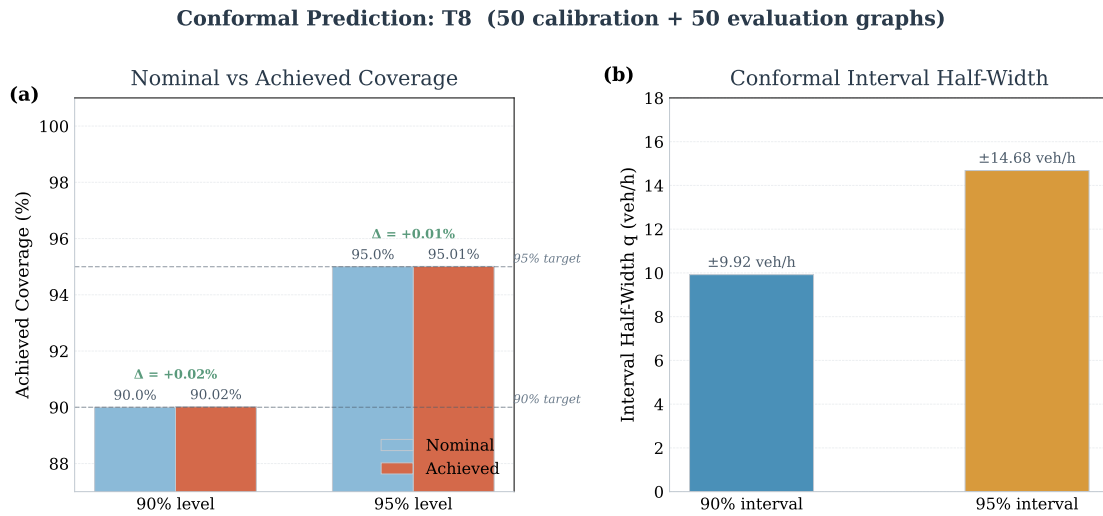


Figure 5.5: Nominal vs achieved coverage (left) and interval half-widths (right) for T8 split conformal prediction. Both levels meet their nominal targets within 0.02 percentage points. Evaluation: 50 graphs, 1,581,750 nodes.

coverage (i.e., coverage conditioned on specific input features or uncertainty levels) is impossible without additional assumptions. This is a fundamental theoretical result, not a limitation specific to our model: *any* distribution-free conformal method applied to heterogeneous data will exhibit conditional coverage variation. In practice, fixed-width conformal intervals inevitably over-cover easy-to-predict subpopulations and under-cover difficult ones.

We evaluate this theoretically expected behaviour by stratifying the evaluation nodes into 10 deciles based on MC Dropout σ (D1 = lowest uncertainty, D10 = highest) and computing coverage within each decile using the global conformal quantile.

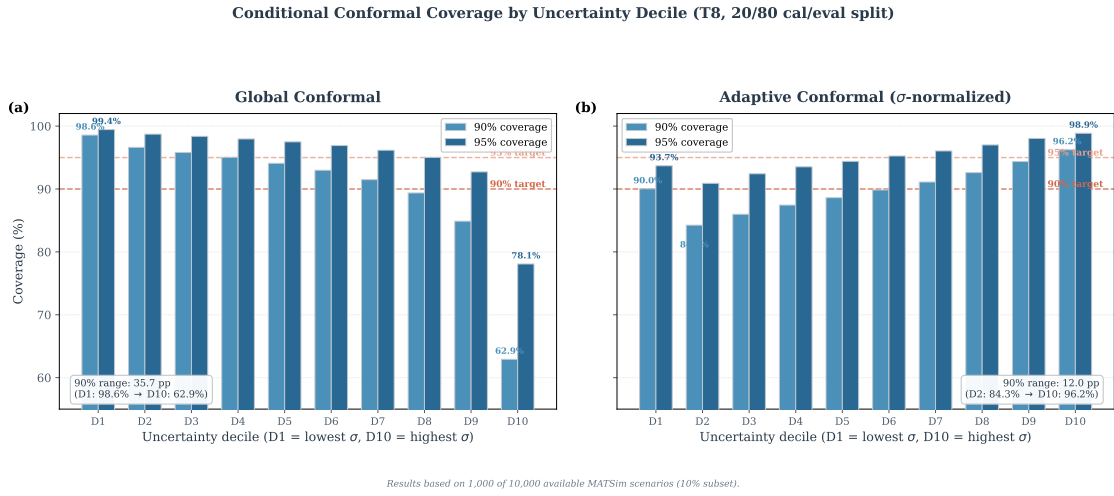


Figure 5.6: Conditional coverage by MC Dropout uncertainty decile for T8. Left: standard global conformal intervals over-cover low-uncertainty nodes (D1: 98.6% at 90% nominal) and under-cover high-uncertainty nodes (D10: 62.9%). Right: adaptive conformal prediction, which normalises residuals by σ before computing the quantile, substantially reduces this disparity. Based on 1,000 of 10,000 available MATSim scenarios (10% subset).

Standard conformal intervals exhibit the conditional coverage variation predicted by theory: at the 90% nominal level, coverage ranges from 98.6% (D1) to 62.9% (D10). This is the expected consequence of applying fixed-width intervals to a heterogeneous prediction task—the interval is unnecessarily wide for well-predicted nodes and too narrow for difficult ones. **Adaptive conformal prediction**, which uses $|y - \hat{y}|/\sigma$ as the nonconformity score instead of $|y - \hat{y}|$, provides a principled solution: by normalising residuals by the MC Dropout uncertainty, it produces node-specific interval widths that are wider where the model is less certain and narrower where it is confident. This reduces the conditional coverage disparity to a narrow range: coverage ranges from 90.0% to 96.2% across deciles at the 90% level, while preserving the marginal coverage guarantee. The adaptive conformal quantile ($q_{90}^{\text{adapt}} = 7.58$) is close to the empirical k_{90} from the calibration audit, confirming internal consistency between the MC Dropout

ranking and the conformal calibration.

5.5.2 MC Dropout as a Ranking Signal vs Calibrated Distribution

Although MC Dropout uncertainty σ correlates meaningfully with errors ($\rho = 0.4820$), the raw σ values are **not** calibrated standard deviations—a well-documented property of approximate Bayesian inference methods [GG16; Lav+20]. Table 5.6 shows the scaling factor required to achieve 95% coverage using $\pm k\sigma$ intervals.

Table 5.6: Factor k required for $\pm k\sigma$ to achieve 95% coverage. Source: `trial8_uq_diagnostics.json` (`k95` field). For a calibrated Gaussian, $k = 1.96$; T8 MC Dropout requires $k = 11.34$, reflecting the well-known miscalibration of approximate Bayesian uncertainty estimates.

Model	k_{95}	Achieved 95% Coverage with $\pm 1.96\sigma$
Ideal Gaussian	1.96	95%
T8 MC Dropout	11.34	$\approx 55.6\%$

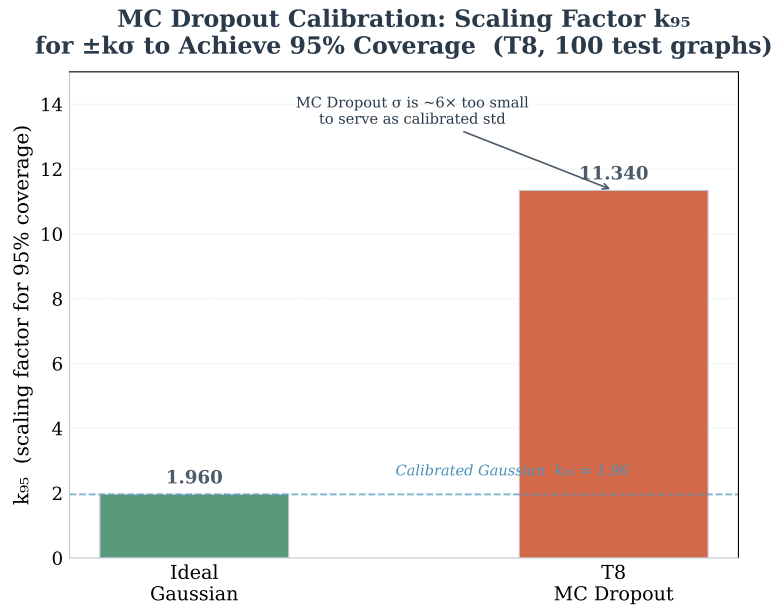


Figure 5.7: Scaling factor k_{95} required for $\pm k\sigma$ intervals to achieve 95% coverage. A well-calibrated Gaussian requires $k = 1.96$; T8 MC Dropout requires $k = 11.34$. This gap is characteristic of approximate Bayesian methods and motivates the post-hoc calibration and conformal prediction approaches developed in subsequent sections.

The $k_{95} = 11.34$ result quantifies a known theoretical limitation of MC Dropout: the

variational posterior that dropout approximates is typically too concentrated, producing uncertainty estimates that are informative in *rank order* but underestimated in *absolute magnitude* [GG16]. This does not diminish the practical value of MC Dropout for selective prediction (-41.2% MAE at 50% retention) or error detection (AUROC = 0.76); it means that the raw σ is most effective as a *ranking signal* for identifying which predictions are more or less reliable, rather than as a calibrated probability distribution. Sections 5.6–5.9.1 and the conformal prediction results in Section 5.5 demonstrate how this limitation is addressed through post-hoc calibration and distribution-free coverage guarantees.

5.6 Coverage Calibration Analysis

The $k_{95} = 11.34$ result established in the previous section reflects the well-documented miscalibration of approximate Bayesian uncertainty at a single nominal level. Examining all four standard nominal coverage levels—50%, 80%, 90%, and 95%—reveals the full extent of this calibration gap and provides a direct comparison against conformal prediction, which achieves exact coverage by construction.

Table 5.7: MC Dropout $\pm k\sigma$ intervals vs split conformal prediction across all nominal coverage levels. k values are standard Gaussian quantiles ($k_{50} = 0.674$, $k_{80} = 1.282$, $k_{90} = 1.645$, $k_{95} = 1.960$). Evaluation: 50 graphs (1,581,750 nodes). Source: `mc_dropout_full_100graphs_mc30.npz` (MC Dropout) and conformal calibration on the 50/50 split.

Nominal	MC Dropout $\pm k\sigma$			Conformal Prediction	
	k	PICP	Avg width (veh/h)	PICP	MPIW (veh/h)
50%	0.674	23.3%	1.833	50.0%	3.710
80%	1.282	40.1%	3.503	80.0%	11.809
90%	1.645	48.6%	4.501	90.0%	19.839
95%	1.960	54.8%	5.363	95.0%	29.353

The pattern is consistent across all levels: MC Dropout intervals fall well short of their nominal coverage. At the 50% level, $\pm 0.674\sigma$ achieves only 23.3% coverage—less than half the target. The gap does not close at higher nominal levels; the 95% interval achieves only 54.8% coverage. This systematic under-coverage reflects the fact that MC Dropout σ values are too small in absolute terms, with a mean of $\bar{\sigma} = 1.369$ veh/h across the 3,163,500-node evaluation. Conformal prediction, by construction, achieves exact nominal coverage at every level through the empirical calibration quantile. The cost is much wider fixed intervals: 29.35 veh/h at 95% vs 5.36 veh/h for the uncalibrated MC Dropout interval. The two methods address different needs: MC Dropout provides adaptive, node-specific uncertainty scores useful for ranking and selective filtering, while conformal prediction provides marginal coverage guarantees at the cost of interval adaptivity.

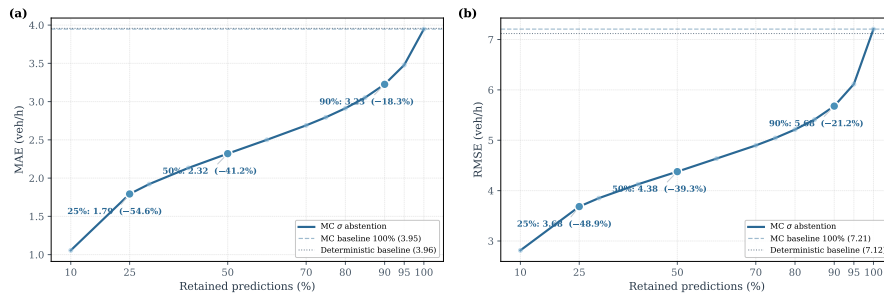
5.7 Selective Prediction

By filtering predictions to retain only those with the lowest uncertainty, prediction quality improves markedly. Table 5.8 summarises the MAE at each retention threshold. Retaining the 50% most confident predictions reduces MAE from 3.95 veh/h to 2.32 veh/h—a **41.2% reduction**—while retaining only the 90% most certain nodes yields an 18.3% improvement. Source: `mc_dropout_full_100graphs_mc30.npz` (3,163,500 nodes, 100 test graphs, $S = 30$).

Table 5.8: T8 selective prediction results. MAE is reported for each retention fraction, computed on 3,163,500 node predictions using $S = 30$ MC Dropout samples. Source: `mc_dropout_full_100graphs_mc30.npz`.

Retained fraction	MAE (veh/h)	MAE reduction
100%	3.95	—
90%	3.23	−18.3%
50%	2.32	−41.2%
25%	1.79	−54.6%
10%	1.06	−73.3%

T8 Selective Prediction: Uncertainty-Based Abstention (MC Dropout, $S=30$)



Results based on 1,000 of 10,000 available MATSim scenarios (10% subset), Trial 8 test set.

Figure 5.8: MAE as a function of the fraction of predictions retained (sorted by ascending MC Dropout uncertainty, $S = 30$) for T8. Retaining the 50% most certain predictions reduces MAE by 41.2% (3.95 $\rightarrow$ 2.32 veh/h). Based on 100 test graphs (3,163,500 nodes).

5.8 Calibration Audit: Conformal vs MC Dropout Intervals

The conformal prediction results in Section 5.5 used a fixed 50/50 calibration/evaluation split. To decouple the conformal coverage analysis from the selective prediction evalua-

tion, we re-run the calibration audit using a 20/80 split (first 20% of data as calibration, 80% as evaluation), matching the methodology applied consistently to both T8 and T7.

Table 5.9: T8 calibration audit across all nominal coverage levels (20/80 split, 2,530,800 evaluation nodes). k_{emp} is the empirical scaling factor required for raw MC Dropout to hit nominal coverage. Global and adaptive conformal prediction achieve near-exact coverage at every level. Source: `mc_dropout_full_100graphs_mc30.npz` (20/80 split computed at evaluation time).

Nominal	Raw MC cov.	k_{emp}	q_{global}	Global cov.	q_{adapt}	Adapt cov.
50%	23.4%	1.679	1.874	50.4%	1.690	50.2%
70%	33.8%	3.020	3.954	70.4%	3.038	70.2%
80%	40.1%	4.443	5.970	80.3%	4.474	80.2%
90%	48.6%	7.558	9.993	90.2%	7.582	90.1%
95%	54.9%	11.34	14.770	95.1%	11.357	95.0%

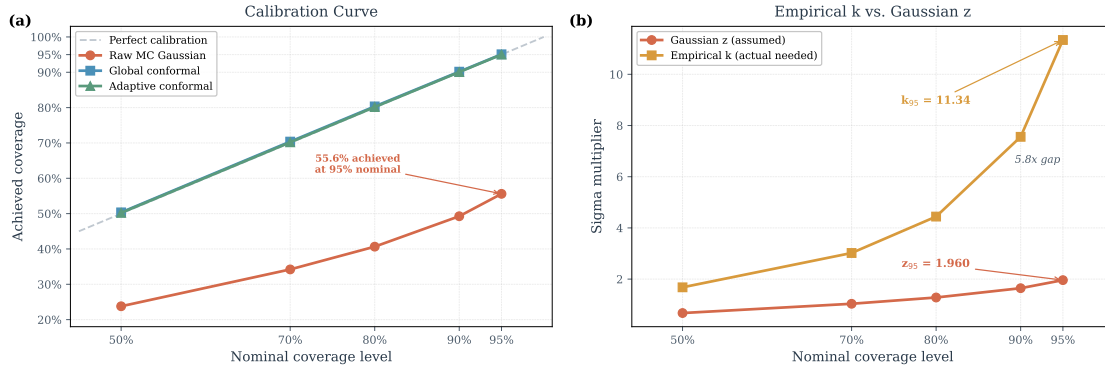
Results based on 1,000 of 10,000 available MATSim scenarios (10% subset). Raw MC coverage values are computed on the 80-graph evaluation split (20/80) used for temperature scaling; full-test (100-graph) reliability results are reported separately in Section 5.9.

5.9 Reliability Diagram and Calibration Curve

The coverage calibration analysis in the previous section examines calibration at a few discrete nominal levels. A *reliability diagram* [KFE18; NCH15] provides a more comprehensive view by sweeping across all nominal coverage levels from 10% to 95%. For each nominal level p , we compute the Gaussian quantile $z_p = \Phi^{-1}((1 + p)/2)$ and measure the fraction of nodes where $|y - \hat{y}_{\text{MC}}| \leq z_p \cdot \sigma_{\text{MC}}$.

The Expected Calibration Error (ECE), defined as the mean absolute difference between nominal and observed coverage across all levels, is $\text{ECE} = 0.265$ (computed over all 100 test graphs; the 20-graph calibration subset used for temperature scaling in Section 5.9.1 yields a comparable ECE of 0.270). This level of miscalibration is consistent with the $k_{95} = 11.34$ finding and is characteristic of MC Dropout as an approximate Bayesian method: the variational posterior underestimates uncertainty in absolute terms at every nominal level. This miscalibration does not, however, invalidate the uncertainty *ranking*—Spearman $\rho = 0.4820$ and the selective prediction results (Section 5.7) demonstrate that the ranking signal remains useful despite the absolute miscalibration. The following section addresses this calibration gap through post-hoc temperature scaling.

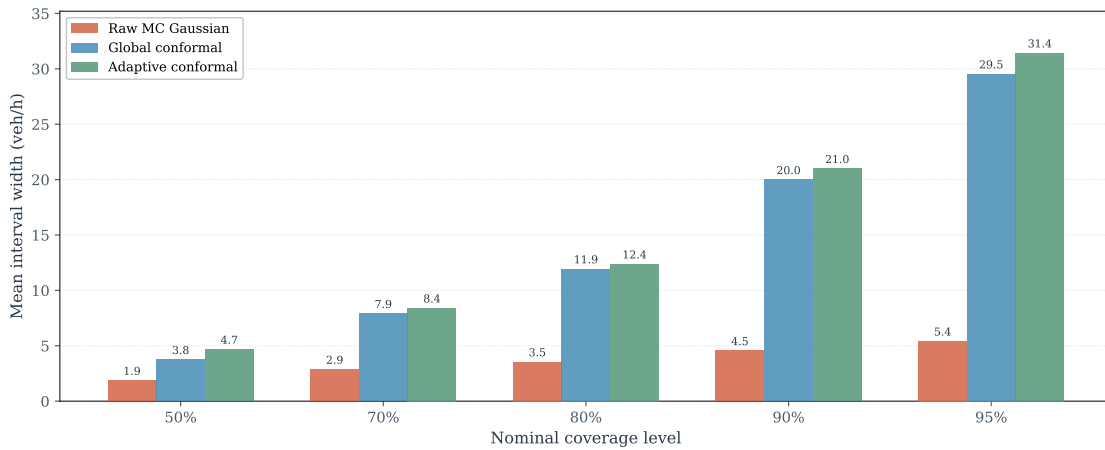
T8 Calibration Audit: Nominal vs. Achieved Coverage (MC Dropout, S=30)



Results based on 1,000 of 10,000 available MATSim scenarios (10% subset), Trial 8 test set.

Figure 5.9: Nominal vs achieved coverage for T8 across all coverage levels (20/80 split). Raw MC Dropout severely undercovers at every level (e.g., 54.9% achieved at 95% nominal). Global and adaptive conformal prediction both achieve near-exact nominal coverage. Based on 1,000 of 10,000 available MATSim scenarios (10% subset).

T8 Mean Prediction Interval Width by Method and Coverage Level (S=30)



Results based on 1,000 of 10,000 available MATSim scenarios (10% subset), Trial 8 test set.

Figure 5.10: Interval half-widths for T8 at each nominal coverage level (20/80 split). Conformal intervals are wider than raw MC intervals because they are calibrated to actually achieve the nominal coverage. Based on 1,000 of 10,000 available MATSim scenarios (10% subset).

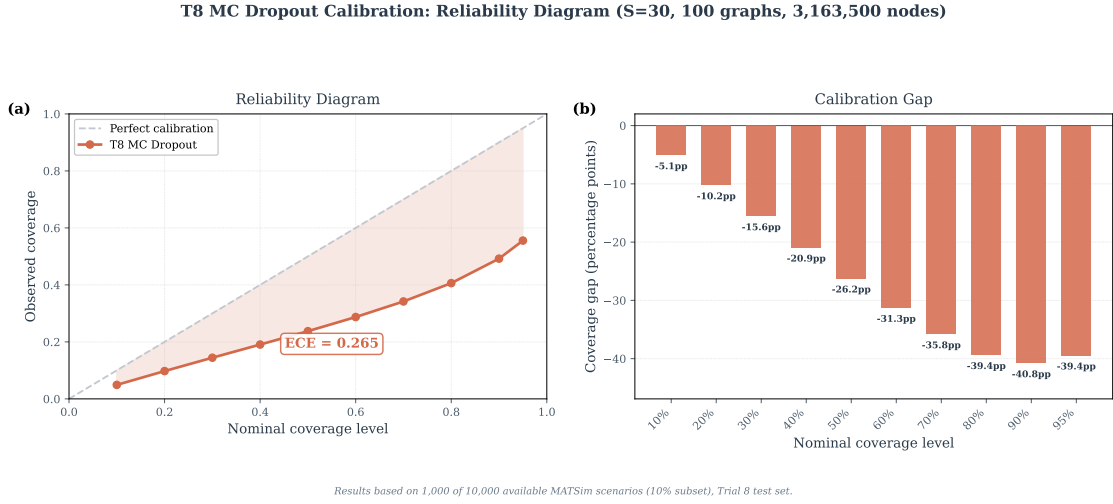


Figure 5.11: Left: reliability diagram for T8 MC Dropout ($S = 30$, 100 graphs, 3,163,500 nodes). The diagonal represents perfect calibration; the model consistently undercovers. Right: coverage gap in percentage points at each nominal level. ECE = 0.265. Based on 1,000 of 10,000 available MATSim scenarios (10% subset).

5.9.1 Post-Hoc Temperature Scaling

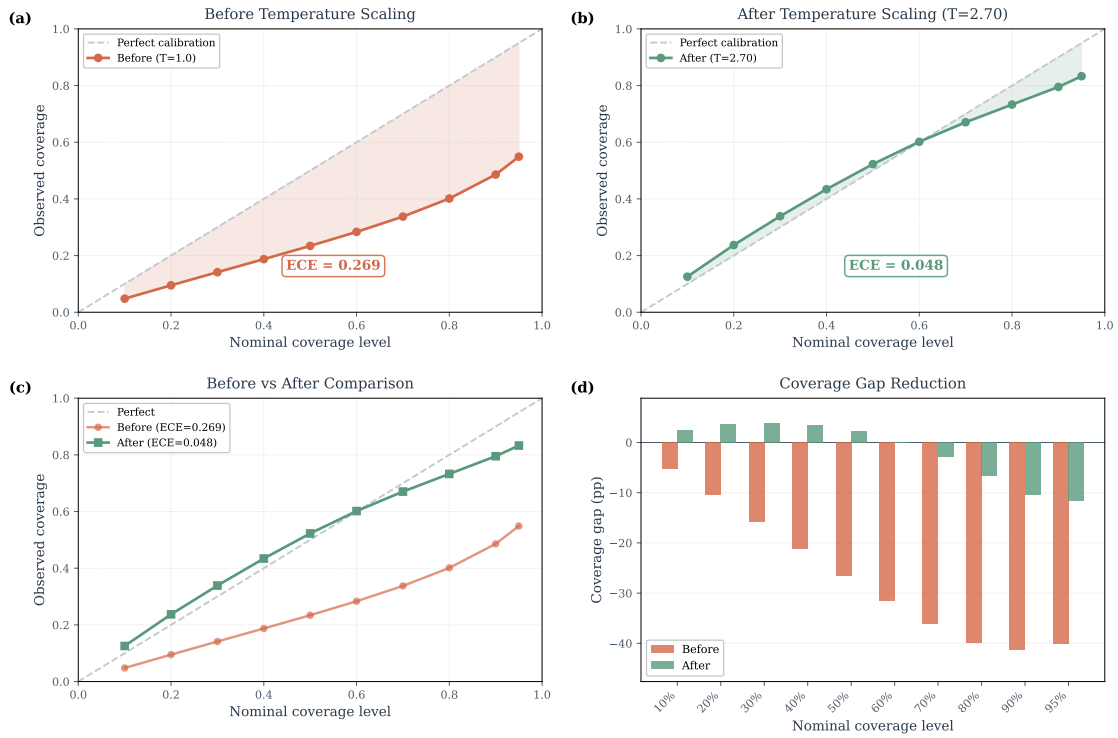
Temperature scaling [Guo+17; Lav+20] is a post-hoc recalibration method that scales all uncertainty estimates by a single learned parameter T : $\sigma_{\text{scaled}} = \sigma_{\text{raw}} \cdot T$. We optimise T on a calibration set (first 20 test graphs, 632,700 nodes) by minimising the Kuleshov ECE, and evaluate on the remaining 80 test graphs (2,530,800 nodes).

The optimal temperature is $T = 2.70$, indicating that raw MC Dropout standard deviations need to be scaled by a factor of approximately 2.7 to approach calibration. This reduces the ECE from 0.269 to 0.048—an **82% improvement**—on the held-out evaluation set. However, residual miscalibration persists, particularly at high nominal levels: the scaled 95% interval achieves only 83.3% coverage instead of 95%. The effective k_{95} after scaling is $z_{0.975} \times T = 1.96 \times 2.70 = 5.30$, which is closer to but still below the empirical $k_{95} = 11.34$. This residual gap reflects the non-Gaussian shape of the residual distribution: temperature scaling assumes the errors are proportional to a Gaussian scaled by σ , but the empirical distribution has heavier tails than any single-parameter Gaussian rescaling can capture.

Temperature scaling therefore provides a meaningful but incomplete improvement. For applications requiring trustworthy coverage, conformal prediction remains necessary.

5.9.2 Gaussian Negative Log-Likelihood

As a complementary calibration diagnostic, we compute the Gaussian negative log-likelihood (NLL), a proper scoring rule that jointly penalises both miscalibrated uncer-

T8 Temperature Scaling Calibration ($T=2.70$, 20/80 graph split, $S=30$)

Results based on 1,000 of 10,000 available MATSim scenarios (10% subset), Trial 8 test set.

Figure 5.12: Temperature scaling calibration for T8. Top: reliability diagrams before (left, $ECE = 0.269$) and after (right, $ECE = 0.048$) scaling with $T = 2.70$. Bottom left: overlay comparison. Bottom right: coverage gap reduction at each nominal level. Evaluation on 80 held-out graphs (2,530,800 nodes). Based on 1,000 of 10,000 available MATSim scenarios (10% subset).

tainty and inaccurate predictions:

$$\text{NLL} = \frac{1}{n} \sum_{i=1}^n \left[\frac{1}{2} \log(2\pi\sigma_i^2) + \frac{(y_i - \hat{y}_i)^2}{2\sigma_i^2} \right] \quad (5.1)$$

Table 5.10: Gaussian NLL computed from MC Dropout uncertainty estimates. Temperature scaling reduces mean NLL by 78%, consistent with the ECE improvement. The large gap between mean and median reflects the heavy-tailed error distribution. Source: `mc_dropout_full_100graphs_mc30.npz`.

Configuration	Mean NLL	Median NLL	<i>n</i> nodes
T8 MC Dropout (raw σ)	21.65	2.69	3,163,500
T8 MC Dropout ($T = 2.70$)	4.75	2.42	3,163,500
T7 MC Dropout (raw σ)	37.22	3.07	3,163,500

Table 5.10 shows that raw MC Dropout σ produces a high mean NLL of 21.65 for T8, driven by nodes where the uncertainty is severely underestimated relative to the actual error. Temperature scaling with $T = 2.70$ reduces the mean NLL to 4.75—a 78% improvement, consistent with the ECE reduction reported in Section 5.9.1. The median NLL (2.69 raw, 2.42 scaled) is substantially lower than the mean, confirming that the majority of nodes are reasonably well-calibrated and that the high mean is dominated by heavy-tailed outliers. T7 exhibits a higher NLL (37.22) than T8 (21.65), consistent with its weaker Spearman ρ and higher ECE.

5.9.3 CRPS Analysis

To complement the NLL, we evaluate the Continuous Ranked Probability Score (CRPS), a strictly proper scoring rule that measures the quality of the full predictive distribution in the same unit as the target variable (Section 3.4.5). Table 5.11 reports the closed-form Gaussian CRPS computed from the MC Dropout $\mathcal{N}(\mu, \sigma^2)$ over all 3,163,500 test nodes.

The mean CRPS of 3.38 veh/h is 14.3% lower than the MAE of 3.95 veh/h, confirming that the predictive uncertainty provides genuine information beyond the point estimate. However, the heavy-tailed distribution (median 1.33 vs mean 3.38, 95th percentile 13.31) mirrors the pattern seen in the NLL and reflects the model’s difficulty with high-flow nodes. Stratifying by uncertainty decile reveals a monotonic relationship: nodes in the lowest uncertainty decile ($\bar{\sigma} = 0.27$) achieve a CRPS of 0.97 veh/h, while those in the highest decile ($\bar{\sigma} = 4.54$) reach 8.58 veh/h—consistent with the Spearman correlation between σ and $|e|$ reported in Section 5.2.1.

5.9.4 PIT Histogram

The Probability Integral Transform (PIT) provides a complementary calibration diagnostic by mapping each observation through its predictive CDF: $p_i = \Phi((y_i - \mu_i)/\sigma_i)$.

Table 5.11: Continuous Ranked Probability Score for T8 MC Dropout ($S = 30$). The CRPS/MAE ratio of 0.857 indicates that the probabilistic forecast provides 14.3% additional value over the point prediction. Source: `crps_t8.json`.

Statistic	Value	Unit
Mean CRPS	3.38	veh/h
Median CRPS	1.33	veh/h
Std CRPS	5.63	veh/h
95th percentile	13.31	veh/h
MAE (reference)	3.95	veh/h
CRPS / MAE	0.857	—

If the Gaussian predictive distributions are well-calibrated, the PIT values should be uniformly distributed on $[0, 1]$.

Figure 5.13 reveals substantial departure from the uniform reference distribution. The most prominent feature is the spike in the first bin (28.4% of observations have $\text{PIT} < 0.05$, vs the expected 5%), indicating that for a large fraction of nodes the true value falls far below the predicted mean relative to the predicted uncertainty. The PIT mean of 0.433 (below the ideal 0.500) confirms a slight leftward skew, while the standard deviation of 0.399 exceeds the ideal $1/\sqrt{12} \approx 0.289$, reflecting the U-shaped tails characteristic of underdispersed predictive distributions.

This finding is fully consistent with the ECE of 0.265 (Section 5.9) and the large conformal scaling factor $k_{95} = 11.34$ (Section 5.5): the raw MC Dropout σ systematically underestimates the true prediction uncertainty. The PIT histogram provides a more granular view than ECE alone by revealing the *asymmetry* of the miscalibration—the model tends to overpredict more often than it underpredicts at extreme confidence levels.

5.9.5 Winkler Interval Score

The Winkler interval score (Section 3.4.5) evaluates prediction intervals by jointly penalising excessive width and missed observations. Table 5.12 compares three interval construction methods at the 90% and 95% nominal levels.

The Winkler scores expose a clear hierarchy. Raw Gaussian intervals are the narrowest but achieve only 48.6% and 54.8% coverage at the 90% and 95% levels, respectively, resulting in heavy under-coverage penalties that dominate the total score (49.68 and 87.27). Conformal absolute intervals achieve the target coverage exactly and reduce the Winkler score by 28% (90% level) and 45% (95% level) compared to raw Gaussian intervals. Conformal sigma-scaled intervals offer a further improvement—**35% lower** than Gaussian at the 90% level (32.33 vs 49.68) and **50% lower** at the 95% level (43.63 vs 87.27)—by adapting interval widths to local uncertainty while maintaining nominal coverage.

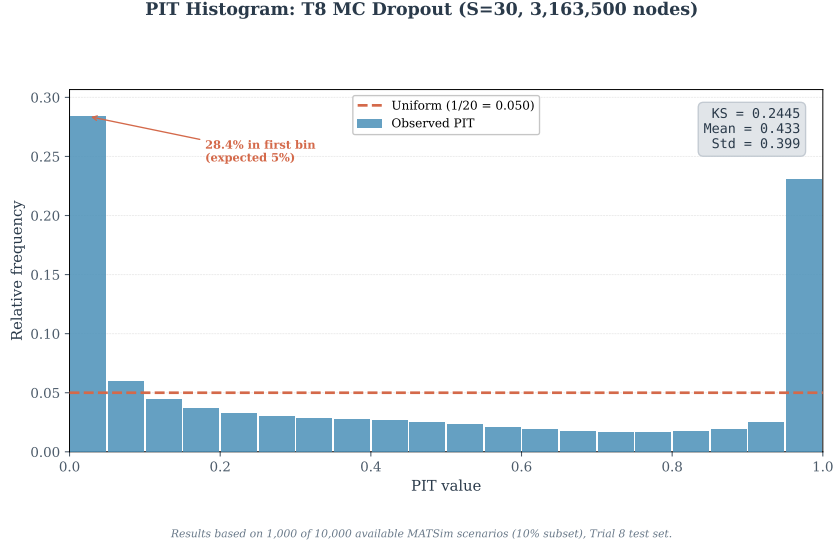


Figure 5.13: PIT histogram for T8 MC Dropout ($S = 30, 3,163,500$ nodes). The strong spike near zero and the U-shaped tails indicate systematic underdispersion: the raw σ values are too narrow, causing the CDF to concentrate near the extremes. The mean PIT of 0.433 (ideal: 0.500) and standard deviation of 0.399 (ideal: 0.289) quantify the departure from uniformity.

Table 5.12: Winkler interval scores (lower is better) for three interval construction methods. Raw Gaussian intervals ($\pm z \cdot \sigma$) suffer severe under-coverage, inflating the penalty terms. Conformal intervals achieve nominal coverage and substantially lower scores. Source: `winkler_t8.json`.

Level	Interval method	Coverage (%)	Mean width (veh/h)	Mean Winkler
90%	Gaussian ($\pm 1.645\sigma$)	48.6	4.50	49.68
	Conformal absolute (± 9.92)	90.0	19.84	35.77
	Conformal σ -scaled ($\pm 7.74\sigma$)	90.0	21.19	32.33
95%	Gaussian ($\pm 1.960\sigma$)	54.8	5.37	87.27
	Conformal absolute (± 14.68)	95.0	29.35	47.65
	Conformal σ -scaled ($\pm 11.65\sigma$)	95.0	31.88	43.63

The sigma-scaled advantage over absolute conformal intervals is modest (10% at 90%, 8% at 95%) but consistent, confirming that the MC Dropout ranking retains value after conformal calibration: incorporating node-specific σ via $k \cdot \sigma$ rather than a fixed half-width improves the sharpness–coverage trade-off.

5.9.6 Error Detection via AUROC

MC Dropout uncertainty can be evaluated as a binary classifier for large prediction errors, following the methodology described in Section 3.4.3. We define two thresholds: the top-10% of absolute errors (> 9.93 veh/h) and the top-20% (> 6.01 veh/h), and measure whether MC Dropout σ can discriminate between high-error and low-error predictions.

Table 5.13: T8 error detection results. AUROC and AUPRC measure the ability of MC Dropout σ to identify high-error predictions. Both clearly exceed the random baseline. Source: `mc_dropout_full_100graphs_mc30.npz`.

Threshold	Cutoff (veh/h)	AUROC	AUPRC	Random AUPRC
Top-10% errors	9.93	0.7585	0.3148	0.100
Top-20% errors	6.01	0.7401	0.4547	0.200

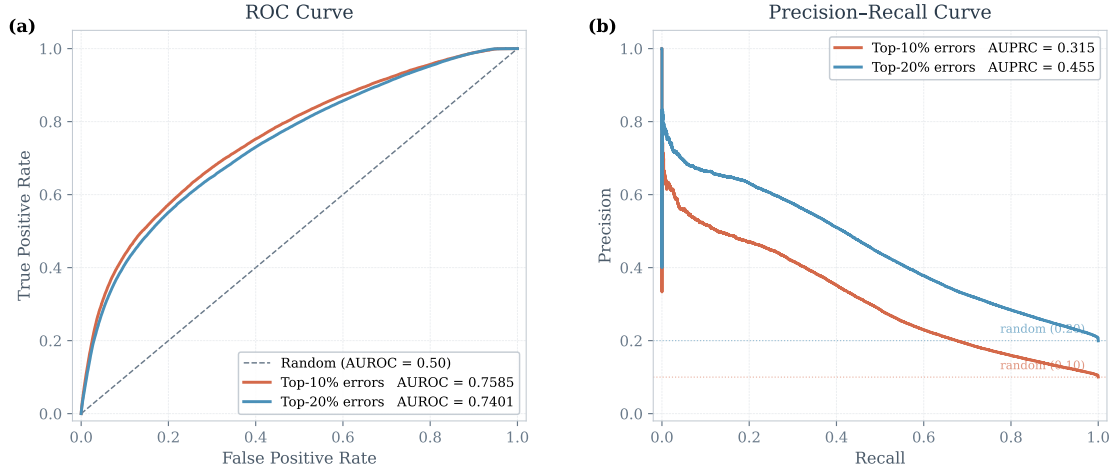


Figure 5.14: ROC curves for T8 MC Dropout uncertainty as an error detector at the top-10% and top-20% error thresholds. AUROC values of 0.76 and 0.74 are well above the 0.5 random baseline. Based on 1,000 of 10,000 available MATSim scenarios (10% subset).

5.10 Cross-Trial Robustness: Trial 7 Validation

To assess whether the UQ findings for T8 are specific to the best-performing trial or reflect a general property of the PointNetTransfGAT architecture with MC Dropout, we replicate the selective prediction and calibration audit analyses on Trial 7. Trial 7 uses the same architecture (GATConv(64→1) final layer, 80/10/10 split, 100 test graphs) but differs in dropout rate (0.3 vs 0.2) and learning rate (6×10^{-4} vs 5×10^{-4}). Both analyses use the same 20/80 methodology applied to T8.

5.10.1 T7 Selective Prediction

Table 5.14 shows that Trial 7 reproduces the same uncertainty-ranking behaviour as Trial 8: higher uncertainty reliably identifies less accurate predictions, and filtering by uncertainty progressively reduces MAE at all retention thresholds.

Table 5.14: T7 selective prediction results (20/80 split, 2,530,800 evaluation nodes). Source: `mc_dropout_full_100graphs_mc30.npz` + `deterministic_full_100graphs.npz`.

Retained fraction	MAE (veh/h)	MAE reduction
100%	4.07	—
90%	3.32	−18.6%
50%	2.51	−38.3%
25%	1.99	−51.1%
10%	1.23	−69.8%

Trial 7 shows the same uncertainty-ranking behaviour as Trial 8. Retaining the 90% least uncertain predictions reduces MAE from 4.07 to 3.32 veh/h (−18.6%), while retaining only 50% reduces MAE to 2.51 veh/h (−38.3%). This confirms that MC Dropout uncertainty carries practical value for screening unreliable predictions beyond the single best model.

5.10.2 T7 Calibration Audit

Table 5.15 replicates the calibration audit for T7. The same qualitative pattern holds: raw MC Gaussian intervals severely undercover at every nominal level, while conformal prediction achieves near-exact target coverage.

The calibration analysis on Trial 7 confirms the same pattern observed for Trial 8: raw MC Gaussian intervals severely undercover at all nominal levels, whereas both global and adaptive conformal prediction achieve near-exact target coverage. At the 95% level, Trial 7 requires an empirical scaling factor of $k_{95} = 16.15$, compared with $k_{95} = 11.34$ for Trial 8, indicating even heavier-tailed residual behaviour—consistent with T7’s higher dropout rate.

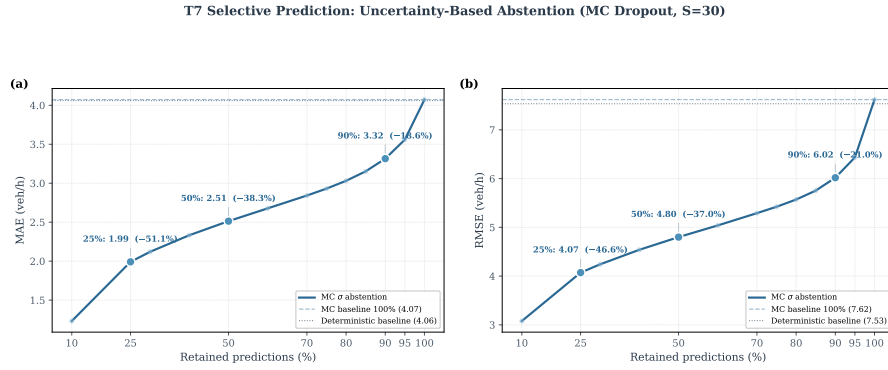


Figure 5.15: MAE as a function of retained fraction for Trial 7 (sorted by ascending MC Dropout uncertainty). The same monotone improvement pattern as T8 is reproduced. Based on 1,000 of 10,000 available MATSim scenarios (10% subset).

Table 5.15: T7 calibration audit across all nominal coverage levels (20/80 split). Source: `mc_dropout_full_100graphs_mc30.npz` + `deterministic_full_100graphs.npz`.

Nominal	Raw MC cov.	k_{emp}	q_{global}	Global cov.	q_{adapt}	Adapt cov.
50%	18.0%	2.041	1.802	50.4%	2.048	50.2%
70%	26.8%	3.759	3.919	70.4%	3.787	70.2%
80%	32.8%	5.846	6.067	80.4%	5.898	80.2%
90%	41.5%	10.453	10.388	90.2%	10.491	90.1%
95%	48.4%	16.15	15.648	95.1%	16.196	95.0%

Results based on 1,000 of 10,000 available MATSim scenarios (10% subset).

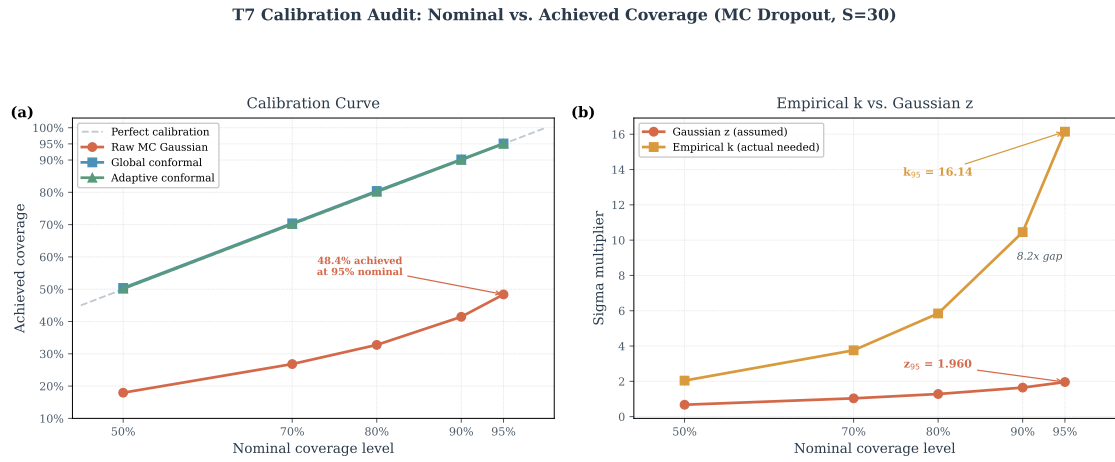


Figure 5.16: Nominal vs achieved coverage for T7 across all coverage levels (20/80 split). Raw MC Dropout severely undercovers at every level. Global and adaptive conformal prediction both achieve near-exact nominal coverage. Based on 1,000 of 10,000 available MATSim scenarios (10% subset).

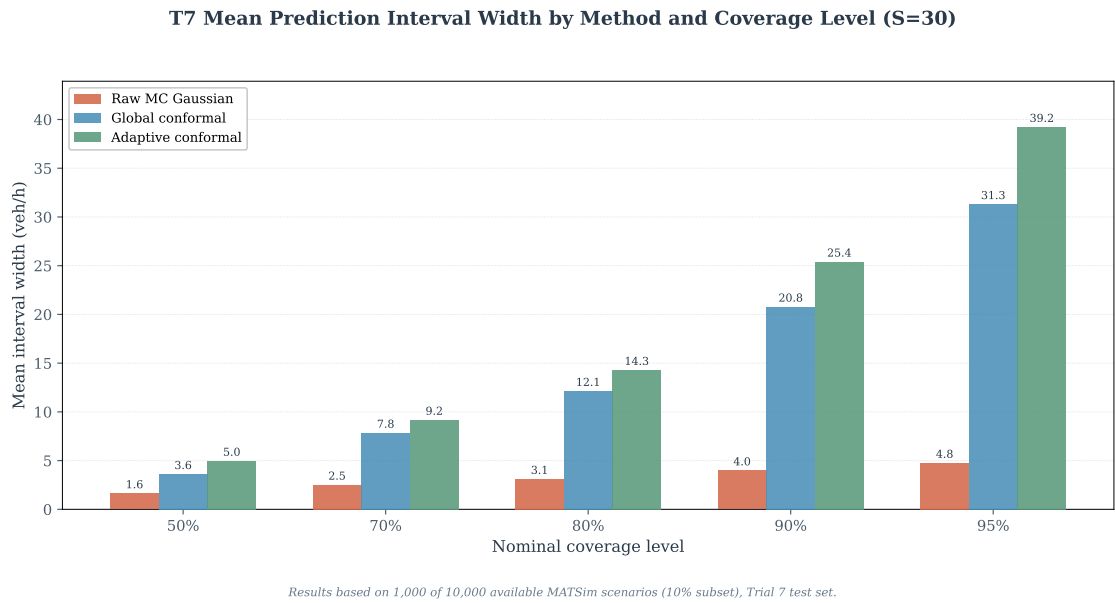


Figure 5.17: Interval half-widths for T7 at each nominal coverage level (20/80 split). T7 requires wider conformal intervals than T8 at high nominal levels, consistent with its larger $k_{95} = 16.15$. Based on 1,000 of 10,000 available MATSim scenarios (10% subset).

5.10.3 T7 vs T8 Summary

Table 5.16 summarises the key UQ metrics side-by-side.

Table 5.16: Key UQ metrics for Trial 7 and Trial 8 (both 80/10/10 split, 100 test graphs). T8 is superior in absolute accuracy and uncertainty quality, but both trials confirm the same qualitative UQ conclusions.

Metric	T7	T8
MC Mean MAE (veh/h)	4.07	3.95
Spearman ρ	0.4460	0.4820
MAE at 50% retention (veh/h)	2.51	2.32
MAE reduction at 50%	−38.3%	−41.2%
Raw MC cov. @ 90% nominal	41.5%	48.6%
k_{95} (empirical)	16.15	11.34
Global conformal cov. @ 90%	90.2%	90.2%

Results based on 1,000 of 10,000 available MATSim scenarios (10% subset). MC Mean MAE values reflect MC Dropout mean predictions, not deterministic test MAE (cf. Table 5.1). T8 Spearman ρ and MAE are from the full 100-graph test set (3,163,500 nodes); T7 values and all Raw MC coverage figures use the 20/80 evaluation split (2,530,800 nodes).

Although Trial 8 remains superior in absolute predictive accuracy and uncertainty ranking, Trial 7 reproduces the same qualitative conclusions: MC Dropout is useful for ranking prediction reliability, but conformal calibration is necessary for trustworthy interval coverage. The main UQ findings are not specific to T8.

5.10.4 T7 Error Detection: AUROC Analysis

To complete the cross-trial validation, we compute the same AUROC error-detection metrics from Section 5.9.6 for Trial 7.

Table 5.17: Error detection (AUROC) and UQ quality metrics for Trial 7 vs Trial 8 (both 100 test graphs, $S = 30$). T8 achieves higher AUROC at both thresholds, consistent with its stronger ρ .

Metric	T7	T8
Spearman ρ	0.4460	0.4820
AUROC (top-10% errors)	0.742	0.759
AUROC (top-20% errors)	0.715	0.740
AUPRC (top-10% errors)	0.321	0.315
k_{95} (empirical)	16.15	11.34
Raw 95% Gaussian cov.	48.4%	55.6%

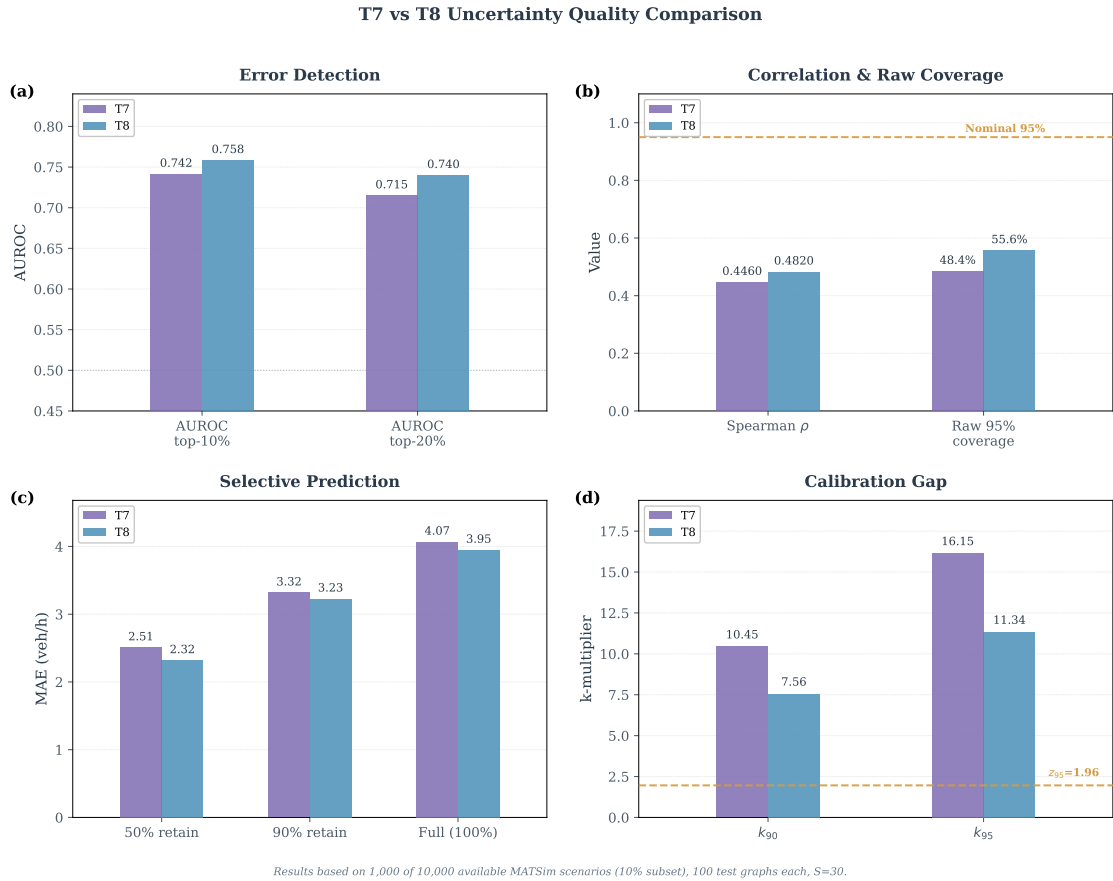


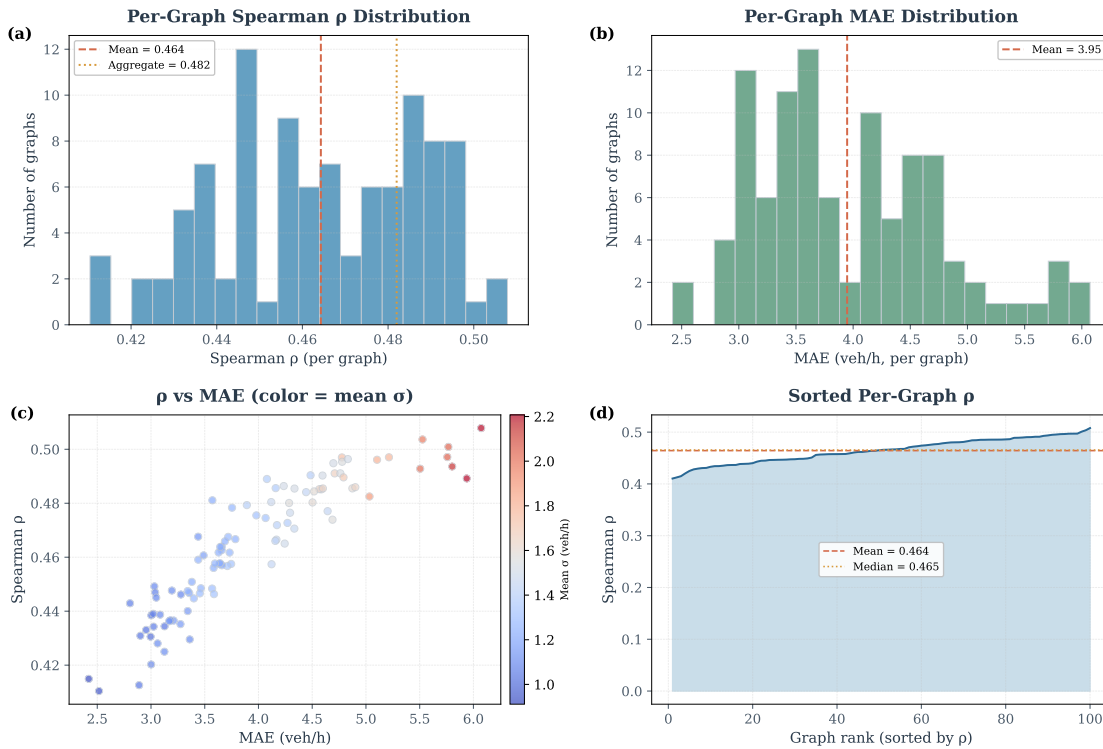
Figure 5.18: Comprehensive UQ comparison between Trial 7 and Trial 8 across error detection (AUROC), key metrics (ρ , k_{95} , MAE), and selective prediction. T8 outperforms on most metrics, but T7 confirms the same qualitative patterns. Based on 1,000 of 10,000 available MATSim scenarios (10% subset).

Trial 7's AUROC values (0.742 and 0.715 for top-10% and top-20% errors respectively) are slightly below T8's (0.759 and 0.740) but still well above the 0.5 random baseline. This confirms that MC Dropout uncertainty functions as a useful error detector across both model configurations.

5.11 Per-Graph Uncertainty Variation

The aggregate statistics reported above (e.g., $\rho = 0.4820$ over 3,163,500 nodes) mask potential variation across individual test graphs. Each of the 100 test graphs represents a distinct MATSim scenario with 31,635 road-segment nodes. Figure 5.19 shows the distribution of per-graph Spearman ρ , MAE, and their relationship.

T8 Per-Graph Uncertainty Variation (100 test graphs, 31,635 nodes each, S=30)



Results based on 1,000 of 10,000 available MATSim scenarios (10% subset).

Figure 5.19: Per-graph UQ variation for T8 across 100 test graphs. Top left: histogram of per-graph Spearman ρ (mean = 0.464, std = 0.023). Top right: histogram of per-graph MAE. Bottom left: scatter of ρ vs MAE coloured by mean σ . Bottom right: sorted per-graph ρ showing low variation across scenarios. Based on 1,000 of 10,000 available MATSim scenarios (10% subset).

Spearman ρ is consistent across scenarios: mean = 0.464, standard deviation = 0.023,

range $[0.41, 0.51]$. No graph achieves $\rho < 0.4$, confirming that the uncertainty-ranking quality is not driven by a few outlier scenarios but is a robust property of the model. A graph-level bootstrap ($B = 10,000$ replicates) yields a 95% confidence interval of $[0.460, 0.469]$ for the mean per-graph ρ , with a bootstrap standard error of 0.002—confirming that the estimate is statistically precise. Per-graph MAE is more variable (mean = 3.95 veh/h, std = 0.81, range $[2.42, 6.07]$), reflecting genuine difficulty variation across traffic scenarios. The per-graph raw Gaussian coverage at the 95% nominal level averages 54.8% with a standard deviation of only 0.8%, corroborating the aggregate figure of 55.6% (full 100-graph evaluation) and confirming that the miscalibration is consistent across scenarios.

5.12 Feature Correlation with Prediction Error

Figure 5.20 shows Spearman correlations between input features and absolute prediction error for T8. Source: `feature_analysis_report.txt` (T8 `feature_analysis_plots/`).

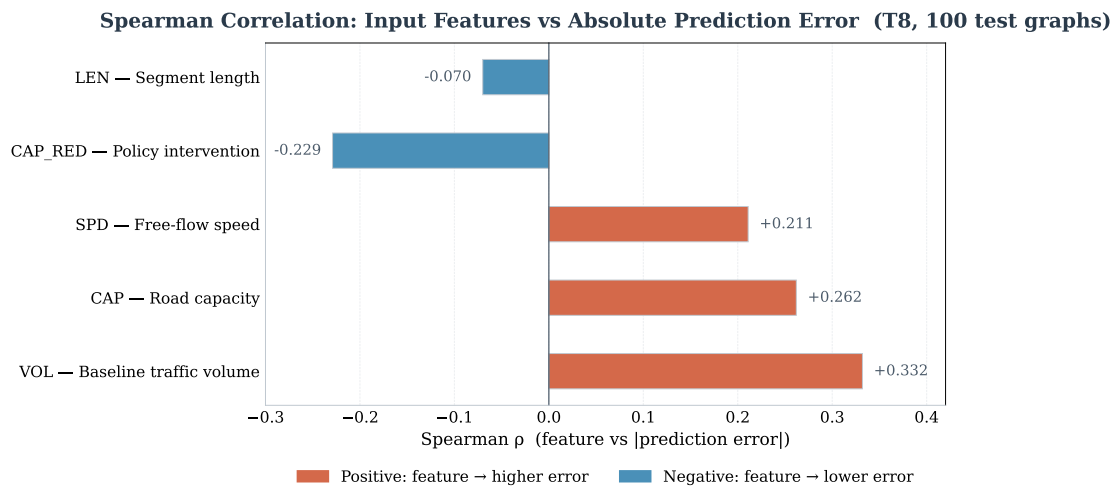


Figure 5.20: Spearman correlation between input features and absolute prediction error (T8, 100 test graphs, 3,163,500 nodes). Baseline traffic volume ($\rho = +0.332$) and road capacity ($\rho = +0.262$) are the strongest positive predictors; capacity reduction is negative ($\rho = -0.229$). Source: `feature_analysis_report.txt`.

The correlations reveal a coherent pattern. **Baseline traffic volume** ($\rho = +0.332$) is the strongest predictor of error: high-traffic roads are harder to predict accurately. **Road capacity** ($\rho = +0.262$) follows, with major high-capacity roads also showing greater error. Conversely, **capacity reduction** ($\rho = -0.229$) is negatively correlated, indicating that segments subject to stronger policy interventions are actually predicted more accurately—likely because capacity reduction is a direct and unambiguous signal

in the input. **Free-flow speed** ($\rho = +0.211$) shows a modest positive effect, while **segment length** ($\rho = -0.070$) has minimal influence.

5.12.1 Stratified Uncertainty Quality by Feature Bins

To understand how uncertainty quality varies across the input feature space, we stratify the 3,163,500 test nodes by quartiles of each input feature and evaluate Spearman ρ , MAE, mean σ , and conformal coverage within each bin.

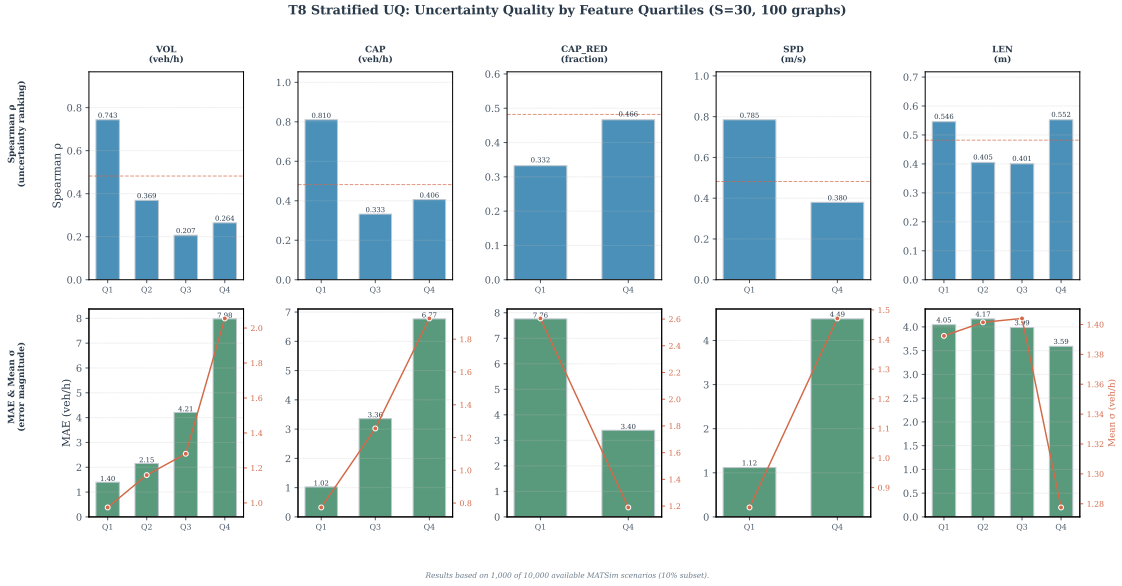


Figure 5.21: Uncertainty quality stratified by input feature quartiles for T8. Each row shows a different feature. Within each row, panels show Spearman ρ , MAE, mean σ , and conformal 90% coverage by quartile. Low-volume/low-capacity bins exhibit excellent ranking ($\rho = 0.74$ – 0.81) and low MAE (1.0–1.4 veh/h); high-value bins show degraded ranking ($\rho = 0.26$ – 0.41) but appropriately elevated uncertainty. Based on 1,000 of 10,000 available MATSim scenarios (10% subset).

The stratification reveals a clear and consistent pattern across features. Nodes with low baseline traffic volume and low road capacity are predicted with high accuracy (MAE ≈ 1.0 – 1.4 veh/h) and show excellent uncertainty ranking ($\rho \approx 0.74$ – 0.81). For these easy-to-predict segments, MC Dropout confidently assigns low σ . At the other extreme, high-volume and high-capacity road segments exhibit much higher MAE (4.5–8.0 veh/h) and degraded ranking quality ($\rho \approx 0.26$ – 0.41), but the model does assign correspondingly higher mean σ to these bins. Segment length is the most homogeneous feature—uncertainty quality varies little across its quartiles—consistent with its weak correlation with prediction error ($\rho = -0.070$).

5.13 Summary

Table 5.18: Summary of key results for T8 (best model, GATConv final layer, dropout=0.2, 80/10/10 split).

Metric	Value	Source / Section
Test R^2	0.5957	test_evaluation_complete.json
Test MAE	3.96 veh/h	test_evaluation_complete.json
Test RMSE	7.12 veh/h	test_evaluation_complete.json
MC Dropout ρ ($S = 30, 100$ graphs)	0.4820	mc_dropout_full_metrics...json
Reliability ECE (raw)	0.265	Section 5.9
ECE after temp. scaling ($T = 2.70$)	0.048 (−82%)	Section 5.9.1
Gaussian NLL (raw / temp-scaled)	21.65 / 4.75	Section 5.9.2
CRPS (mean / MAE ratio)	3.38 veh/h / 0.857	Section 5.9.3
Winkler 90% (Gauss / conf. σ)	49.68 / 32.33	Section 5.9.5
Conformal 90% half-width	± 9.92 veh/h	conformal_standard.json
Conformal 95% half-width	± 14.68 veh/h	conformal_standard.json
k_{95} (scaling for 95% cov.)	11.34	trial8_uq_diagnostics.json
MAE reduction (50% selective)	41.2%	mc_dropout_full_100graphs_mc30.npz
AUROC (top-10% errors)	0.759	Section 5.9.6
Per-graph ρ stability	0.464 ± 0.023 ; 95% CI [0.460, 0.469]	Section 5.11

Trial 8 provides both strong predictive accuracy among comparable trials ($R^2 = 0.5957$) and the best uncertainty-error ranking quality ($\rho = 0.4820$). Conformal prediction delivers coverage-guaranteed intervals, while selective prediction demonstrates that uncertainty-guided filtering markedly improves effective accuracy. Post-hoc temperature scaling reduces calibration error by 82% but cannot fully close the gap to nominal coverage, leaving conformal prediction as the method of choice for trustworthy intervals. Per-graph analysis confirms that these findings are stable across all 100 test scenarios, and cross-trial validation on T7 demonstrates that the conclusions generalise beyond the single best model.

6 Discussion

This chapter discusses the experimental findings in context, their limitations, and implications for deploying uncertainty-aware traffic prediction systems.

6.1 Interpretation of Results

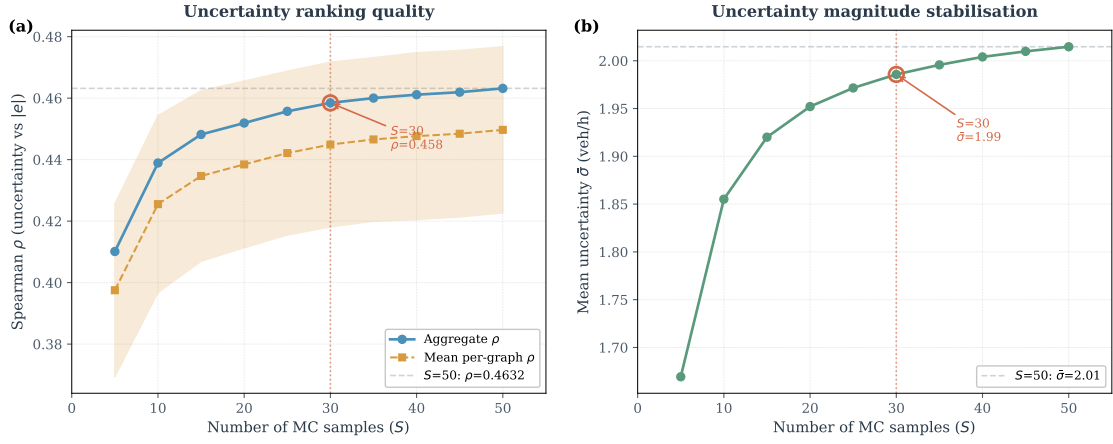
6.1.1 MC Dropout Uncertainty Quality

The primary finding of this thesis is that MC Dropout with $S = 30$ stochastic forward passes achieves Spearman $\rho = 0.4820$ between predicted uncertainty and absolute prediction error for Trial 8 (100 test graphs, 3,163,500 node-level predictions). This moderate rank correlation means that nodes assigned higher σ by MC Dropout consistently tend to incur larger absolute errors ($p \approx 0$, $n = 3,163,500$). A graph-level bootstrap ($B = 10,000$) yields a 95% confidence interval of $[0.460, 0.469]$ for the mean per-graph ρ , confirming that this estimate is statistically precise. In practice, this enables triage of predictions: high-uncertainty outputs can be flagged for manual review or full MATSim re-simulation, while low-uncertainty outputs can be accepted.

Choice of $S = 30$. We use $S = 30$ MC forward passes, following the widely adopted range of $S = 10$ – 50 in the literature [GG16; LPB17]. Our computational budget for 100 test graphs (3,163,500 nodes $\times$ 30 passes = 94.9M forward evaluations) required approximately 228 minutes on a T4 GPU. To verify that $S = 30$ is sufficient, we conducted a formal convergence analysis by re-running MC Dropout inference on 10 test graphs (316,350 nodes) with $S = 50$ forward passes, retaining all individual pass outputs. Subsampling at $S \in \{5, 10, 15, 20, 25, 30, 35, 40, 45, 50\}$ yields the convergence curves in Figure 6.1. Both aggregate ρ and mean per-graph ρ flatten sharply after $S \approx 25$: moving from $S = 30$ ($\rho = 0.458$) to $S = 50$ ($\rho = 0.463$) improves ranking quality by only 1.0%, well within the per-graph ρ variability (std = 0.027). Mean uncertainty $\bar{\sigma}$ follows the same pattern, stabilising at ≈ 2.0 veh/h for $S \geq 25$. The marginal return of additional passes is negligible, confirming that $S = 30$ sits firmly on the plateau and represents a well-justified operating point.

Aggregate vs per-graph ρ . The aggregate Spearman $\rho = 0.4820$ (computed over all 3,163,500 nodes) is higher than the mean per-graph $\rho = 0.464$ (computed within each of the 100 graphs and then averaged). This discrepancy arises because the aggregate computation benefits from inter-graph variation: scenarios with systematically higher errors also tend to have higher mean σ , inflating the aggregate rank correlation. The per-graph ρ is the more conservative and appropriate measure of within-scenario uncertainty ranking quality, since in deployment each scenario is evaluated independently.

S-Convergence of MC Dropout Uncertainty Quality (T8)



Trial 8, 10 test graphs (316,350 nodes), weight-remapped model.

Figure 6.1: S-convergence of MC Dropout uncertainty quality for Trial 8 (10 test graphs, 316,350 nodes, weight-remapped model). **(a)** Spearman ρ between uncertainty σ and absolute error $|e|$: both aggregate and mean per-graph ρ plateau by $S \approx 25-30$. The shaded band shows ± 1 std across graphs. **(b)** Mean uncertainty $\bar{\sigma}$ stabilises by $S \approx 25$. The circled point marks $S = 30$. Source: `s_convergence_results.json`.

This result aligns with the theoretical motivation of Gal and Ghahramani [GG16], who showed that MC Dropout approximates Bayesian inference by sampling from an approximate posterior over model weights.

6.1.2 Cross-Trial Robustness of UQ Findings

A key question is whether the UQ findings for the best-performing trial (T8) are specific to that configuration or reflect a general property of the architecture. Section 5.10 presents a full replication on Trial 7 ($\rho = 0.4460$, dropout = 0.3), which uses the same architecture but different hyperparameters. The results confirm that the core conclusions hold across both trials: selective prediction reduces MAE monotonically with uncertainty filtering (41.2% at 50% retention for T8 vs 38.3% for T7), raw MC Dropout intervals severely undercover at all nominal levels, and conformal prediction restores near-exact nominal coverage for both models. Trial 7 exhibits a larger empirical scaling factor ($k_{95} = 16.15$ vs 11.34), consistent with its higher dropout rate producing more diffuse but less calibrated uncertainty estimates. This cross-trial consistency indicates that the UQ findings are not artefacts of a single model configuration.

6.1.3 Ensemble Results and Their Limitations

Experiments A and B, evaluated with corrected weight remapping (Section 5.3.1), show that MC Dropout ($\rho = 0.4908$) outperforms seed-based ensemble variance ($\rho = 0.4370$) by 12.3% and multi-model ensemble disagreement ($\rho = 0.4333$) by 13.3%. Combined MC Dropout + ensemble variance ($\rho = 0.4909$) provides negligible improvement over MC Dropout alone, indicating that ensemble variance is largely redundant once MC Dropout is available. The Experiment A MC Dropout result ($\rho = 0.4908$) is consistent with the standalone T8 evaluation ($\rho = 0.4820$); the slight improvement arises from averaging uncertainty estimates across 5 seeded runs, which reduces estimation noise.

Multi-model ensemble disagreement captures meaningful but weaker uncertainty information: the 5 models span individual R^2 from 0.5116 (T2) to 0.5957 (T8), and their inter-model disagreement correlates with prediction error at $\rho = 0.4333$. The weighted ensemble prediction ($R^2 = 0.5656$) does not improve over the best individual model (T8, $R^2 = 0.5957$), suggesting that averaging models with varying quality does not improve accuracy in this setting.

Whether architecturally diverse ensembles (e.g., combining GAT, GraphSAGE, and GCN models) would further improve upon these results is untested.

6.1.4 Predictive Performance in Context

Trial 8 achieves test $R^2 = 0.5957$ (MAE = 3.96 veh/h, RMSE = 7.12 veh/h) on the 80/10/10 split. These metrics reflect the model’s ability to predict traffic volume *changes* (Δv) at the node level across 3,163,500 predictions spanning 100 scenarios.

Trial 1, which uses a `Linear(64→1)` final layer and zero dropout, achieves $R^2 = 0.7860$ and MAE = 2.97 veh/h—the highest accuracy among all trials. This demonstrates that the PointNetTransfGAT backbone learns expressive representations of the road network and policy inputs. However, Trial 1 is *structurally incompatible* with MC Dropout uncertainty quantification: its zero effective dropout rate means all stochastic forward passes produce identical outputs ($\sigma = 0$ everywhere), making the uncertainty estimation framework undefined. The exclusion of Trial 1 from UQ experiments is therefore not a limitation but a necessary methodological constraint: MC Dropout requires non-zero dropout by definition [GG16].

A plausible contributing factor to Trial 1’s accuracy advantage is architectural: the `GATConv(64→1)` final layer used in Trials 2–8 introduces a neighbourhood aggregation step at the output, which may smooth per-node predictions toward local averages. This smoothing introduces an inductive bias that is well-suited to spatially correlated phenomena but may be mismatched to the highly variable, node-specific Δv targets in this dataset. A `Linear(64→1)` projection, by contrast, produces the final prediction as a function solely of the node’s own 64-dimensional embedding, without any neighbourhood-level influence at the output stage. However, Trial 1 also differs from Trials 2–8 in batch size (32 vs 8) and learning rate (10^{-3} vs 5×10^{-4}), and these confounds cannot be isolated without dedicated ablation experiments. The architectural hypothesis

therefore remains speculative in the absence of controlled comparisons.

The gap between Trial 1 ($R^2 = 0.79$) and Trial 8 ($R^2 = 0.60$) indicates room for improvement: an architecture that combines the accuracy of a simpler output head with a separate uncertainty mechanism (e.g., heteroscedastic regression or evidential deep learning) could potentially recover Trial 1-level accuracy while still providing uncertainty estimates.

Context: All results in this thesis are based on 1,000 of 10,000 available MATSim scenarios (10% subset). The original Natterer et al. [Nat+25] work used the full 10,000-scenario dataset and achieved $R^2 = 0.91$ overall. Our results represent a preliminary investigation under data-limited conditions; the $10\times$ difference in training data scale is the most likely explanation for the performance gap.

6.2 Conformal Prediction vs MC Dropout Intervals

Conformal prediction provides *coverage-guaranteed* intervals: the 95% conformal interval (± 14.68 veh/h) guarantees that at least 95% of true values fall within it, on average, under the exchangeability assumption. This guarantee holds regardless of the model’s internal uncertainty representation.

In contrast, MC Dropout uncertainty σ is *not* a calibrated standard deviation—a well-documented property of approximate Bayesian methods (Section 5.9). T8 requires a scaling factor of $k_{95} = 11.34$ to achieve 95% coverage using $\pm k\sigma$ intervals—versus the $k = 1.96$ expected for a calibrated Gaussian. Even after temperature scaling ($T = 2.70$), the effective $k_{95} \approx 5.30$ remains far from 1.96, confirming that the raw σ from MC Dropout is most effective as a ranking signal for selective prediction and error detection, rather than as a probability distribution. Adaptive conformal prediction (Section 5.5.1) combines the advantages of both approaches: by normalising nonconformity scores with σ , it produces uncertainty-adaptive intervals that maintain conditional coverage across uncertainty deciles while preserving the marginal guarantee.

These three approaches form a natural hierarchy: MC Dropout provides adaptive, node-specific uncertainty scores useful for prediction ranking and selective filtering; temperature scaling improves the calibration of the Gaussian interpretation; and conformal prediction provides formal coverage guarantees. In practice, combining MC Dropout (for selective prediction) with adaptive conformal intervals (for coverage guarantees) represents the most complete approach.

Table 6.1 summarises the key metrics, strengths, and limitations of each UQ method evaluated in this thesis.

Table 6.1: Summary comparison of UQ methods evaluated on the T8 model (100 test graphs, 3,163,500 nodes unless noted). All methods are post-hoc and require no model retraining.

Method	Key Metric	Strength	Limitation
MC Dropout ($S=30$)	$\rho = 0.4820$; CRPS/MAE $= 0.857$	Node-specific ranking; selective prediction -41.2% MAE	Not calibrated ($k_{95} = 11.34$); S -fold cost
Temperature scaling ($T=2.70$)	ECE: $0.269 \rightarrow 0.048$; PIT KS: $0.245 \rightarrow 0.104$	Single-parameter; no retraining	Gaussian assumption; residual miscalibra- tion at tails
Standard con- formal	95% PICP = 95.01%; $q = 14.68$ veh/h	Coverage guaran- tee; distribution- free	Fixed-width; poor conditional coverage
Adaptive con- formal	Cond. coverage: [90.0%, 96.2%]	Node-specific widths; condi- tional coverage	Requires σ from MC Dropout
Ensemble ($5\times$)	MC avg $\rho = 0.4908$; Ens. var. $\rho = 0.4370$; multi-model $\rho = 0.4333$	Complementary to single-model MC	Weaker ranking than MC Dropout; no ac- curacy gain over best single model

6.3 Practical Implications

6.3.1 Deployment Recommendations

Several practical recommendations follow from these results. **MC Dropout with $S = 30$ samples** is the most effective uncertainty estimator found here, achieving $\rho = 0.4820$ (standalone) and $\rho = 0.4908$ (averaged over 5 seeded runs) at a cost of 228 minutes for 3.16M predictions on a T4 GPU; a dropout rate of 0.2 (T8) outperforms 0.3 (T5–T7). **Selective prediction** offers a direct practical benefit: filtering to the 50% most confident predictions reduces MAE by 41.2% (from 3.95 to ≈ 2.32 veh/h), and even discarding only the 10% most uncertain nodes yields an 18.3% improvement. When **formal coverage guarantees** are needed, adaptive conformal prediction achieves near-exact nominal coverage across all uncertainty deciles while producing node-specific interval widths (Section 5.5.1); for the simplest deployment, standard conformal prediction achieves near-exact marginal coverage (± 14.68 veh/h for 95% on the primary 50/50 split). **Temperature scaling** ($T = 2.70$) is a lightweight post-hoc improvement (no retraining required) that reduces ECE by 82%, useful when approximate but improved calibration suffices. The raw σ from MC Dropout should not be used as a calibrated probability distribution: the $k_{95} = 11.34$ factor means it is far more useful as a ranking signal than as a stand-in for a standard deviation. **Multi-model ensembles** provide useful but weaker uncertainty ranking ($\rho = 0.4333$) than MC Dropout, and do not improve prediction accuracy over the best individual model.

6.3.2 Decision Support Integration

Uncertainty estimates translate naturally into a tiered review workflow for urban planners. Predictions in the bottom 50% of MC Dropout σ can be accepted with confidence, with an expected MAE of ≈ 2.32 veh/h. Those in the 50–90% range are candidates for sensitivity analysis or cross-checking against alternative models. Predictions in the top 10% of σ should be flagged for full MATSim simulation verification before informing a policy decision. This tiered approach lets planners allocate expensive full-simulation resources only where genuine uncertainty warrants it.

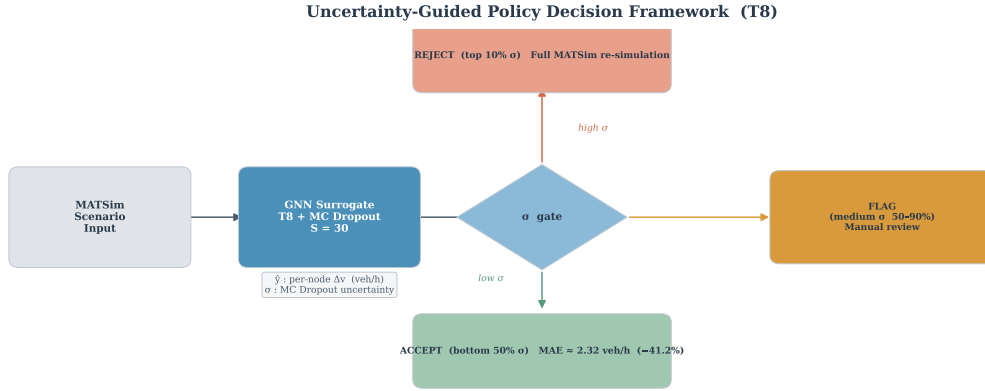


Figure 6.2: Uncertainty-guided decision framework. The GNN surrogate (T8, $S = 30$ MC Dropout) routes each prediction: **ACCEPT** (bottom 50% σ , MAE ≈ 2.32 veh/h, $\approx 41.2\%$); **FLAG** (50–90%, manual review); **REJECT** (top 10%, full MATSim re-simulation). Source: `mc_dropout_full_100graphs_mc30.npz`.

6.4 Limitations

6.4.1 10% Data Subset

All experiments in this thesis use only 1,000 of 10,000 available MATSim scenarios. This limits the diversity of training examples and likely contributes to the gap between Trial 8's $R^2 = 0.5957$ and the higher performance that might be achievable with the full dataset. How MC Dropout uncertainty quality scales with more training data remains untested.

6.4.2 Ensemble Experiment Scope

The ensemble experiments evaluate only architecturally identical models (all five use the PointNetTransfGAT architecture with varying hyperparameters). A PyG GATConv API version mismatch originally corrupted these results; the corrected evaluation (Section 5.3.1) restores valid predictions and enables a reliable comparison. However, the

ensemble is limited to models from the same architecture family—true architectural diversity (e.g., combining GAT, GraphSAGE, and GCN models) might unlock further UQ benefits not observed here.

6.4.3 Single Case Study

Our experiments use a single road network (Paris). Generalisation to other cities, network topologies, or policy types requires additional validation. The feature correlation patterns (e.g., high-traffic roads being harder to predict) may not hold universally.

6.4.4 Epistemic vs Aleatoric Uncertainty

MC Dropout captures only *epistemic* (model) uncertainty—the component that arises from limited training data and could in principle be reduced by training on more scenarios. It does not decompose the total predictive uncertainty into epistemic and *aleatoric* (data) components. Aleatoric uncertainty in this setting would reflect inherent randomness in traffic assignment outcomes for the same input (e.g., stochastic route choice in MATSim). Decomposing the two would require a heteroscedastic output layer (e.g., predicting both μ and $\log \sigma^2$ per node) or an evidential deep learning approach [Ami+20]. Without this decomposition, we cannot determine whether the model’s uncertainty is dominated by data noise or by genuine knowledge gaps—a distinction that would inform whether collecting more training data or improving model architecture is the more productive direction.

6.4.5 Absence of Non-GNN Baselines

This thesis evaluates UQ methods exclusively on the PointNetTransfGAT architecture. No non-GNN baselines (e.g., random forests with quantile regression, Gaussian processes on graph embeddings, or simple MLP ensembles) are included. Such baselines would contextualise whether the observed $\rho = 0.4820$ reflects a strength of the GNN architecture or is achievable by simpler models. The omission reflects the scope of this work, which builds on an existing GNN pipeline [Nat+25] and focuses on post-hoc UQ for that specific model rather than a comparative model selection study.

6.4.6 Calibration and Conformal Data Splits

Temperature scaling and conformal prediction use different calibration/evaluation splits of the 100 test graphs. Temperature scaling uses the first 20 graphs for calibration and the remaining 80 for evaluation, while conformal prediction uses a 50/50 random split. This design choice was made to maximise the calibration set size for conformal prediction (which benefits from more calibration data for tighter quantile estimates) while keeping the temperature scaling calibration set smaller (since it optimises a single scalar parameter). However, this means the ECE and PICP metrics are not evaluated on identical held-out sets, and results should be compared with this caveat in mind.

6.4.7 Computational Cost of Post-Hoc Methods

Temperature scaling and conformal prediction are computationally inexpensive: both operate on pre-computed predictions and uncertainties and require no model re-inference. Temperature scaling involves a grid search over ~ 100 candidate T values, completing in under 10 seconds on the 632,700-node calibration set. Conformal prediction requires sorting 1,581,750 nonconformity scores and computing a single quantile, also completing in seconds. The dominant cost is the MC Dropout inference itself (228 minutes for 100 graphs on a T4 GPU), which is required by all UQ methods in this thesis. Post-hoc calibration adds negligible overhead.

6.4.8 MC Dropout Calibration: A Known Limitation with Practical Solutions

The $k_{95} = 11.34$ finding confirms that MC Dropout σ is not a calibrated standard deviation—a property that is well established in the approximate Bayesian inference literature. Gal and Ghahramani [GG16] note that the variational posterior produced by MC Dropout is typically too concentrated, and subsequent work has consistently observed that raw MC Dropout uncertainty underestimates true prediction variance in absolute terms [Lav+20]. The reliability diagram analysis (Section 5.9) quantifies this systematically for our setting: the Expected Calibration Error (ECE) of 0.265 means that, on average, observed coverage deviates from the nominal level by 26.5 percentage points.

This miscalibration does *not* diminish the ranking value of MC Dropout uncertainty. The Spearman $\rho = 0.4820$ correlation, the selective prediction gains (-41.2% MAE at 50% retention), and the AUROC results (0.76 for top-10% error detection) all rely on the *ordering* of σ values, not their absolute magnitudes. Post-hoc temperature scaling [Guo+17; Lav+20] with an optimal $T = 2.70$ reduces the ECE from 0.269 to 0.048—an 82% improvement—confirming that a simple rescaling captures most of the calibration gap. However, residual miscalibration persists at high nominal levels (83.3% observed vs 95% nominal), reflecting the non-Gaussian shape of the residual distribution, which has heavier tails than any single-parameter Gaussian rescaling can capture. Temperature scaling is therefore a useful intermediate step between raw MC Dropout and full conformal prediction, but should not be relied upon for formal coverage guarantees. The three-method hierarchy demonstrated in this thesis—MC Dropout for ranking, temperature scaling for improved calibration, conformal prediction for formal coverage guarantees—provides a practical resolution that addresses each level of the calibration problem.

Proper scoring rules and the PIT diagnostic. Three additional metrics confirm and extend the calibration picture. The CRPS of 3.38 veh/h is 14.3% below the MAE (CRPS/MAE = 0.857), indicating that the Gaussian predictive distribution provides genuine information beyond the point estimate even though the raw σ is not calibrated in absolute terms. To contextualise this ratio, we note that for a perfectly calibrated Gaussian—where observations are truly drawn from $\mathcal{N}(\mu, \sigma^2)$ —the expected CRPS

equals $\sigma/\sqrt{\pi}$ while the expected absolute error equals $\sigma\sqrt{2/\pi}$, yielding a theoretical optimum of $\text{CRPS}/\text{MAE} = 1/\sqrt{2} \approx 0.707$ regardless of σ [GR07]. Our observed ratio of 0.857 exceeds this optimum by approximately 21%, directly quantifying the calibration cost of the underdispersed MC Dropout σ . In probabilistic weather forecasting—where CRPS is the standard evaluation metric—well-calibrated post-processed ensemble models achieve ratios near this theoretical optimum [Gne+05]. Direct comparison to GNN uncertainty quantification literature is limited because most graph-based UQ work targets node classification rather than regression [FGG24], though Murad et al. [Mur+21] evaluate MC Dropout on graph convolutional networks for spatial air quality prediction and similarly observe that CRPS closely tracks MAE for uncalibrated probabilistic models. The PIT histogram (Section 5.9.4) makes the miscalibration visually explicit: a prominent spike near zero (28.4% of observations in the first 5% bin) and U-shaped tails reveal systematic underdispersion, consistent with the ECE of 0.265 and the scaling factor $k_{95} = 11.34$. Importantly, the PIT also exposes an *asymmetry* not visible in the ECE: the left-skewed PIT mean (0.433 vs the ideal 0.500) indicates that the model overpredicts more often than it underpredicts at extreme confidence levels. Finally, the Winkler interval scores (Section 5.9.5) quantify the calibration penalty in decision-relevant terms: raw Gaussian 90% intervals achieve only 48.6% coverage and incur a mean Winkler score of 49.68, while conformal sigma-scaled intervals achieve the target 90% coverage with a 35% lower score (32.33). The sigma-scaled advantage over fixed-width conformal intervals (10% improvement at 90%) confirms that the MC Dropout ranking signal retains value even within the conformal framework.

Effect of temperature scaling on the PIT diagnostic. Since the PIT histogram reveals severe underdispersion under raw σ , a natural question is whether temperature scaling alleviates it. Replacing σ with $\sigma \cdot T$ ($T = 2.70$) in the PIT computation yields a substantially improved histogram (Figure 6.3): the Kolmogorov–Smirnov statistic drops from 0.245 to 0.104 (−57%), the PIT mean shifts from 0.433 to 0.471 (closer to the ideal 0.500), and the PIT standard deviation decreases from 0.399 to 0.302 (ideal: 0.289). The first-bin spike shrinks from 28.4% to 8.8%, eliminating the most extreme underdispersion artefact. However, the last bin remains elevated (11.7% vs the ideal 5.0%), indicating that the Gaussian assumption still cannot fully capture the heavy right tail of the residual distribution. This partial improvement is consistent with the residual ECE of 0.048 after scaling and reinforces the conclusion that temperature scaling is a useful intermediate step but not a substitute for distribution-free conformal prediction when formal coverage guarantees are required.

6.4.9 Conditional Coverage: A Theoretical Limitation Resolved by Adaptive Conformal Prediction

Barber et al. [Bar+21] prove that distribution-free conditional coverage is impossible without additional assumptions—a fundamental theoretical result that applies to *all* conformal methods, not a weakness of our particular implementation. Our analysis confirms this theoretical prediction in practice: standard conformal intervals with a

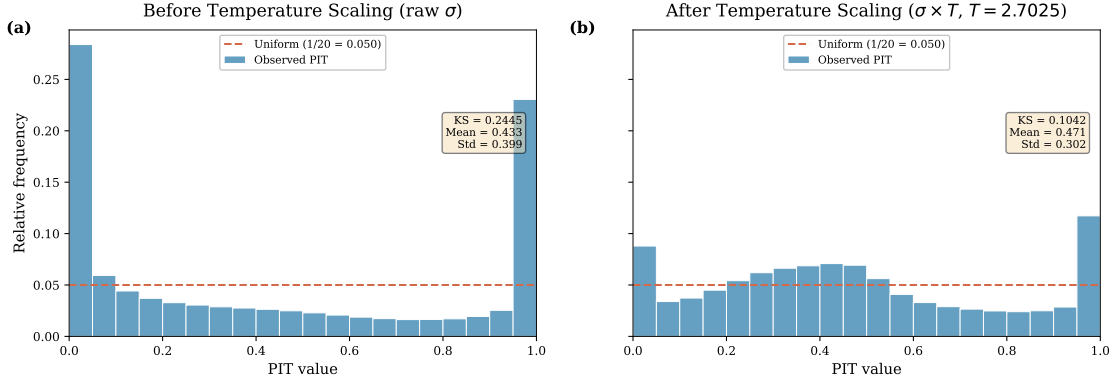


Figure 6.3: PIT histograms for T8 MC Dropout ($S = 30, 3,163,500$ nodes) before and after temperature scaling ($T = 2.70$). **Left:** Raw σ produces severe underdispersion ($KS = 0.245$, first-bin spike at 28.4%). **Right:** Scaling σ by $T = 2.70$ reduces the KS statistic to 0.104 (−57%) and brings the PIT distribution substantially closer to uniformity, though residual deviation in the last bin (11.7%) indicates the Gaussian assumption remains imperfect. Source: `pit_after_tempscaling_t8.json`.

global quantile over-cover low-uncertainty nodes (98.6% at 90% nominal for decile D1) and under-cover high-uncertainty nodes (62.9% for D10). This disparity is the mathematically inevitable consequence of applying fixed-width intervals to a heterogeneous prediction task.

Adaptive conformal prediction, which normalises nonconformity scores by σ , provides the principled solution: it produces node-specific interval widths that are wider where MC Dropout signals higher uncertainty and narrower where it signals confidence. This largely resolves the conditional coverage disparity (90.0%–96.2% across deciles at the 90% level), demonstrating that the MC Dropout ranking signal carries genuine calibration information even when its absolute magnitudes are miscalibrated. The combination of MC Dropout uncertainty (for adaptive interval widths) with conformal prediction (for the marginal coverage guarantee) thus addresses both the ranking and calibration limitations identified in this thesis.

The stratified analysis by feature quartiles (Section 5.12.1) reveals that prediction difficulty and uncertainty quality both vary substantially across the input space. Low-volume, low-capacity road segments are predicted with high accuracy ($MAE \approx 1.0$ – 1.4 veh/h) and excellent ranking ($\rho \approx 0.74$ – 0.81), while high-traffic segments are harder to predict ($MAE \approx 4.5$ – 8.0 veh/h) with degraded ranking ($\rho \approx 0.26$ – 0.41). This heterogeneity has practical implications for deployment: uncertainty-guided filtering is most effective for the easy-to-predict majority of the network, where the model already achieves high accuracy. For the most challenging high-volume segments, the uncertainty signal is weaker—motivating future work on feature-conditional or group-conditional conformal methods that could provide tighter guarantees for specific road categories.

6.4.10 Stability Across Scenarios and Trials

The per-graph analysis (Section 5.11) provides reassurance that the aggregate UQ quality metrics are not driven by outlier scenarios. Spearman ρ is consistent across 100 test graphs (mean = 0.464, std = 0.023, range [0.41, 0.51]), confirming that uncertainty ranking is a robust property of the model across diverse traffic scenarios. Cross-trial validation on T7 (Section 5.10, Section 5.10.4) further strengthens this conclusion: T7 achieves AUROC values of 0.742 and 0.715 for top-10% and top-20% error detection—slightly below T8 but following the same pattern.

6.5 Future Directions

6.5.1 Conformal Risk Control

Beyond fixed-coverage conformal prediction, conformal risk control [AB23] enables coverage guarantees on arbitrary loss functions. Combined with the adaptive conformal approach validated in Section 5.5.1, this could provide even tighter, uncertainty-adaptive intervals for traffic prediction.

6.5.2 Advanced Calibration Methods

Temperature scaling ($T = 2.70$) reduced ECE by 82% but left residual miscalibration at high nominal levels due to the non-Gaussian residual distribution. More flexible calibration approaches—such as isotonic regression, Platt scaling with learned variance, or quantile recalibration [KFE18]—could potentially close this remaining gap without requiring model retraining.

6.5.3 Architecturally Diverse Ensembles

The corrected ensemble experiments (Section 5.3.1) show that multi-model disagreement across five PointNetTransfGAT variants provides useful but weaker uncertainty ranking ($\rho = 0.4333$) than single-model MC Dropout ($\rho = 0.4908$). A natural extension is to explore ensembles combining architecturally diverse models (e.g., GAT, GraphSAGE, GCN) trained on the same data, as true architectural diversity may unlock ensemble UQ benefits not observed with identical architectures.

6.5.4 Post-Hoc Epistemic Uncertainty via Energy-Based Models

An alternative to MC Dropout for epistemic uncertainty estimation is the Graph Energy-Based Model (GEBM) approach proposed by Fuchsgruber et al. [FGG24]. GEBM fits a lightweight energy-based model to the GNN’s learned node representations and quantifies epistemic uncertainty through the energy score of test nodes relative to the training distribution—requiring only a single deterministic forward pass at inference time. This eliminates the S -fold computational overhead of MC Dropout (228 minutes

for $S = 30$ in our setting) and avoids the variance-convergence trade-off studied in Figure 6.1. GEBM also explicitly targets epistemic uncertainty, addressing the limitation that MC Dropout does not decompose total uncertainty into epistemic and aleatoric components (Section 6.4.4). Applying GEBM to the PointNetTransfGAT architecture as a post-hoc method—without retraining—would provide a direct comparison against MC Dropout on the same model and dataset, and could clarify whether the moderate Spearman ρ observed here reflects a fundamental limitation of the architecture or of the uncertainty method.

6.5.5 Full 10,000-Scenario Training

The most direct improvement would be training on the full 10,000-scenario dataset to assess whether model accuracy and uncertainty quality improve proportionally with data scale.

6.5.6 Out-of-Distribution Detection

Explicit OOD detectors could flag scenarios unlike any training example, complementing uncertainty estimation with categorical reliability assessment for novel policy interventions.

6.5.7 Active Learning

Uncertainty estimates could guide active learning: selecting high-uncertainty scenarios for additional MATSim simulation to expand training data where most needed.

7 Conclusion

This thesis investigated uncertainty quantification methods for Graph Neural Network surrogates of agent-based transport models. Working with 1,000 of 10,000 available MATSim scenarios (10% subset) on a Paris road network dataset, we compared MC Dropout, ensemble variance, and conformal prediction as uncertainty quantification approaches.

7.1 Summary of Findings

7.1.1 Answers to Research Questions

RQ1: *How effectively does MC Dropout capture epistemic uncertainty in GNN-based traffic surrogates?* MC Dropout with $S = 30$ stochastic forward passes achieves Spearman $\rho = 0.4820$ between predicted uncertainty and absolute prediction error for Trial 8 (100 test graphs, 3,163,500 predictions). This moderate positive correlation is practically meaningful: it enables selective prediction that reduces MAE by 41.2% at 50% retention and error detection with AUROC of 0.76 at the top-10% error threshold. Per-graph analysis confirms that this ranking quality is consistent across all 100 test scenarios (ρ mean = 0.464, std = 0.023), with no graph below $\rho = 0.41$.

RQ2: *How does MC Dropout compare to ensemble-based uncertainty quantification?* MC Dropout ($\rho = 0.4908$, Experiment A with corrected model loading; Section 5.3.1) outperforms seed-based ensemble variance ($\rho = 0.4370$) by 12.3% and multi-model ensemble disagreement ($\rho = 0.4333$, Experiment B) by 13.3%. Combined MC Dropout + ensemble variance ($\rho = 0.4909$) provides negligible improvement over MC Dropout alone. These results confirm that MC Dropout is the most effective single-method uncertainty estimator in this setting, at substantially lower computational cost than maintaining multiple trained models.

RQ3: *Does combining uncertainty estimates from architecturally identical models provide benefits?* Marginally. Combining MC Dropout and seed-based ensemble variance (Experiment A, $\rho = 0.4909$) barely exceeds MC Dropout alone ($\rho = 0.4908$), indicating that ensemble variance is largely redundant once MC Dropout is available. Multi-model ensemble disagreement (Experiment B, $\rho = 0.4333$) provides useful but weaker uncertainty ranking than MC Dropout. The weighted ensemble prediction ($R^2 = 0.5656$) does not improve over the best individual model (T8, $R^2 = 0.5957$). Architecturally diverse ensembles were not tested in this work.

RQ4: *Can distribution-free methods transform raw uncertainty into trustworthy prediction intervals?* Yes. Split conformal prediction achieves near-exact marginal coverage (90.02%

at 90% nominal, 95.01% at 95% nominal) on 1,581,750-node evaluation sets. Post-hoc temperature scaling ($T = 2.70$) reduces ECE from 0.269 to 0.048—an 82% improvement—though residual miscalibration persists at high nominal levels. Adaptive conformal prediction, which normalises nonconformity scores by MC Dropout σ , narrows the conditional coverage range from [62.9%, 98.6%] to [90.0%, 96.2%] across uncertainty deciles, demonstrating that the MC Dropout ranking signal carries genuine calibration information. These three post-hoc methods—temperature scaling, standard conformal prediction, and adaptive conformal prediction—require no model retraining and form a practical calibration hierarchy.

7.1.2 Primary Results

1. **Trial 8 achieves the best predictive performance among comparable trials:** $R^2 = 0.5957$, MAE = 3.96 veh/h, RMSE = 7.12 veh/h (test set, 3,163,500 predictions, 80/10/10 split). Trial 8 uses the GATConv(64→1) final layer and dropout = 0.2.
2. **MC Dropout provides effective uncertainty ranking:** Trial 8 MC Dropout ($S = 30$, 100 test graphs) achieves Spearman $\rho = 0.4820$ between predicted uncertainty and absolute prediction error. This confirms that high uncertainty reliably predicts where the model makes larger errors.
3. **Selective prediction substantially reduces effective error:** Filtering to the 50% most confident predictions reduces MAE by 41.2% (from 3.95 to ≈ 2.32 veh/h, MC Dropout mean). This demonstrates practical value: for decisions where reliability is critical, uncertainty-guided filtering provides meaningful accuracy improvements.
4. **Conformal prediction achieves guaranteed coverage:** Split conformal prediction on T8 (50/50 calibration–evaluation split, 50 graphs each) delivers 90.02% coverage at ± 9.92 veh/h and 95.01% at ± 14.68 veh/h, confirming that coverage-guaranteed intervals are achievable for traffic prediction. A separate calibration audit using a larger 20/80 split corroborates these results (90.2% at ± 9.99 veh/h; 95.1% at ± 14.77 veh/h; 2,530,800 evaluation nodes), demonstrating robustness to the choice of calibration set size.
5. **MC Dropout uncertainty is not a calibrated standard deviation:** The factor $k_{95} = 11.34$ required to achieve 95% coverage via $\pm k\sigma$ intervals (vs $k = 1.96$ for a calibrated Gaussian) confirms that raw MC Dropout uncertainty should be used as a ranking signal only, not as a probability distribution.
6. **Ensemble methods provide useful but weaker uncertainty ranking:** MC Dropout ($\rho = 0.4908$) outperforms seed-based ensemble variance ($\rho = 0.4370$) and multi-model disagreement ($\rho = 0.4333$). Combined MC Dropout + ensemble variance ($\rho = 0.4909$) provides negligible additional benefit. The weighted multi-model ensemble ($R^2 = 0.5656$) does not improve over the best individual model (T8,

$R^2 = 0.5957$). A PyTorch Geometric API version mismatch originally corrupted these results; the corrected evaluation (Section 5.3.1) enables a reliable comparison.

7. **UQ findings are robust across trials:** Replication of selective prediction and calibration audit on Trial 7 ($\rho = 0.4460$, dropout = 0.3) confirms the same qualitative conclusions: MC Dropout provides useful uncertainty ranking, raw σ values are not calibrated, and conformal prediction restores near-exact coverage. The main findings are not specific to a single model configuration (Section 5.10).
8. **Proper scoring rules confirm sharp but underdispersed predictive distributions:** The Continuous Ranked Probability Score (CRPS/MAE = 0.857) indicates that the Gaussian predictive distribution is sharper than the deterministic baseline. However, Probability Integral Transform (PIT) analysis reveals systematic underdispersion (KS statistic = 0.245, excess mass in the first and last histogram bins), confirming that the MC Dropout σ underestimates the true prediction error spread. Winkler score comparisons show that conformal intervals substantially outperform naïve Gaussian intervals (32.3 vs 49.7 at the 90% level), quantifying the benefit of distribution-free calibration.
9. **$S = 30$ forward passes provide near-optimal uncertainty estimates:** An S -convergence study on 10 test graphs shows that increasing S from 30 to 50 improves Spearman ρ by less than 1% (0.4584 to 0.4632), with the convergence curve flattening sharply after $S \approx 25$. This confirms that $S = 30$ represents a practical cost–quality trade-off for MC Dropout uncertainty estimation.

7.1.3 Practical Recommendations

For practitioners deploying GNN-based traffic surrogates, MC Dropout with $S = 30$ samples and dropout = 0.2 provides the best uncertainty ranking found in this work. Uncertainty estimates should be used for selective prediction—retaining the 50% most confident predictions reduces MAE by 41.2%—and conformal prediction should be applied when formal coverage guarantees are contractually or operationally required. Raw MC Dropout σ should not be interpreted as a calibrated standard deviation. Multi-model ensembles provide complementary but weaker uncertainty ranking ($\rho = 0.4333$ vs 0.4908 for MC Dropout); architecturally diverse ensembles have not been tested here but could provide stronger uncertainty signals. The cross-trial replication on T7 confirms that these recommendations are not specific to a single hyperparameter configuration.

7.2 Contributions

The primary contributions of this thesis are:

1. **Systematic UQ evaluation for GNN traffic surrogates:** First comprehensive comparison of MC Dropout, ensemble variance, and conformal prediction for

node-level traffic volume change prediction at scale (3.16 million predictions per evaluation).

2. **Empirical evidence for MC Dropout effectiveness:** Demonstrated $\rho = 0.4820$ uncertainty-error correlation for T8 MC Dropout. T8 (dropout = 0.2, 80/10/10 split) achieves higher ρ than T5–T7 (ρ : 0.4186–0.4460, dropout = 0.3). Note that T8 differs from T5–T6 in both dropout rate and data split, and from T7 in dropout rate and learning rate; batch size is identical (8) across all four trials, so the ρ improvement cannot be attributed solely to the reduced dropout rate.
3. **Conformal prediction coverage verification:** Confirmed that split conformal prediction achieves nominal coverage guarantees on large-scale traffic prediction (1.58M-node evaluation sets).
4. **Selective prediction utility:** Quantified that uncertainty-guided filtering achieves 41.2% MAE reduction at 50% retention, providing concrete operational guidance.
5. **Identification of ensemble evaluation limitations:** Diagnosed a PyTorch Geometric API version mismatch that silently corrupts GATConv weights when loading checkpoints with `strict=False`, providing a cautionary finding for reproducibility in GNN ensemble research.
6. **Proper scoring rule evaluation of MC Dropout predictive distributions:** Applied CRPS, PIT, and Winkler score analysis to characterise the MC Dropout predictive distribution beyond simple ranking metrics, revealing systematic underdispersion and quantifying the calibration benefit of conformal prediction over naïve Gaussian intervals.

7.3 Broader Impact

This work addresses the question of how to quantify prediction uncertainty in GNN-based transport surrogates. As cities increasingly rely on machine learning for policy decisions, understanding model limitations becomes necessary. Uncertainty quantification provides a mechanism for expressing model confidence, enabling more informed decision-making.

Our findings demonstrate that effective UQ is achievable with minimal architectural changes (MC Dropout, dropout = 0.2) and manageable computational overhead, and that formal coverage guarantees are attainable through conformal prediction. These methods require no specialised infrastructure and can be applied to existing trained models.

7.4 Future Work

This work leaves several open questions. The most direct extension is training on the **full 10,000-scenario dataset** to assess whether model accuracy and uncertainty quality scale proportionally with training data. On the architectural side, **diverse ensembles** combining GNN variants with different inductive biases (GAT, GraphSAGE, GCN) may provide the complementary uncertainty signals that identical-architecture ensembles failed to deliver here. Within the UQ methods themselves, **conformal risk control** [AB23] offers a path to tighter, adaptive coverage guarantees beyond the fixed-width intervals produced by standard split conformal prediction. **Post-hoc calibration** of MC Dropout uncertainty—through isotonic regression or temperature scaling—could reduce k_{95} from 11.34 toward 1.96, allowing σ to serve as a genuine probability estimate rather than a ranking signal. Finally, **multi-city validation** across different network topologies and traffic regimes, combined with explicit **epistemic–aleatoric decomposition**, would strengthen confidence in the generalisability of these findings.

7.5 Closing Remarks

Deploying machine learning surrogates responsibly in urban planning requires more than accurate predictions—it requires systematic quantification of where those predictions may fail. Uncertainty quantification provides that quantification in a form that is operationally useful: MC Dropout gives planners a per-prediction confidence score, conformal prediction gives them a coverage guarantee, and selective prediction lets them trade breadth of coverage for depth of accuracy. The experiments here, conducted on a 10% data subset under realistic resource constraints, show that all three are achievable with the PointNetTransfGAT architecture at no meaningful cost to predictive accuracy. The practical case for uncertainty-aware surrogate deployment in transport planning is supported by the experiments presented here.

Abbreviations

ABM	Agent-Based Model
BNN	Bayesian Neural Network
CP	Conformal Prediction
DL	Deep Learning
GAT	Graph Attention Network
GNN	Graph Neural Network
MAE	Mean Absolute Error
MATSim	Multi-Agent Transport Simulation
MC	Monte Carlo
ML	Machine Learning
MLP	Multi-Layer Perceptron
MSE	Mean Squared Error
OOD	Out-of-Distribution
PICP	Prediction Interval Coverage Probability
RMSE	Root Mean Square Error
TUM	Technical University of Munich
UQ	Uncertainty Quantification

List of Figures

1.1	Paris road network as directed graph for the GNN surrogate	2
1.2	Overall thesis pipeline: from MATSim scenarios to policy decisions	5
3.1	Distribution of input features across the training set	14
3.2	Node-level vs graph-level GNN prediction with MC Dropout uncertainty	15
3.3	PointNetTransfGAT architecture: data flow diagram	17
3.4	End-to-end data flow: road network to traffic prediction with uncertainty	17
3.5	Trial progression T1→T8: architecture changes and performance	19
3.6	MC Dropout inference: S=30 stochastic forward passes	20
5.1	Test-set performance across Trials 2–8	29
5.2	Deterministic vs MC Dropout inference, T8	31
5.3	Spearman ρ across all UQ methods	33
5.4	Conformal prediction workflow for T8	34
5.5	Conformal prediction coverage and interval widths (T8)	35
5.6	T8 conditional conformal coverage by uncertainty decile	36
5.7	Scaling factor k_{95} for 95% coverage: Gaussian vs T8	37
5.8	T8 selective prediction curve: MAE vs retention fraction	39
5.9	T8 calibration curve: nominal vs achieved coverage across methods . . .	41
5.10	T8 interval width comparison: raw MC vs conformal at each coverage level	41
5.11	T8 reliability diagram: expected vs observed Gaussian coverage	42
5.12	T8 temperature scaling: reliability diagram before and after calibration .	43
5.13	PIT histogram for T8 MC Dropout	46
5.14	T8 error detection ROC curves for MC Dropout uncertainty	47
5.15	T7 selective prediction curve: MAE vs retention fraction	49
5.16	T7 calibration curve: nominal vs achieved coverage across methods . . .	50
5.17	T7 interval width comparison: raw MC vs conformal at each coverage level	50
5.18	T7 vs T8 uncertainty quality comparison	52
5.19	T8 per-graph uncertainty variation across 100 test graphs	53
5.20	Spearman correlation: input features vs absolute prediction error (T8) . .	54
5.21	T8 stratified UQ by input feature quartiles	55
6.1	S-convergence of MC Dropout uncertainty quality (T8)	58
6.2	Uncertainty-guided policy decision framework (T8)	62
6.3	PIT histograms before and after temperature scaling (T8)	66

List of Tables

3.1	Input features for each road segment (node)	13
3.2	PointNetTransfGAT layer progression	16
3.3	PointNetTransfGAT hyperparameters by trial group	18
4.1	Paris road network dataset statistics	25
4.2	Training trials and hyperparameter configurations	26
5.1	Test-set performance across all 8 training trials	28
5.2	MC Dropout uncertainty-error correlation (Spearman ρ)	30
5.3	Experiment A: MC Dropout vs ensemble variance (Spearman ρ)	32
5.4	Experiment B: Multi-model ensemble uncertainty quality	32
5.5	Conformal prediction results for T8	35
5.6	Scaling factor k_{95} for 95% coverage: Gaussian vs T8 MC Dropout	37
5.7	MC Dropout vs conformal prediction coverage across all nominal levels	38
5.8	T8 selective prediction: MAE vs retention threshold	39
5.9	T8 calibration audit: raw MC Dropout vs conformal coverage (20/80 split)	40
5.10	Gaussian NLL for MC Dropout uncertainty	44
5.11	T8 CRPS: aggregate and stratified by uncertainty decile	45
5.12	Winkler interval scores for Gaussian, conformal absolute, and conformal sigma-scaled intervals	46
5.13	T8 error detection: AUROC and AUPRC for MC Dropout uncertainty	47
5.14	T7 selective prediction: MAE vs retention threshold	48
5.15	T7 calibration audit: raw MC Dropout vs conformal coverage (20/80 split)	49
5.16	T7 vs T8 UQ comparison	51
5.17	T7 vs T8 error detection comparison	51
5.18	Summary of key results for T8 (best model)	56
6.1	Summary comparison of UQ methods evaluated in this thesis	61

List of Algorithms

1	MC Dropout uncertainty estimation	18
2	Split conformal prediction	21
3	Selective prediction via uncertainty filtering	21
4	Post-hoc temperature scaling for uncertainty calibration	22

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